

Pre-Calculus Compendium



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Preface

Learning mathematics is like learning a new language. To start with, it looks like lots of strange symbols and difficult, but with time and effort, you will experience that it becomes gradually easier. *Many mathematical terminologies are related and built upon each other.* To understand the meaning of a specific terminology will help you to understand many more.

Being frustrated over your own progression and your work on the subject is a natural part of the learning process. Remember: learning is not just a good effort, but it also requires intense concentration.

This compendium is a summary of some chosen fundamental mathematical topics. Some examples are meant as additional explanation to teaching material and show how it can be used to solve practical exercises.

Each chapter has an exercise section followed by a solution to the exercises.

When learning a new subject is not just to learn a procedure to solve exercises. It is at least as important to ask yourself *why* and try to understand *how things are connected*: it will make learning easier. The more you understand the underlying mathematics, the easier it is to use it to solve problems in other disciplines as well.

The compendium is organized as follows:

Part 1: Algebra Chapter 1 - Fundamentals

Part 2: Functions

Part 3: Trigonometry, Integration and some applications, Vectors

Test yourself

Before you start reading the content of these notes, test yourself!

NOTE: the exercises below should be solved without the help of a calculator.

Exercise 0.1 Compute:

a) $4 + 3 \cdot 2^4 + 2(4^2 - 3)$, b) $3^2(4 + 2) - 4(4 + 2^2)$, c) $(-2)^5 \cdot (5^2 - 4^2 - 3^2) + 13 \cdot (-1)^{13}$

Exercise 0.2 Compute:

a) $\frac{25}{16} \cdot \frac{32}{50}$, b) $\frac{3}{7} \cdot \frac{28}{15}$, c) $\frac{\frac{1}{3}}{\frac{5}{6}}$, d) $\frac{4}{7} : 7$

Exercise 0.3 Compute:

a) $1 + \frac{5}{12} - \frac{7}{18}$, b) $\left(2 + \frac{1}{2}\right) \cdot \frac{2}{5}$, c) $\left(\frac{1}{3} + \frac{2}{5}\right) : \frac{22}{5}$

Exercise 0.4 Compute:

a) $\frac{\frac{56}{15}}{\frac{64}{21}}$, b) $\frac{1 + \frac{1}{2}}{2 - \frac{1}{3}}$, c) $\frac{\frac{1}{3} - \frac{5}{6}}{\frac{2}{3} + \frac{5}{12}}$

Exercise 0.5 Compute:

a) $\frac{\frac{x}{2}}{\frac{2x}{5}}$, b) $\frac{\frac{x}{3}}{\frac{3}{9}}$, c) $\frac{\frac{a}{2} - \frac{1}{3}}{\frac{a}{6} - \frac{1}{4}}$, d) $\frac{\frac{3}{x} + \frac{3}{2}}{3 + \frac{3}{2x}}$

Exercise 0.6 Simplify the following expressions:

a) $ab(1 + 2b) - 2a(b^2 - b)$ b) $(2x + 1)(2x + 1) - (2x + 1)(2x - 1)$

c) $(a - 3b)^2 - (a + 3b)(a - 3b)$ d) $\left(\frac{1}{2}a - b\right)\left(\frac{1}{2}a + b\right) - \frac{1}{2}(2b - a)^2 + 3b\left(b - \frac{2}{3}a + \frac{1}{12b}a^2\right)$

Exercise 0.7 Write as simple as possible:

$$\text{a)} \quad \frac{3a^2 - 6ab + 3b^2}{6(a - b)}, \quad \text{b)} \quad \frac{a^2 - 6ab + 9b^2}{a^2 - 9b^2}$$

Exercise 0.8 Use binomial perfect squares and differences to simplify the following expressions:

$$\begin{array}{ll} \text{a)} & (x + 5)^2 - (x + 5)(x - 5) \quad \text{b)} \quad (x - 3)^2 - (x + 3)^2 \\ \text{c)} & (3a + 2)(2 + 3a) \quad \text{d)} \quad (\sqrt{5} + \sqrt{2})(\sqrt{5} - \sqrt{2}) - (\sqrt{5} - 1)^2 \end{array}$$

Exercise 0.9 Factorize the following expressions:

$$\text{a)} \quad 4x^2 + 2x, \quad \text{b)} \quad x^2 - 81, \quad \text{c)} \quad 2t^2 - 8, \quad \text{d)} \quad x^2 + 2x + 1$$

Exercise 0.10 Factorize the following expression using the zeros of the polynomials

$$\text{a)} \quad x^2 - 4x + 3, \quad \text{b)} \quad x^2 - x - 2, \quad \text{c)} \quad a^2 + 2a - 15, \quad \text{d)} \quad y^2 + 11y + 28$$

Exercise 0.11 Shorten the following fractions

$$\text{a)} \quad \frac{x^2 - 1}{2x + 2}, \quad \text{b)} \quad \frac{3x^2 - 12}{6x - 12}, \quad \text{c)} \quad \frac{x^3 - x^2}{x^2 - 1}, \quad \text{d)} \quad \frac{2x^2y - 4xy^2}{x^3 - 4xy^2}$$

Exercise 0.12

- a) The price of a good was raised from 10 to 20 kr. What's the percent of price rise?
- b) A family of two adults and two children paid a total of 220 kr ticket price for an arrangement. Another family, one adult and three children, paid 190 kr.
What is the price of a child ticket and how much is an adult ticket?
- c) A cinema theater has 80 places. An adult ticket costs 100 kr and a child ticket costs 60 kr. At one show, the theater was sold out and all seats taken. The total income for the cinema was 6 000 kr.
How many adults and how many children were at that show?
- d) If $(x + y)^2 = 100$ and $x^2 + y^2 = 60$, what is the product $x \cdot y$?
- e) The sum of two positive number is 10 and the sum of their squares is 52. What are the two positive numbers?

Solutions

0.1

a) 78 **b)** 22 **c)** -13

0.2

a) 1 **b)** $\frac{4}{5}$ **c)** $\frac{2}{5}$ **d)** $\frac{2}{21}$

0.3

a) $\frac{37}{36}$ **b)** 1 **c)** $\frac{1}{6}$

0.4

a) $\frac{49}{40}$ **b)** $\frac{9}{10}$ **c)** $-\frac{6}{13}$

0.5

a) $\frac{5}{4}$ **b)** $\frac{3x}{5}$ **c)** $\frac{6(a-1)}{2a-9}$ **d)** $\frac{2+x}{2x+1}$

0.6

a) $3ab$ **b)** $4x+2$ **c)** $6b(3b-a)$ **d)** 0

0.7

a) $\frac{3a^2-6ab+3b^2}{6(a-b)} = \frac{\cancel{3}(a^2-2ab+b^2)}{2 \cdot \cancel{3}(a-b)} = \frac{(\cancel{a-b})(a-b)}{2(\cancel{a-b})} = \underline{\underline{\frac{a-b}{2}}}$

b) $\frac{a^2-6ab+9b^2}{a^2-9b^2} = \frac{(a-3b)^2}{(\cancel{a-3b})(a+3b)} = \underline{\underline{\frac{a-3b}{a+3b}}}$

0.8 **a)** $10(x+5)$ **b)** $-12x$ **c)** $9a^2+12a+4$ **d)** $2\sqrt{5}-3$

0.9 **a)** $2x(2x+1)$ **b)** $(x-9)(x+9)$ **c)** $2(t-2)(t+2)$ **d)** $(x+1)^2$

0.10

a) $(x-1)(x-3)$ **b)** $(x+1)(x-2)$ **c)** $(a-3)(a+5)$ **d)** $(y+4)(y+7)$

0.11

a) $\frac{x-1}{2}$ **b)** $\frac{x+2}{2}$ **c)** $\frac{x^2}{x+1}$ **d)** $\frac{2y}{x+2y}$

0.12 **a)** 20% **b)** 40 kr and 70kr **c)** 30 adult and 50 child **d)** 20 **e)** 4 and 6

Chapter 1

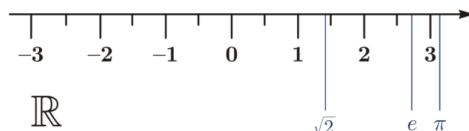
Fundamentals

This chapter is a repetition of fundamental concepts and principles. These are:

- Number sets, intervals
- Computing with letters (algebra) and fractions
- Rules for computing with powers of numbers
- Absolute value
- Geometric and arithmetic series

1.1 The real line and real numbers

Real numbers are the set of all numbers corresponding to points on an infinitely long number line. It is denoted as $\mathbb{R}$ or $\mathbf{R}$.



1.2 Sets and number sets

A *set* is a collection of objects, called *elements*. Elements can, in principle, be whatever, for instance numbers, persons, cars or even other sets.

In mathematics we use the following short-hand notation:

notation	meaning
$x \in M$	x is an element of the set M
$x \notin M$	x is not in the set M
$\emptyset$	empty set (set with no elements)

We recall the following sets of numbers:

- Set of all *natural* numbers, $\mathbb{N} = \{1, 2, 3, \dots\}$
- Set of all *integer* numbers, $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
- Set of all *rational* numbers, i.e. numbers that can be written as a fraction (ratio) of two integers, $\mathbb{Q} = \{\frac{p}{q} \mid p, q \text{ integers}, q \neq 0\}$
- Set of all *real* numbers, $\mathbb{R}$ that contains all numbers on the real line.

Numbers that are not rational are called *irrational* numbers. For example, $\sqrt{2}$ is an irrational number.

Example 1.1

Some rational numbers: 0.18 (is the same as $\frac{9}{5}$), $\frac{2}{3}$, $-\frac{3}{7}$, 5

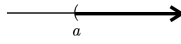
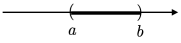
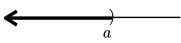
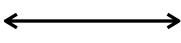
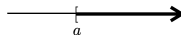
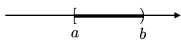
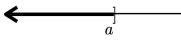
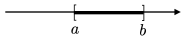
Some irrational numbers: $\sqrt{2}$, π

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1.3 Intervals

Intervals are portions of the real line. Intervals can be:

- *open*, if none of the endpoints belong to the interval. These are indicated with parentheses. For instance $(0, 3)$ is the set of real numbers $0 < x < 3$. In norwegian school system the parentheses are often replaced by $<$, $>$ symbols, and the notation is used often also in calculus courses (grunnkurs) at norwegian universities. Thus $<0, 3>$ is the same as $(0, 3)$.
- *half-open* if only one of the endpoints belong to the interval. For instance, the interval $[0, 3)$ is the set of real numbers x such that $0 \leq x < 3$. Note that 0 is included (we use a square bracket), 3 is not, and we use parenthesis. Similarly, $(0, 3]$ is the same as $0 < x \leq 3$.
- *closed* if both the endpoints belong to the interval. These are denoted with square brackets. For instance $[0, 3]$ is the same as $0 \leq x \leq 3$.

Open interval	Notation	Norwegian school notation	Grafical notation
$x > a$	(a, ∞)	$<a, \infty>$	
$a < x < b$	(a, b)	$<a, b>$	
$x < a$	$(-\infty, a)$	$<-\infty, a>$	
$x \in \mathbb{R}$	$(-\infty, \infty)$	$<-\infty, \infty>$	
Half-open interval	Notation	Norwegian school notation	Grafical notation
$x \geq a$	$[a, \infty)$	$[a, \infty>$	
$a \leq x < b$	$[a, b)$	$[a, b>$	
$x \leq a$	$(-\infty, a]$	$<-\infty, a]$	
Closed interval	Notation	Norwegian school notation	Grafical notation
$a \leq x \leq b$	$[a, b]$	$[a, b]$	

1.4 Order of operations

It is important to know the convention for the order of operations in algebra. Without the convention, the same expression might have different results!!

The order of operations is:

1. Parenthesis
2. Exponents
3. Multiplication and Division (from left to right)
4. Addition and Subtractions (from left to right)

This can be remembered in two ways: “Please Excuse My Dear Aunt Sally” or PEMDAS.

Example 1.2

Compute without calculator:

$$12 \cdot 5 - 3 \cdot 2^2 - 7(5 - 3)^3 + 2(-3)^2 + (-2)^3$$

Parentheses: $12 \cdot 5 - 3 \cdot 2^2 - 7 \cdot 2^3 + 2(-3)^2 + (-2)^3$

Powers: $12 \cdot 5 - 3 \cdot 4 - 7 \cdot 8 + 2 \cdot 9 - 8$

Multiplications: $60 - 12 - 56 + 18 - 8$

Additions and Subtractions: $60 - 12 - 56 + 18 - 8 = \underline{\underline{2}}$

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1.5 Algebra and fractions

For most people, Algebra is the same as doing computations with alphabetical letters. In Mathematics it is rather common that such letters are used, among others, in formulas, equations and inequations, identities and functions.

$$a + 2a = 3a$$

$$a \cdot a^2 = a^3$$

- **A term**

$2x^3$ is a *term*, where 2 is the *coefficient*, x is the *variable* and 3 is the *exponent*.

Two terms are separated with a + or a -: $2x^3 - 3x$

- **A factor**

$x(x + 2)$ is a term made of two factors, x and $x + 2$.

Two factors are separated by a multiplication sign (often omitted): xy

Properties

	Addition	Multiplication
Commutative law	$a + b = b + a$	$a \cdot b = b \cdot a$
Associative law	$a + (b + c) = (a + b) + c$	$a \cdot (b \cdot c) = (a \cdot b) \cdot c$
Distributive law	$a \cdot (b + c) = ab + ac$	
Opposite and inverse	$a + (-a) = -a + a = 0$	$a \cdot \frac{1}{a} = \frac{1}{a} \cdot a = 1, \quad a \neq 0$

The number 0 and 1:

Addition	Multiplication
$a + 0 = 0 + a = a$	$a \cdot 1 = 1 \cdot a = a$

1.6 Parentheses rules

$$a(b + c) = ab + ac$$

$$(a + b)(c + d) = ac + ad + bc + bd$$

1.7 Computing with fractions and broken fractions

$$\begin{aligned} \bullet \quad \frac{b+c}{a} &= \frac{b}{a} + \frac{c}{a} \\ \bullet \quad \frac{a}{b} \cdot \frac{c}{d} &= \frac{a \cdot c}{b \cdot d} \\ \bullet \quad a : \frac{c}{d} &= a \cdot \frac{d}{c} = \frac{a \cdot d}{c} \end{aligned}$$

$$\text{Remember: } \frac{a}{b+c} \neq \frac{a}{b} + \frac{a}{c}$$

$$\text{Remember: } a \cdot \frac{c}{d} = \frac{a \cdot c}{d}$$

$$\text{Remember: } \frac{a}{c} = a : \frac{c}{d} = a \cdot \frac{d}{c} = \frac{a \cdot d}{c}$$

$$\bullet \quad \frac{a}{b} : \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c} = \frac{ad}{bc}$$

$$\text{Remember: } \frac{\frac{a}{b}}{\frac{c}{d}} = \frac{a}{b} \cdot \frac{d}{c} = \frac{ad}{bc}$$

1.8 Factorization

Factorization is a process in which a mathematical expression is decomposed in smaller objects (factors) that, when multiplied together, give back the original expression.

$$\text{Example 1: } x + 3x^2 + 5x - x^2 = 2x^2 + 6x = 2x(x + 3)$$

$$\text{Example 2: } 4a^2b - 12ab^2 + 8b^3 = 4b(a^2 - 3ab + 2b^2)$$

1.9 Lowest common denominator

The *lowest common denominator* is the smallest number divisible by all the denominators in a mathematical expression.

$$\text{Example 1: } \frac{1}{2a} + \frac{2}{3a} = \frac{1 \cdot 3}{2a \cdot 3} + \frac{2 \cdot 2}{3a \cdot 2} = \frac{3+4}{6a} = \frac{7}{\underline{\underline{6a}}}$$

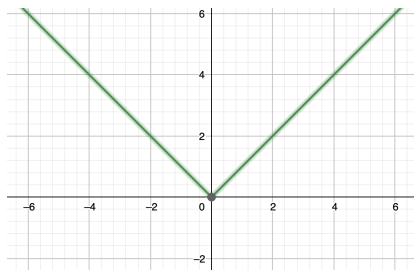
$$\text{Example 2: } \frac{x-1}{x+1} + \frac{2}{x} = \frac{(x-1)x}{(x+1)x} + \frac{2(x+1)}{x(x+1)} = \frac{x^2-x+2x+2}{x(x+1)} = \frac{x^2+x+2}{\underline{\underline{x(x+1)}}}$$

1.10 Absolute value

The absolute value of a real number is the numerical value of the number without the sign. The geometrical meaning of that absolute value is that of a distance.

$$|x| = \begin{cases} x, & \text{for } x \geq 0 \\ -x, & \text{for } x < 0 \end{cases}$$

Graph of $|x|$:



Remember: $\sqrt{x^2} = |x|$

The absolute value $|x|$ can be thought as the distance of the point x on the real line from the point 0.

In a similar manner, $|x - y|$ is the distance between two points x and y on the real line. Thus, to solve the inequality $|x - 1| < 2$ we need to find all the points on the real line that have distance less than 2 from 1, namely $-1 < x < 3$.

Some computational rules:

$$|x| = a \Leftrightarrow x = \pm a$$

$$|x| = 0 \Leftrightarrow x = 0$$

$$|x| < a \Leftrightarrow -a < x < a$$

$$|x| > a \Leftrightarrow x < -a \cup x > a$$

$$|ab| = |a| |b|$$

$$\left| \frac{a}{b} \right| = \frac{|a|}{|b|} \quad b \neq 0$$

One can show that:

$$|a \pm b| \leq |a| + |b|$$

$$||a| - |b|| \leq |a + b|$$

Example 1.3

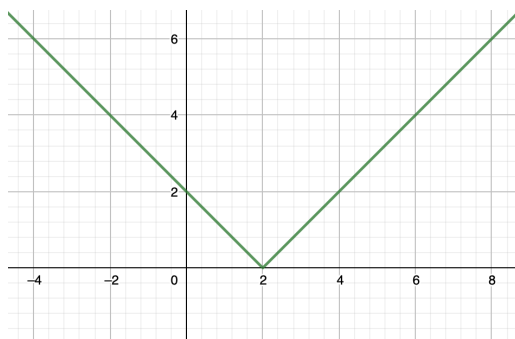
- Compute $|3 - |-7||$
- Solve the equation $|x - 2| = 3$
- Draw the graph of $y = |x - 2|$

Solutions:

a) $|3 - |-7|| = |3 - 7| = 4$

b) $x = 5$ and $x = -1$

c) $y = |x - 2| = \begin{cases} x - 2 & x \geq 2 \\ -(x - 2) & x < 2 \end{cases}$



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Example 1.4

Solve the following equations and inequalities:

a) $|x - 5| = 3$ b) $|x - 5| < 3$ c) $|x| = x$

Solutions:

a) $x - 5 = \pm 3 \Rightarrow \underline{x = 2 \vee x = 8}$

b) $|x - 5| < 3 \Rightarrow \underline{-3 < x - 5 < 3} \Rightarrow \underline{2 < x < 8}$

c) If $x \geq 0$ then $x = x$. If $x < 0$ then $-x = x$ hence $x = 0$. Therefore the solution is $x \geq 0$.

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1.11 Powers with integer exponents

a^n is called *power* and is defined as $a^n = \underbrace{a \cdot a \cdots a \cdot a}_{n \text{ times}}$, where a is the *base* and n , the *exponent*, is a natural number.

If $a \neq 0$, then $a^0 = 1$ and $a^{-n} = \frac{1}{a^n}$.

Power rules

$$a^m \cdot a^n = a^{m+n}$$

$$(a \cdot b)^n = a^n \cdot b^n$$

$$\frac{a^m}{a^n} = a^{m-n}$$

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$(a^n)^m = a^{nm} = (a^m)^n$$

$$\sqrt[p]{a^p} = a^{p/q}$$

Remember: for $a \neq 0$

$$a^0 = 1$$

$$a^{-n} = \frac{1}{a^n}$$

$$a^{\frac{1}{n}} = \sqrt[n]{a}$$

The square root is written as $\sqrt{}$ and $\sqrt{a} = a^{\frac{1}{2}}$. The n th root is written as $\sqrt[n]{}$ and $\sqrt[n]{a} = a^{\frac{1}{n}}$

Example 1.5

Write as simple as possible: a) $\left(\frac{x}{\sqrt{x}}\right)^2$ b) $\frac{x^3}{\sqrt{x}}$

Solutions:

$$\text{a) } \left(\frac{x}{\sqrt{x}}\right)^2 = \frac{x^2}{x} = x$$

$$\text{b) } \frac{x^3}{\sqrt{x}} = x^{(3-\frac{1}{2})} = x^{\frac{5}{2}} = x^{\frac{4+1}{2}} = x^2 \sqrt{x}$$

◇

1.12 Perfect squares and difference of squares

$$(a \pm b)^2 = a^2 \pm 2ab + b^2 \quad \text{Perfect squares. In norwegian 1st and 2nd square rules (1. og 2. kvadratsetning)}$$

$$a^2 - b^2 = (a + b)(a - b) \quad \text{Difference of squares. In norwegian: 3. kvadratsetning}$$

1.13 Geometric series

A geometric serie is a sum of terms $a_1 + a_2 + \cdots + a_n + \cdots$ such that the *ratio* between two successive terms is constant,

$$\frac{a_2}{a_1} = \frac{a_3}{a_2} = \cdots = \frac{a_n}{a_{n-1}} = k \quad (\text{quotient}).$$

The sum of the first n terms in the series is then $S = a_1 + a_1 k + a_1 k^2 + \cdots + a_1 k^{n-1}$ and can be written as $S = a_1 \frac{1 - k^n}{1 - k}$ provided $k \neq 1$, where n is the number of terms in the series.

Geometric series, n th term	$a_n = a_1 k^{n-1}$	k is the series' quotient
Sum of the first n terms in a geometric series	$S = a_1 \frac{1 - k^n}{1 - k}$	True for $k \neq 1$. For $k = 1$, then $S = na_1$
Sum of an infinite geometric series (convergent)	$S = \frac{a_1}{1 - k}$	For $-1 < k < 1$. If $a_1 = 0$ then $S = 0$
Rent formula	$K_n = K_0(1 + \frac{p}{100})^n$	Value K_n after n units (years, months, ...) with investment K_0 and percentual rate p

Example 1.6

A patient has to receive 20 injections at weekly intervals. The first dose he receives is $5 \mu\text{l}$ (microliters).

- a) How much should each injection increase (at a fast rate) such that the last dose is $100\,000 \mu\text{l}$?
b) How many microliters has the patient received after the 20th injection?

Solution:

a) $K_n = 100\,000$, $K_0 = 5$, $n = 20$.

$$100\,000 = 5(1 + \frac{p}{100})^{19} \Rightarrow 1 + \frac{p}{100} = (20\,000)^{\frac{1}{19}} \approx 1.684 \Rightarrow p \approx 0.684 \text{ or } p = 68.4\%.$$

b) We want to find the sum of the first 20 terms:

$$S = 5 + 5(1.684) + \cdots + 5(1.684)^{19} = 5 \frac{1 - (1.684)^{20}}{1 - 1.684} \approx \underline{\underline{2.459 \cdot 10^5 \mu\text{l}}} \quad \diamond$$

The summation symbol $\sum$

The symbol $\sum$ is the greek letter Sigma corresponding to the latin letter S. Thus $\sum$ stands as an abbreviation for “sum” and $\sum_{i=1}^n a_i = a_1 + a_2 + \cdots + a_n$.

Example 1.7

Write the sum $3(2)^4 + 3(2)^5 + \cdots + 3(2)^{10}$ using the summation symbol. Compute the sum.

Solution:

$$3(2)^4 + 3(2)^5 + \cdots + 3(2)^{10} = \sum_{i=4}^{10} 3(2)^i = 3 \sum_{i=1}^7 2^{i+3}$$

$$\text{The sum is } S = 3(2)^4(1 + 2 + \cdots + 2^6) = 3(2)^4 \frac{1 - 2^7}{1 - 2} = \underline{\underline{6096}} \quad \diamond$$

$$\text{NOTE: } a_1 + a_1 k + a_1 k^2 + \cdots + a_1 k^{n-1} = a_1 \sum_{i=1}^n k^{i-1} = a_1 \frac{1 - k^n}{1 - k}$$

Example 1.8

Compute the sum of the geometric series:

$$5 + 10 + 20 + \cdots + 640$$

Solution:

The series quotient is $k = 2$ and $a_1 = 5$.

To find the number of terms in the series, we can use $\frac{a_n}{a_1} = k^{n-1}$

$$\frac{640}{5} = 128 = 2^{n-1} \Rightarrow 2^{n-1} = 2^7 \Rightarrow n = 8, \text{ therefore } S = 5 \frac{1 - 2^8}{1 - 2} = \underline{\underline{1275}} \quad \diamond$$

1.14 Arithmetic series

A geometric serie is a sum of terms $a_1 + a_2 + \cdots + a_n + \cdots$ such that the *difference* between two successive terms is constant,

$$a_2 - a_1 = a_3 - a_2 = \cdots = a_n - a_{n-1} = d$$

There are two main ways to find the sum of a n -terms arithmetic series.

In the first method, we can observe that if we sort elements in a different way:

$$\begin{array}{cccc} a_1 & a_2 & a_3 & \cdots \\ + & + & + & \\ a_n & a_{n-1} & a_{n-3} & \cdots \end{array} \quad \underbrace{\hspace{10em}}_{n/2 \text{ terms}}$$

the sum of the elements in each column is always $a_1 + a_n$. We check it for $a_2 + a_{n-1}$: $a_2 = a_1 + d$ and $a_{n-1} = a_n - d$. Hence $a_2 + a_{n-1} = a_1 + d + a_n - d = a_1 + a_n$. Similary for the other columns. As there are $n/2$ such columns, we have

$$\sum_{i=1}^n a_i = (a_1 + a_n) + (a_2 + a_{n-1}) + \cdots = \frac{n}{2}(a_1 + a_n)$$

For the second method, note that $a_2 = a_1 + d$, $a_3 = a_2 + d = a_1 + 2d$, proceeding in the same way, $a_n = a_1 + (n-1)d$. Hence

$$\sum_{i=1}^n a_i = a_1 + (a_1 + d) + (a_2 + 2d) + \cdots + (a_1 + (n-1)d) = na_1 + \frac{n}{2}(n-1)d$$

Arithmetic series, n th term	$a_n = a_1 + (n-1)d$	d is the series' difference
Sum of the first n terms in a arithmetic series	$S = \frac{n}{2}(a_1 + a_n) = na_1 + \frac{n}{2}(n-1)d$	True for any d

REMARK: An infinite arithmetic series does not converge, that is it sum goes to infinity.

Example 1.9

Compute the sum of the arithmetic series

$$5 + 9 + 13 + \cdots + 49$$

Solution:

The difference d is: $d = 9 - 5 = 13 - 9 = 4$. To find n we can use $a_n = a_1 + (n-1)d$.

$$49 = 5 + 4(n-1) \Rightarrow n-1 = \frac{49-5}{4} = 11 \Rightarrow n = 12.$$

$$\text{Hence } 5 + 9 + 13 + \cdots + 49 = \frac{12}{2}(5 + 49) = \underline{\underline{324}} \quad \diamond$$

Exercises Chapter 1

The exercises should be solved without the use of calculator

Exercise 1.1

Compute:

$$\text{a) } 3 \cdot 2^4 - 3(4^2 - 4) \quad \text{b) } 3^2(4+2) - 4(2^2 + 2) \quad \text{c) } (-3)^5 \cdot (5^2 - 4^2 - 3^2) + 12 \cdot (-1)^{19}$$

Exercise 1.2

Compute:

$$\text{a) } \frac{36}{8} \cdot \frac{32}{144} \quad \text{b) } \frac{3}{7} \cdot \frac{35}{30} \quad \text{c) } \frac{\frac{2}{5}}{\frac{7}{15}} \quad \text{d) } \frac{2}{3} : 6$$

Exercise 1.3

Compute:

$$\text{a) } \frac{2}{x} - \frac{1}{2x} - \frac{4}{3x} \quad \text{b) } \frac{1}{x+1} - \frac{1}{x-1} + \frac{3}{x^2-1} \quad \text{c) } \left(\frac{1}{\sqrt{x}} + \frac{1}{x} \right) : \frac{1}{x}$$

Exercise 1.4

Compute:

$$\text{a) } \frac{\frac{15}{a}}{\frac{a}{3}} \quad \text{b) } \frac{b + \frac{b}{2}}{2b - \frac{b}{3}} \quad \text{c) } \frac{\frac{a}{2} - \frac{a}{5}}{\frac{a}{5} + \frac{a}{2}}$$

Exercise 1.5

Compute:

$$\text{a) } \frac{x-1}{1-\sqrt{x}} \quad \text{b) } \frac{x^2-2}{x-\sqrt{2}} \quad \text{c) } \frac{\frac{a}{2x-1} - \frac{a}{2x+1}}{\frac{1}{4x^2-1}}$$

Exercise 1.6

Simplify:

$$\begin{array}{ll} \text{a) } a(1+b^2) - 2a(b^2-b) - ab(2-b) & \text{b) } (3x+2)(3x+2) - (3x+2)(3x-2) \\ \text{c) } (a-2b)^2 - (a+2b)(a-2b) & \text{d) } (2+\sqrt{x})(2-\sqrt{x}) - (2-\sqrt{x})^2 - 4\sqrt{x} \end{array}$$

Exercise 1.7

Simplify:

a) $\frac{4a^2 - 4ab + b^2}{(2a - b)}$

b) $\frac{a^2 - 4ab + 4b^2}{a^2 - 4b^2}$

Exercise 1.8

Use perfect squares and difference of squares and compute:

a) $(x+3)^2 - (x+3)(x-3)$

b) $(x-a)^2 - (x+a)^2$

c) $(3a+2)(2-3a) - (3a-2)^2$

d) $(\sqrt{3} + \sqrt{2})(\sqrt{3} - \sqrt{2}) - (\sqrt{2} - 1)^2$

Exercise 1.9

Factorize the following expressions:

a) $9x^2 + 3x$

b) $4x^2 - 49$

c) $4t^2 - 9$

d) $x^2 + 6ax + 9a^2$

Exercise 1.10

Factorize with help of zeros:

a) $x^2 - 5x + 4$

b) $x^2 - 2x - 3$

c) $a^2 - a - 12$

d) $b^2 + b - 6$

Exercise 1.11

Simplify the fractions:

a) $\frac{x^2 - 9}{2x + 6}$

b) $\frac{x^2 - 36}{4x - 24}$

c) $\frac{x - x\sqrt{x}}{1 - \sqrt{x}}$

d) $\frac{3x^2y - 9xy^2}{x^3 - 9xy^2}$

Exercise 1.12Write as simple as possible: ($a > 0$):

a) $\frac{\sqrt{a^3} \cdot a^4}{(a^2)^3}$

b) $\frac{\sqrt{a^3}}{\sqrt[3]{a^2} \sqrt{a}}$

c) $\frac{a \cdot \sqrt{a} \cdot \sqrt[3]{a}}{\sqrt[3]{a^4}}$

d) $\frac{\sqrt[3]{a^2} \cdot \sqrt{a} \cdot \sqrt[6]{a} \cdot \sqrt[4]{a}}{a \sqrt[3]{\sqrt[4]{a^3} \cdot a^4}}$

Exercise 1.13

Compute the sums of the following series:

a) $1 + \frac{2}{3} + \frac{4}{9} + \frac{8}{27} + \dots$

b) $1 + \frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3} + \dots$ der $x > 1$

Exercise 1.14

a) Solve: $\sqrt{(x-1)^2} = 2.$

b) Draw the graph: $y = |x-1|.$

c) Find the maximal value of $f(x) = 5 - |x-1|.$

Solutions to the exercises Chapter 1

1.1 a) 12 b) 30 c) -12

1.2 a) 1 b) $\frac{1}{2}$ c) $\frac{6}{7}$ d) $\frac{1}{9}$

1.3

a) $\frac{6 \cdot 2}{6 \cdot x} - \frac{3 \cdot 1}{3 \cdot 2x} - \frac{2 \cdot 4}{2 \cdot 3x} = \frac{1}{6x}$ b) $\frac{1}{x^2 - 1}$ c) $\sqrt{x} + 1$

1.4 a) $\frac{1}{5}$ b) $\frac{b + \frac{b}{2}}{2b - \frac{b}{3}} = \frac{\frac{3b}{2}}{\frac{5b}{3}} = \frac{9}{10}$ c) $\frac{3}{7}$

1.5 a) $-(1 + \sqrt{x})$ b) $x + \sqrt{2}$ c) $2a$

1.6 a) a b) $4(3x + 2)$ c) $-4b(a - 2b)$ d) $-2x$

1.7 a) $2a - b$ b) $\frac{a - 2b}{a + 2b}$

1.8 a) $6(x + 3)$ b) $-4ax$ c) $-6a(3a - 2)$ d) $2(\sqrt{2} - 1)$

1.9 a) $3x(3x + 1)$ b) $(2x - 7)(2x + 7)$ c) $(2t - 3)(2t + 3)$ d) $(x + 3a)^2$

1.10 a) $(x - 1)(x - 4)$ b) $(x - 3)(x + 1)$ c) $(a - 4)(a + 3)$ d) $(b - 2)(b + 3)$

1.11 a) $\frac{x - 3}{2}$ b) $\frac{x + 6}{4}$ c) x d) $\frac{3x^2y - 9xy^2}{x^3 - 9xy^2} = \frac{3xy(x - 3y)}{x(x - 3y)(x + 3y)} = \frac{3y}{x + 3y}$

1.12

a) $\frac{\sqrt{a^3} \cdot a^4}{(a^2)^3} = \frac{a^{\frac{7}{2}}}{a^6} = a^{-\frac{5}{2}} = \frac{1}{a^{\frac{5}{2}}} = \frac{1}{a^2 \sqrt{a}}$ b) $\frac{\sqrt{a^3}}{\sqrt[3]{a^2} \sqrt{a}} = a^{\frac{3}{2} - (\frac{2}{3} + \frac{1}{2})} = a^{\frac{9 - 4 - 3}{6}} = a^{\frac{2}{6}} = \sqrt[3]{a}$

c) $\frac{a \cdot \sqrt{a} \cdot \sqrt[3]{a}}{\sqrt[3]{a^4}} = a^{1 + \frac{1}{2} + \frac{1}{3} - \frac{4}{3}} = \sqrt{a}$ d) $\frac{\sqrt[3]{a^2} \cdot \sqrt{a} \cdot \sqrt[6]{a} \cdot \sqrt[4]{a}}{a \sqrt[3]{\sqrt[4]{a^3} \cdot a^4}} = a^{(\frac{2}{3} + \frac{1}{2} + \frac{1}{6} + \frac{1}{4}) - (1 + \frac{7}{12})} = a^0 = 1$

1.13 a) $S = 1 + \frac{2}{3} + \frac{4}{9} + \frac{8}{27} + \dots$ $S = a_1 \frac{1 - k^n}{1 - k} = \lim_{n \rightarrow \infty} 1 \frac{1 - (\frac{2}{3})^n}{1 - \frac{2}{3}} = \frac{3}{1 - \frac{2}{3}} = 3$

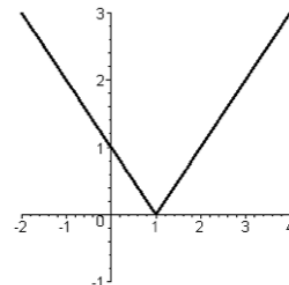
b) $1 + \frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3} + \dots$ der $x > 1$ $S = a_1 \frac{1 - k^n}{1 - k} = \lim_{n \rightarrow \infty} 1 \frac{1 - (\frac{1}{x})^n}{1 - \frac{1}{x}} = \frac{x}{x - 1}$ der $\lim_{n \rightarrow \infty} (\frac{1}{x})^n = 0$ for $x > 1$

1.14

a) $|x - 1| = 2 \Rightarrow x - 1 = \pm 2 \Rightarrow \underline{\underline{x = 3 \vee x = -1}}$

b) $y = |x - 1| = \begin{cases} x - 1 & x > 1 \\ 1 - x & x < 1 \end{cases}$

c) Største verdien er 5 for $x = 1$.



Chapter 2

Functions, equations and inequalities

In this chapter, we will review central topics related to functions, equations and inequalities.

2.1 What is a function?

A function is a rule that associates to each element x from a set D_f called *domain* to a unique element y in a set called *range*,

$$y = f(x), \quad \text{where } x \in D_f \text{ and } y \in V_f$$

x and y are called *independent* and *dependent* variables, respectively.

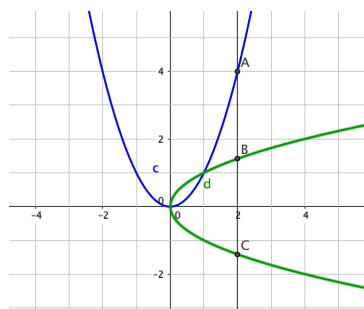
The necessary condition for a relation $y = f(x)$ to be a function is that for any x in the domain, there is *one and only one* y in the range:

$$x_1 = x_2 \Rightarrow y_1 = f(x_1) = f(x_2) = y_2$$

Vertical line test: Any line parallel to the y -axis intersects a function at most in one point.

Example 2.1

$y = x^2$ is a function while $y^2 = x$ does not satisfy the function requirement. Using the vertical line test, the graph of $y = x^2$ (in blue) has only an intersection (point A) with the vertical line $x = 2$, while $y^2 = 2$ (in green) has two intersections (points B, C).



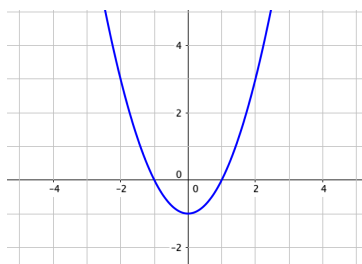
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2.2 Graph of a function

Let f be a function. The set of all couples $(x, f(x))$ that we obtain by taking all possible x in the domain of f , is called *graph* of the function $y = f(x)$.

Example 2.2

The graph of $y = f(x) = x^2 - 1$ is shown here:
As we can observe, the relation $(x, x^2 - 1)$ is a function (it satisfies the vertical line test).



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2.3 Some important properties

- **Monotonicity**

- | | |
|---|--|
| (i) A function f is <i>monotonically increasing</i> if: | $x_2 > x_1 \Rightarrow f(x_2) \geq f(x_1)$ |
| (ii) A function f is <i>strictly monotonically increasing</i> if: | $x_2 > x_1 \Rightarrow f(x_2) > f(x_1)$ |
| (iii) A function f is <i>monotonically decreasing</i> if: | $x_2 > x_1 \Rightarrow f(x_2) \leq f(x_1)$ |
| (iv) A function f is <i>strictly monotonically decreasing</i> if: | $x_2 > x_1 \Rightarrow f(x_2) < f(x_1)$ |

- **Continuity**

A function $y = f(x)$ is *continuous* if the graph of f does not have interruptions (you can draw the graph of f without lifting pen from paper).

The concept of continuity will be studied further in Chapter 5.

- **One-to-one functions (injective functions)**

A function is *one-to-one* (or *injective*) if for each y in the range of f there is only one x such that $y = f(x)$. In other words, if $y = f(x_1)$ and $y = f(x_2)$, then $x_1 = x_2$.

To check injectivity, one can use the *horizontal line test*: if a horizontal line (i.e. parallel to the x -axis) intersects f in at most one point, then f is injective.

- **Composite functions**

An example of a composite function is $y = \sqrt{x^2 + 1}$, as it can be thought as $y = \sqrt{g(x)}$, where $g(x) = x^2 + 1$.

- **Piecewise functions**

These are functions that are defined differently in different intervals. An example is:

$$f(x) = \begin{cases} x^2 & x \geq 0 \\ -2x & x < 0 \end{cases}$$

- **Even and odd functions and symmetry properties**

(At this time not in the syllabus)

2.4 Some important functions classes

Polynomial functions: $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ (polynomial of degree n)

First degree and second degree functions are examples of polynomial functions

Rational functions: $y = \frac{f(x)}{g(x)}$, where $f(x)$ and $g(x)$ are polynomials and $g(x) \neq 0$.

Exponential functions: $y = a^x$, $a > 0$

Logarithmic functions: $y = \log_a x$, with $x > 0$, $a > 0$

The base-10 logarithm function $y = \log x$ and the natural logarithm $y = \ln x$ are examples of logarithmic functions

Trigonometric functions: $y = \sin x$, $y = \cos x$, $y = \tan x$, ...

2.5 First-degree functions $f(x) = ax + b$

A first-degree function is a function of type $y = ax + b$, that means a polynomial function of degree at most 1 (the highest power of x is at most 1). a is the *slope* and b is the *intercept*.

- One-point formula: $y - y_0 = a(x - x_0)$
(a straight line with slope a and passing through the point (x_0, y_0))
- Two-points formula: $\frac{y - y_0}{x - x_0} = \frac{y_1 - y_0}{x_1 - x_0}$
(a straight line through the points (x_0, y_0) and (x_1, y_1) , where the slope is $a = \frac{y_1 - y_0}{x_1 - x_0}$)

The graph of $y = ax + b$ is a straight line:

$a > 0 \Rightarrow$ the function is strictly monotonically increasing

$a < 0 \Rightarrow$ the function is strictly monotonically decreasing

$a = 0 \Rightarrow$ the function is constant: $y = b$

2.6 Equations

An equation is obtained by imposing the equality of two mathematical expressions, with the expressions containing at least an unknown. If there is only one unknown, this is usually denoted as x .

- An equations does not change if we add or subtract the same term to the expressions on both sides of '='.
This can be used to move terms. Moving a term from one side to the other changes its sign: $\pm \rightarrow \mp$.
- An equations does not change if we multiply or divide by the same term $\neq 0$ the expressions on both sides of '='.

An equations that is always satisfied, independently of the value of the unknowns, is called an *identity*, for instance $(x + 1)(x - 1) = x^2 - 1$.

2.7 Linear equations

Linear equations (or first degree equations) are equations where the unknown x appears with power 1. To solve a first order equations we isolate the x on one side and the numeric values on the other side. To this purpose, we must add or subtract the same term on both sides, or multiply or divide both sides by the same number.

Example 2.3

Solve the equations:

$$\text{a) } 8x + 7 = 88 - x \quad \text{b) } \frac{x}{2} + \frac{x}{3} = 11 - x \quad \text{c) } 1 - \frac{1}{2}(3 - x) + \frac{2x}{5} = 4$$

Solution:

a) $8x + 7 = 88 - x$ $8x + x = 88 - 7$ $9x = 81$ $x = \frac{81}{9} = \underline{\underline{9}}$	b) $6\left(\frac{x}{2} + \frac{x}{3} = 11 - x\right)$ $3x + 2x = 66 - 6x$ $11x = 66$ $\underline{\underline{x = 6}}$	c) $10\left(1 - \frac{1}{2}(3 - x) + \frac{2x}{5} = 4\right)$ $10 - 5(3 - x) + 2(2x) = 40$ $10 - 15 + 5x + 4x = 40$ $9x = 45 \Rightarrow \underline{\underline{x = 5}}$
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2.8 Second degree (or quadratic) equations $ax^2 + bx + c = 0$

A second degree (or quadratic) equations is an equations in which the highest power of the unknown is 2. The first step is to collect all the x^2 , x and numeric values so that the equation can be written in the form $ax^2 + bx + c = 0$, with a, b, c real numbers and $a \neq 0$. (For $a = 0$ the equation reduces to a linear one, see above)

Solution of the quadratic equation $ax^2 + bx + c = 0$:

$$\bullet \quad x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad \begin{cases} b^2 - 4ac > 0 & 2 \text{ distinct real solutions} \\ b^2 - 4ac = 0 & \text{double solution} \\ b^2 - 4ac < 0 & \text{no real solution} \end{cases}$$

• If $b = 0$, the equation reduces to $ax^2 + c = 0$ and has solution $x_{1,2} = \pm \sqrt{-\frac{c}{a}}$ if $ac < 0$

• If $c = 0$, the equations reduces to $ax^2 + bx = 0$ and has solutions $x_1 = 0$ and $x_2 = -\frac{b}{a}$

• A quadratic function $f(x) = ax^2 + bx + c$ that has zeros can be factorized using its zeros:
 $ax^2 + bx + c = a(x - x_1)(x - x_2)$
 where x_1 and x_2 can be computed with any of the formulas above.

Summarizing, there are three main cases:

Case	Solution	Example
No constant term ($c = 0$)	$ax^2 + bx = 0$ $x(ax + b) = 0$ $x = 0$ or $ax + b = 0$ $x = 0$ or $x = -\frac{b}{a}$	$2x^2 - bx = 0$ $x(2x - 4) = 0$ $x = 0$ or $2x - 4 = 0$ $\underline{\underline{x = 0 \text{ or } x = 2}}$
No first order term ($b = 0$)	$ax^2 + c = 0$ $ax^2 = -c$ $x^2 = -\frac{c}{a}$ $\underline{\underline{x = \pm \sqrt{-\frac{c}{a}}}}$ (if $ac < 0$)	$4x^2 - 9 = 0$ $4x^2 = 9$ $x^2 = \frac{9}{4}$ $\underline{\underline{x = \pm \sqrt{\frac{9}{4}} = \pm \frac{3}{2}}}$
General case	$ax^2 + bx + c = 0$ $x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ (abc-formula) $b^2 - 4ac > 0$: 2 solutions $b^2 - 4ac = 0$: a double solution	$x^2 + x - 6 = 0$ $x = \frac{-1 \pm \sqrt{(-1)^2 - 4 \cdot 1 \cdot (-6)}}{2 \cdot 1}$ $x = \frac{-1 \pm 5}{2}$ $\underline{\underline{x = 2 \text{ or } x = -3}}$ $x^2 + 4x + 4 = 0$ $x = \frac{-4 \pm \sqrt{4^2 - 4 \cdot 1 \cdot 4}}{2 \cdot 1}$ $x = \frac{-4 \pm 0}{2}$ $\underline{\underline{x = 2}}$

Example 2.4

Solve the following quadratic equation: $4x^2 + 10x - 24 = 0$

Solution:

We use the abc-formula. We have $a = 4$, $b = 10$ and $c = -24$.

$$x_{1,2} = \frac{-10 \pm \sqrt{10^2 - 4 \cdot 4 \cdot (-24)}}{2 \cdot 4} = \frac{-10 \pm \sqrt{484}}{8} = \frac{-10 \pm 22}{8}$$

This gives the solutions $\underline{\underline{x_1 = -4 \text{ and } x_2 = \frac{3}{2}}}$

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Example 2.5

Solve the equations:

$$\text{a) } x^2 + 3x = 4 \quad \text{b) } x^2 = 5x \quad \text{c) } x^2 - 5 = 0$$

Solution:

a) $x^2 + 3x - 4 = 0$ $x = \frac{-3 \pm \sqrt{3^2 - 4(-4)}}{2}$ $x = \frac{-3 \pm 5}{2}$ $\underline{\underline{x = 1 \vee x = -4}}$	b) $x^2 = 5x$ $x(x - 5) = 0$ $\underline{\underline{x = 0 \vee x = 5}}$	c) $x^2 - 5 = 0$ $x^2 = 5$ $\underline{\underline{x = \pm\sqrt{5}}}$
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◇

2.9 Second degree (or quadratic) functions $f(x) = ax^2 + bx + c$

- If $a > 0$ the the graph is “smiling” (or “happy”), if $a < 0$ the graph is “unhappy”
- The intersection with the y -axis is the point of coordinates $(x = 0, y = c)$
- The zeros of the quadratic functions (intersections with x -axis) are $\left(x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, y = 0\right)$
- Recall that the quadratic function can be factorized if it has zeros:
 $ax^2 + bx + c = a(x - x_1)(x - x_2)$
- The graph is symmetric with respect to the line $x = -\frac{b}{2a}$

To draw the graph of a quadratic function, we can think like this:

1. Is the graph happy or unhappy?
2. Find the symmetry line $x = -\frac{b}{2a}$ and $f(-\frac{b}{2a})$. The point with coordinates $\left(x = -\frac{b}{2a}, y = f(-\frac{b}{2a})\right)$ is the maximum (for $a < 0$) or the minimum (for $a > 0$) of the function
3. Find eventual zeros
4. Find the intersection with the y -axis $(0, c)$

Example 2.6

Draw the graph of $y = x^2 - 4x + 3$

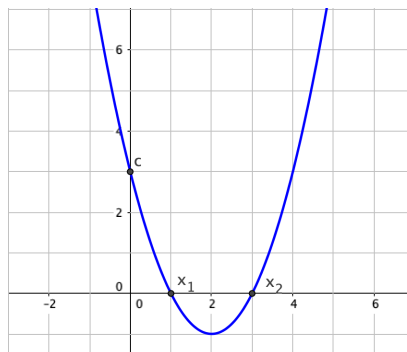
Solution:

1. Zeros:

$x^2 - 4x + 3 = 0$, hence $x = 1$ and $x = 3$

2. Symmetry line: $x = -\frac{b}{2a} = -\frac{-4}{2} = 2$

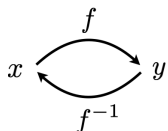
3. $a = 1 > 0$: the graph is happy hence $(2, f(2)) = (2, 2^2 - 4 \cdot 2 + 3 = -1)$ is the local minimum



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2.10 Inverse functions

An inverse function to a function $y = f(x)$, where $x \in D_f$ and $y \in D_y$ is a relation that assigns the y -value back to the x -value.

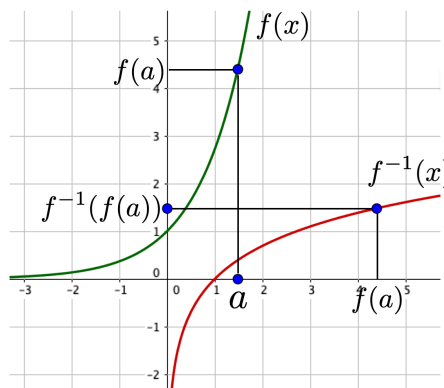


Therefore, for the inverse function:

$D_{f^{-1}} = V_f$ (the domain of f^{-1} is the range of f)

$V_{f^{-1}} = D_f$ (the range of f^{-1} is the domain of f)

For a function to be invertible, it must be one-to-one.



• How can we find the inverse function of $y = f(x)$?

1. Find x in terms of y
2. Exchange x and y

Example 2.7

Find the inverse function of $y = f(x) = x^2 + 1$ for $x > 0$

Solution:

1. We find x in terms of y :

$y = x^2 + 1 \Rightarrow x^2 = y - 1$. Since $x > 0$, we find $x = \sqrt{y - 1}$

2. Exchange x and y :

$y = \sqrt{x - 1}$, hence $f^{-1}(x) = \sqrt{x - 1}$

Note: $D_{f^{-1}} = V_f = [1, \infty)$ and $V_{f^{-1}} = D_f = [0, \infty)$

◇

2.11 Rational equations

A *rational equation* is an equation of type $\frac{p(x)}{q(x)} = 0$, where $p(x)$ and $q(x)$ are polynomials. Thus in rational equations, the unknown x appears both at the numerator and denominator.

To solve rational equations:

- Bring the fractions to common denominator and write in the form $\frac{p(x)}{q(x)} = 0$
- Find the solutions of $p(x) = 0$ (Note: $\frac{A}{B} = 0 \Rightarrow A = 0 \quad (B \neq 0)$)
- Check if any solution of $p(x) = 0$ satisfies also $q(x) = 0$. If so, these *must be discarded* as a fraction is not defined if the denominator is zero.

Example 2.8

Solve the equations: a) $\frac{x+3}{x(x+1)} = 2$, b) $\frac{1}{x} + \frac{1}{2x-1} = 2$

Solution:

$$\text{a) } \frac{x+3}{x(x+1)} - \frac{2x(x+1)}{x(x+1)} = 0 \Rightarrow \frac{-2x^2 - x + 2}{x(x+1)} = 0 \Rightarrow -2x^2 - x + 3 = 0 \Rightarrow \underline{\underline{x = 1, x = -\frac{3}{2}}}$$

b) The 2 is moved to the left side, the common denominator is $x(2x-1)$

$$\frac{2x-1}{x(2x-1)} + \frac{x}{x(2x-1)} - \frac{2x(2x-1)}{x(2x-1)} = 0$$

$$\frac{-4x^2 + 5x - 1}{x(2x-1)} = 0 \Rightarrow -4x^2 + 5x - 1 = 0 \Rightarrow \underline{\underline{x = 1, x = \frac{1}{4}}}$$

In both cases, the solutions were not zeros of the denominator, so we didn't have to discard any solution.

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2.12 Irrational equations

We will consider only some simple irrational equations, of the form $\sqrt{ax+b} = cx+d$.

To solve $\sqrt{ax+b} = cx+d$:

- Square both sides
- Solve the resulting quadratic equation
- Check that the solutions obey the original equation: namely the squaring process might introduce spurious solutions.

Example 2.9Solve the equations: a) $\sqrt{2x+3} = x$ b) $5 + \sqrt{x-3} = x$

Solution:

a) $\sqrt{2x+3} = x$

Square both sides: $2x+3 = x^2$ We obtain the quadratic equation $x^2 - 2x - 3 = 0$

$$x = \frac{2 \pm \sqrt{2^2 - 4(-3)}}{2} = \begin{cases} -1 & \sqrt{2 \cdot (-1) + 3} = 1 \neq -1 \text{ spurious solution, discard} \\ 3 & \sqrt{2 \cdot 3 + 3} = 3 \text{ OK, accept solution} \end{cases}$$

b) We start by isolating the term with the square root on one side: $\sqrt{x-3} = x-5$ Square both sides: $x-3 = (x-5)^2$ Expand the perfect square on right hand side: $x-3 = x^2 - 10x + 25$ Solve the quadratic equation $x^2 - 11x + 28 = 0$.

The equation has two solutions, that have to be checked:

$$(x-4)(x-7) = 0 \Rightarrow x = \begin{cases} 4 & \sqrt{4-3} = 1 \neq -1 = 4-5 \text{ spurious solution, discard} \\ 7 & \sqrt{7-3} = 2 = 7-5 \text{ OK, accept solution} \end{cases}$$

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2.13 Inequalities

When we compare two mathematical expression containing at list an unknown, we have *inequalities*. Equations are thus a special case of inequalities, where we require that the two expression are equal rather than one bigger or smaller than the other.

The rules to solve inequalities are almost the same as for equations: one can add or subtract the same term on both sides, or divide by a positive number, without changing the inequality. However, if you divide by a *negative* number, then the inequality sign must reverse, for the inequality to hold.

Solving an inequality consists in finding the values of x that make the inequality true.

Rule 1: Adding or subtracting the same term on both sides does not change the inequality.

Example: The inequality $x-2 > 6$ has the same solution as the inequality $x > 8$ (we have added 2 to both sides of the first inequality)

Rule 2: If we interchange the left and the right hand side of the inequality, we have to reverse the inequality sign.

Example: The inequality $3-x > 1$ has the same solution as $1 < 3-x$ (we have changed sides and changed “>” to “<”).

And last, but not least, the operation that is the major source of problems, when solving inequalities:

Rule 3a: Multiply/divide both sides by the same positive number on does not change the inequality

Rule 3b: Multiply/divide both sides by the same negative number reverses the inequality.

Example: Assume $c < 0$. Then $a < b$ and $ac > bc$ are equivalent, that is if one is true so is the other.

In what follows, we will look into examples with:

Simple inequalities	$p(x) \leq 0$	(or $<, \geq, >$)
Double inequalities	$m(x) \leq p(x) \leq n(x)$	(or $<, \geq, >$)
Rational inequalities	$\frac{p(x)}{q(x)} \leq 0$	(or $<, \geq, >$)

2.13.1 Simple inequalities

Inequalities of first degree: Like for first degree equations, we collect all the x on one side, all the numerical values on the other side, then solve for x .

Example 2.10

Solve the inequalities: a) $-2x \leq 6$, b) $-\frac{x}{2} \leq 1$

Solution:

a) $-2x \leq 6 \Rightarrow \underline{\underline{x \geq 3}}$. We have divided both sides by -2 and reversed the inequality sign

b) $-\frac{x}{2} \leq 1 \Rightarrow \underline{\underline{x \geq -2}}$. We have multiplied both sides by -2 and reversed the inequality sign

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Example 2.11

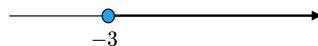
Solve the inequality $4 - 2x \leq 10$

Solution:

$4 - 2x \leq 10 \Rightarrow -2x \leq 6$ (we have subtracted 4 on both sides)

$-2x \leq 6 \Rightarrow x \geq -3$, see previous exercise.

The solution set of the inequality is: $\{x|x \geq -3\}$ (set notation) or $[-3, \infty)$ (interval notation)



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Inequalities of higher degree: For second degree (quadratic) and higher order inequalities, the best method is to collect all terms (a polynomial) on one side, so that we have zero on the other side. Then, factorize the polynomial using its zeros, and analyze the sign of the factors.

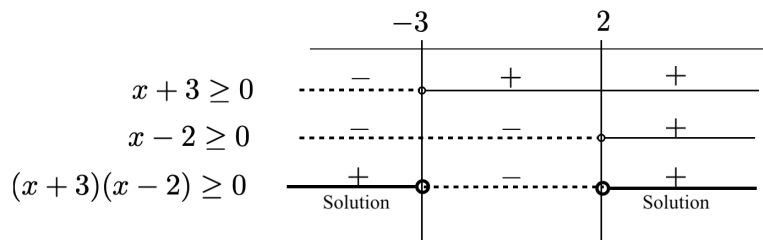
Example 2.12

Solve the inequality $x^2 + x - 6 \geq 0$

Solution:

The quadratic polynomial can be factorized as $(x + 3)(x - 2)$.

We analyze the sign looking at the signs of the individual factors



The solution set is then $\{x | x \leq -3 \vee x \geq 2\}$ or equivalently $(-\infty, -3] \cup [2, \infty)$.

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2.13.2 Double inequalities

A double inequality of the form:

$$A < ax + b < B$$

can be solved by applying the rules described above to all the three parts or can be split in two inequalities taking the intersection of the two solutions.

Example 2.13

Solve the inequality: $2 < 3x - 1 < 8$

Solution:

Method 1:

we add 1 to each part: $3 < 3x < 9$

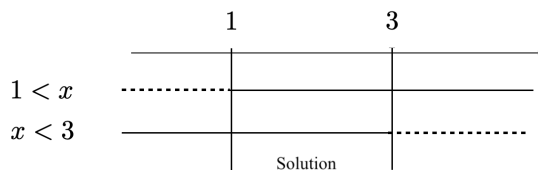
we divide by 3 each part: $1 < x < 3$

Method 2: we split in two inequalities, $2 < 3x - 1$ and $3x - 1 < 8$

$$2 < 3x - 1 \Rightarrow 3 < 3x \Rightarrow 1 < x$$

$$3x - 1 < 8 \Rightarrow x < 3$$

The intersection of the two solution is $\{x | 1 < x < 3\}$ or, equivalently, $(1, 3)$.



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2.13.3 Rational inequalities

Rational inequalities of type $\frac{p(x)}{q(x)}$ are solved in the same manner as polynomial inequalities, by factorizing $p(x)$ and $q(x)$ and analyzing the sign of each factor.

NOTE: the zeros of $q(x)$ must be excluded, as the denominator of a fraction cannot be zero.

Example 2.14

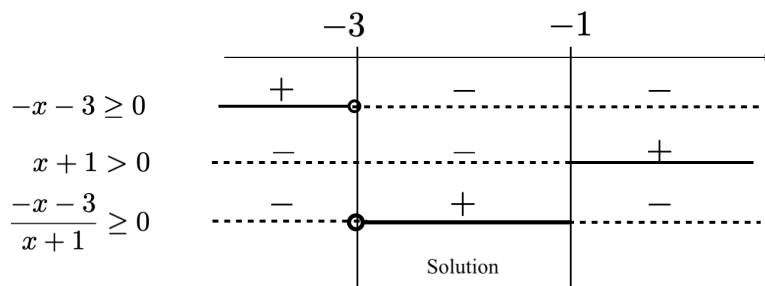
Solve the inequality: $\frac{-x-3}{x+1} \geq 0$

Solution:

$$-x-3 \geq 0 \Rightarrow x+3 \leq 0 \Rightarrow x \leq -3$$

$$x+1 > 0 \Rightarrow x > -1. \text{ (We have excluded the zero } x = -1\text{)}$$

We analyze the signs



The solution set is $\{x \mid -3 \leq x < -1\}$ or equivalently $[-3, -1)$

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2.14 Graphical solution of inequalities

The inequality $f(x) > g(x)$ can be rewritten as $f(x) - g(x) > 0$. The solution consists of all the values x for which the function $y = f(x) - g(x) > 0$, namely that the graph of $y = f(x) - g(x) > 0$ is above the x -axis.

Example 2.15

Solve: $4x + 5 > x^2$ graphically.

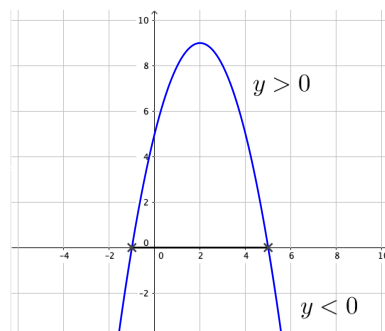
Solution:

First, we rewrite the inequality as

$$-x^2 + 4x + 5 > 0$$

We draw the graph of $y = -x^2 + 4x + 5$ and find the set of x values such that $y > 0$

Solution set: $\{x \mid -1 < x < 5\}$ or equivalently $(-1, 5)$



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Exercises Chapter 2

Exercise 2.1

Which of the following expressions for $y = f(x)$ describes a function?

- a) $y = x^4$ b) $y^2 = x + 1$ c) $y = \sqrt{x + 1}$

Exercise 2.2

- a) Given $f(x) = \sqrt{x^2 + 1}$.

Determine $f(0)$, $f(1)$, $f(\sqrt{3})$ and $f(\sqrt{a-1})$.

- b) Given $f(x) = x^2 + 1$ and $g(x) = \sqrt{x-1}$, where $x > 1$.

Determine $f(g(x))$ and $g(f(x))$ ¹. What can we say about f and g ?²

Exercise 2.3

Given $f(x) = \sqrt{x-2}$.

Determine the domain and range of f . Find $f^{-1}(x)$.

State the function $g(x) = f^2(x) + 2$.

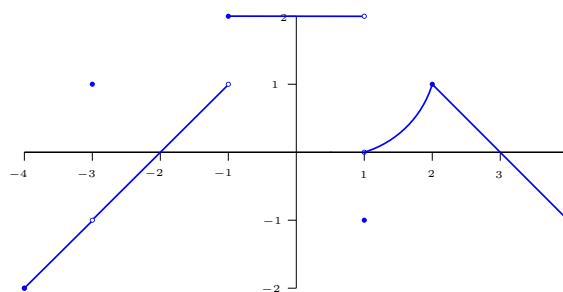
Exercise 2.4

Given $f(x) = \frac{x^2 + x}{x}$. What is the domain of f ? Draw the graph of $f(x)$.

Exercise 2.5

The graph of a function is shown here:

- a) Determine $f(-3)$ and $f(2)$.
 b) Is the function continuous³ in the points $x = -3$ and $x = 2$?



Exercise 2.6

Draw the graph of $f(x) = \frac{|x|}{x}$, where $x \neq 0$.

Exercise 2.7

Solve the equations:

- a) $2(x + 4) - 3x = 5 - 3(x - 3)$ b) $\frac{1}{2}x + \frac{1}{3} - \frac{1}{4}x = \frac{1}{12}$ c) $\frac{3}{2}(b - 1) - \frac{2}{3}(1 - b) = \frac{5}{2}b$

¹ $f(g(x))$ and $g(f(x))$ are called composite functions. They can also be written $f \circ g$ and $g \circ f$.

² See chapter 2.10 Inverse Functions

³ See chapter 5.4 Continuity

Exercise 2.8

Solve the equations:

a) $2x^2 = 50$ b) $3x^2 - 12x = 0$ c) $\frac{2}{5}x^2 - \frac{10}{9} = 0$

Exercise 2.9

Solve the equations:

a) $\frac{x}{5} = \frac{1}{2}$ b) $\frac{1+x}{2} = 3$ c) $\frac{6}{1+4x} = 0.5$

Exercise 2.10

Solve the equations:

a) $(x-5)(x+1) = 0$ b) $x(x+2) = 8$ c) $y^2 - 7y + 12 = 0$

Exercise 2.11

Solve the equations:

a) $\sqrt{x+1} = 1$ b) $2(x-1) = \sqrt{6-x}$ c) $x - \sqrt{x-2} = 2$

Exercise 2.12

Solve the equations:

a) $\sqrt{x+9} - \sqrt{x+2} = 1$ b) $\sqrt{2x-1} = \sqrt{x-1} + 1$

Exercise 2.13

Solve the inequalities:

a) $5x + 4 > 2x - 2$
b) $3(2-x) < 3-x$
c) $x + 3 - 2(3+2x) < 5(2-x)$
d) $2(x-1) - 3(1-x) < x+3$
e) $3(2x+1) - (5-x) > 1 - (x+3)$
f) $\frac{x+4}{3} > \frac{2x+1}{3} + 1$
g) $\frac{7x+4}{4} \leq 2 - \frac{x-3}{2} + \frac{3x}{8}$
h) $\frac{x-4}{3} + \frac{7}{6} > \frac{5-2x}{2}$

Exercise 2.14

Solve the inequalities:

a) $x^2 - 4 > 3x$ b) $x^3 - 4x < x^2 - 4$ c) $\frac{2}{x-1} - 1 < \frac{2}{x}$

$$\text{d) } \frac{4x-2}{x^2+1} < 1 \qquad \text{e) } \frac{x-2}{1+x} > 2 \qquad \text{f) } x > \frac{x+2}{x}$$

Solution to the exercises Chapter 2

2.1 a) Yes **b)** No, since $y = \pm\sqrt{x+1}$ **c)** Yes ($x \geq -11$)

2.2 $f(0) = 1$, $f(1) = \sqrt{2}$, $f(\sqrt{3}) = 2$ and $f(\sqrt{a-1}) = \sqrt{a}$

$f(g(x)) = f(\sqrt{x-1}) = x$ and $g(f(x)) = g(x^2+1) = x$ indicate that f and g are inverse functions.

2.3 $D_f = \{x \in \mathbb{R} | x \geq 2\}$ (all real numbers x such that x is greater than or equal to 2)

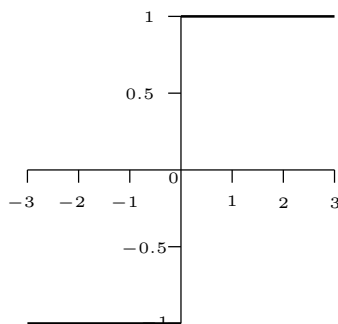
$R_f = \{y \in \mathbb{R} | y \geq 0\}$ (all real numbers y such that y is greater than or equal to 0)

$g(x) = x$ where $x \geq 2$. $f^{-1}(x) = x^2 + 2$.

2.4 $D_f = \{x \in \mathbb{R} | x \neq 0\}$ all real numbers except 0. $f(x) = x + 1$ (draw the graph of the line $y = x + 1$ with the point $(0, 1)$ not lying on the graph).

2.5 $f(-3) = 1$ and $f(2) = 1$. The function is continuous in $x = 2$, but not in $x = -3$.

$$\mathbf{2.6} \quad f(x) = \frac{|x|}{x} = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}$$



2.7 a) $x = 3$ **b)** $x = -1$ **c)** $b = -\frac{13}{2}$

2.8 a) $x = \pm 5$ **b)** $x = 0, x = 4$ **c)** $x = \pm \frac{5}{3}$

2.9 a) $x = 2.5$ **b)** $x = 5$ **c)** $x = 2.75$

2.10 a) $x = 5, x = -1$ **b)** $x = -4, x = 2$ **c)** $x = 3, x = 4$

2.11 a) $x = 0$ **b)** $x = 2$ **c)** $x = 3, x = 2$

2.12 a) $x = 7$ **b)** $x = 1$ and $x = 5$

2.13 a) $5x + 4 > 2x - 2 \iff 3x > -6 \iff x > -2$

b) $3(2 - x) < 3 - x \iff 6 - 3x < 3 - x \iff -2x < -3 \iff x > \frac{3}{2}$

c) $x + 3 - 2(3 + 2x) < 5(2 - x) \iff x + 3 - 6 - 4x < 10 - 5x \iff 2x < 13 \iff x < \frac{13}{2}$

d) $2(x - 1) - 3(1 - x) < x + 3 \iff 2x - 2 - 3 + 3x < x + 3 \iff 4x < 8 \iff x < 2$

e) $3(2x + 1) - (5 - x) > 1 - (x + 3) \iff 6x + 3 - 5 + x > 1 - x - 3 \iff 8x > 0 \iff x > 0$

f) $\frac{x+4}{3} > \frac{2x+1}{3} + 1 \iff x + 4 > 2x + 1 + 3 \iff -x > 0 \iff x < 0$

g) $\frac{7x+4}{4} \leq 2 - \frac{x-3}{2} + \frac{3x}{8} \iff 2(7x+4) \leq 16 - 4(x-3) + 3x \iff 14x + 8 \leq 16 - 4x + 12 + 3x \iff 15x \leq 20 \iff x \leq \frac{20}{15} \iff x \leq \frac{4}{3}$

h) $\frac{x-4}{3} + \frac{7}{6} > \frac{5-2x}{2} \iff 2(x-4) + 7 > 3(5-2x) \iff 2x - 8 + 7 > 15 - 6x \iff 8x > 16 \iff x > 2$

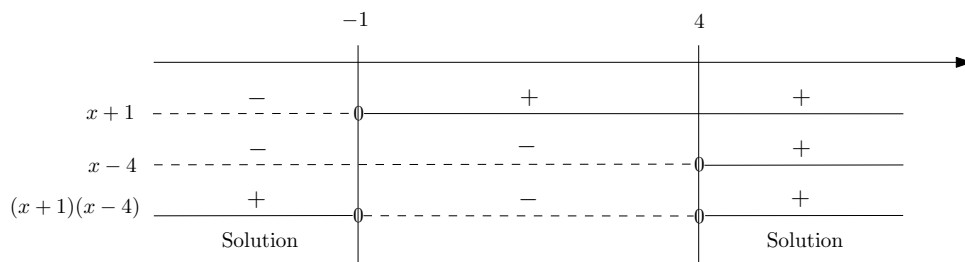
i) $\frac{3-5x}{5} > \frac{1+x}{3} - 2 \iff 3(3-5x) > 5(1+x) - 30 \iff 9 - 15x > 5 + 5x - 30 \iff 34 > 20x \iff x < \frac{34}{20} \iff x < \frac{17}{10}$

j) $\frac{2(x-1)-3}{4} \geq \frac{5+4x}{3} \iff 3 \cdot (2(x-1)-3) \geq 4(5+4x) \iff 6x - 15 \geq 20 + 16x \iff -10x \geq 35 \iff x \leq -\frac{35}{10} \iff x \leq -\frac{7}{2}$

2.14 a)

$$\begin{aligned} x^2 - 4 &> 3x \\ x^2 - 3x + 4 &> 0 \\ (x+1)(x-4) &> 0 \end{aligned}$$

Solution: $\{x | x < -1 \vee x > 4\}$ or: $x \in \langle -\infty, -1 \rangle \vee \langle 4, \infty \rangle$.



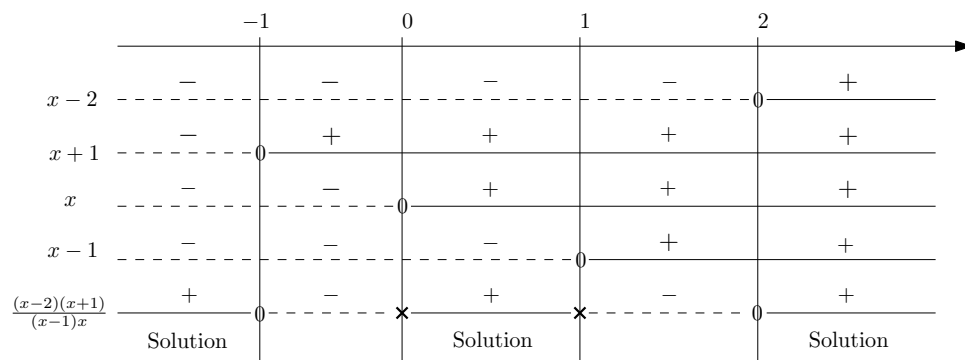
b) $x < -2$ or $1 < x < 2$

c) $\frac{2}{x-1} - 1 < \frac{2}{x}$

$$\frac{2 \cdot x - 1 \cdot x(x-1) - 2 \cdot (x-1)}{(x-1)x} < 0 \Rightarrow \frac{2x - x^2 + x - 2x + 2}{(x-1)x} < 0$$

$$\frac{-x^2 + x + 2}{(x-1)x} < 0 \Rightarrow \frac{-(x-2)(x+1)}{(x-1)x} \Rightarrow \text{or } \frac{(x-2)(x+1)}{(x-1)x} > 0$$

Solution: $x < -1$ or $0 < x < 1$ or $x > 2$

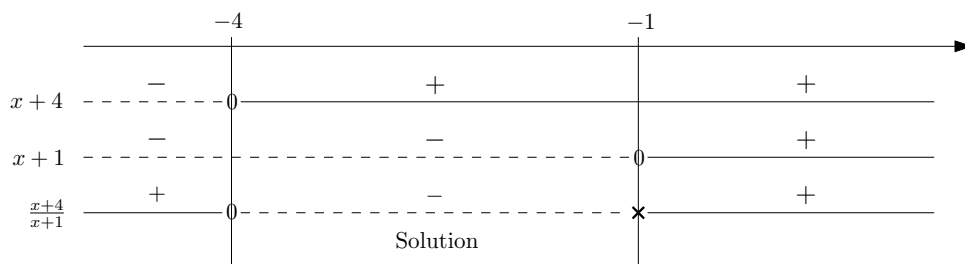


- d) Here, we know that $x^2 + 1$ is positive since x^2 can never become negative. Hence, we can multiply both sides of the inequality by $x^2 + 1$:

$$\frac{4x-2}{x^2+1} < 1 \iff 4x-2 < x^2+1 \iff x^2-4x+3 > 0 \iff (x-3)(x-1) > 0.$$

By drawing the sign chart, we see that $x < 1$ or $x > 3$.

e) $\frac{x-2}{1+x} > 2 \iff \frac{x-2}{x+1} - 2 > 0 \iff \frac{x-2-2(x+1)}{x+1} > 0 \iff \frac{-x-4}{x+1} = -\frac{x+4}{x+1} > 0$
or $\frac{x+4}{x+1} < 0$.



The conclusion is that $\frac{x+4}{x+1} < 0 \iff -4 < x < -1$

- f) $-1 < x < 0$ or $x > 2$

Chapter 3

Exponential function and logarithm

In this chapter we will study exponential functions and what is exponential growth or exponential decay. We will also see how to use logarithms to solve exponential equations, as the exponential function and the logarithmic function are each other's inverse function.

3.1 Exponential growth/decay

When a quantity increases/decreases with a fixed percentage in equal time intervals, we call it an *exponential growth/decay*.

$$f(x) = ca^x, a > 0, c > 0 \quad \begin{cases} a > 1 & \text{Exponential growth, for example } y = 5 \cdot 2^x \\ 0 < a < 1 & \text{Exponential decay, for example } y = 7 \cdot \left(\frac{1}{2}\right)^x \end{cases}$$

(If $a = 1$ the the function is a constant)

a is called *growth* factor and can be written as $a = 1 + \frac{p}{100}$, where p is the interest rate (in %).

Example:

For a value that $\begin{cases} \text{grows with 20\%} & a = 1 + 0.2 = 1.2 \\ \text{decays with 20\%} & a = 1 - 0.2 = 0.8 \end{cases}$

For a value that grows 200%, $a = 1 + 2 = 3$

Example 3.1

The world population was 4.1 billions on 1st January 1980. A good estimate of the population

growth is 2% per year. Find a function that describes the population at time t after 1980.

Solution:

We set up a model using the exponential function $f(t) = c \cdot a^t$.

To find c and a :

We set $t = 0$ for year 1980, therefore $f(0) = 4.1 \Rightarrow f(0) = c \cdot a^0 = c = 4.1$.

The growth fraction is $a = 1 + 0.02 = 1.02$. The function is then:

$$\underline{\underline{f(t) = 4.1 \cdot (1.02)^t}} \quad (\text{billions})$$

that defines the population t years after 1980.

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Continuing on the previous example, what about if we were asked how long did it take for the population to reach 4.9 billions? Then we would have to find t such that:

$$4.1 \cdot (1.02)^t = 4.9$$

To solve the above equation we can use logarithms.

We show the solution here, but you can try for yourself after you have learned logarithms.

$$(1.02)^t = \frac{4.9}{4.01} \quad t \ln(1.02) = \ln\left(\frac{4.9}{4.01}\right) \quad t = \frac{\ln(4.9/4.1)}{\ln(1.02)} \approx 9$$

3.2 The logarithms $f(x) = \log x$ and $f(x) = \ln x$

The *logarithm* of a number b in *base* a is the *exponent* x to which we need to raise the base a to obtain b :

$$x = \log_a b \Leftrightarrow a^x = b \quad (a > 0 \text{ and } b > 0)$$

Thus, to solve for x in the equation $a^x = b$, we use the logarithms:

$$a^x = b \Leftrightarrow x = \log_a b$$

- Logarithms in base e (Euler's constant) are called *natural logarithms* and are denoted with "ln"
- Logarithms in base 10 are sometimes called *Briggs' logarithms* and are denoted with "log" (omitting the base 10 at the foot of the logarithm)

		Important values	Useful rules
10-logarithm (Briggs' logarithm):	$y = 10^x \Leftrightarrow x = \log y$	$\log 10 = 1, \log 1 = 0$	$\log(10^A) = A, 10^{\log A} = A$
Natural logarithm:	$y = e^x \Leftrightarrow x = \ln y$	$\ln e = 1, \ln 1 = 0$	$\ln(e^A) = A, e^{\ln A} = A$

3.3 Rules for computing with logarithms

Assume $a > 0$ (base), $A, B > 0$:

$\log_a(A \cdot B) = \log_a A + \log_a B$	$\log_a\left(\frac{A}{B}\right) = \log_a A - \log_a B$	$\log_a A^n = n \cdot \log_a A$
log of product = sum of logs	log of ratio = difference of logs	log of power = power times log

Let $a, b, c > 0$. Since: $a^x = b \Leftrightarrow x = \log_a b$:

By taking the $\log_c$ on both sides of $a^x = b$:

$$\log_c a^x = \log_c b \Leftrightarrow x \log_c a = \log_c b \Leftrightarrow x = \frac{\log_c b}{\log_c a}$$

Therefore we have:

$$\log_a b = \frac{\log_c b}{\log_c a} \quad \text{change of base rule}$$

Example 3.2

Compute / write as simple as possible:

a) $\log 100$ b) $\log 1000$ c) $\log 0.001$ d) $\log \sqrt{10}$ e) $\log \frac{7}{100}$

Solution:

All the logarithms are in base 10:

a) $\log 100 = \log 10^2 = 2$ b) $\log 1000 = \log 10^3 = 3$ c) $\log 0.001 = \log 10^{-3} = -3$

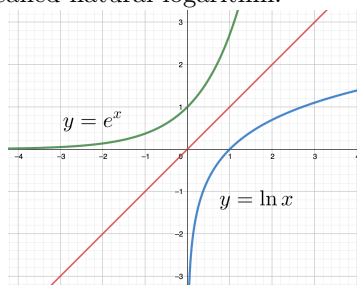
d) $\log \sqrt{10} = \log 10^{\frac{1}{2}} = \frac{1}{2}$ e) $\log \frac{7}{100} = \log 7 - \log 100 = \log 7 - 2$

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3.4 The natural logarithm function

If the base in the logarithm is e , then the logarithm is called natural logarithm.

- $y = e^x$ and $y = \ln x$ are each other's inverse function
- $\lim_{x \rightarrow \infty} e^x = \infty$ and $\lim_{x \rightarrow -\infty} e^x = 0$
($y = 0$ is a horizontal asymptote of $y = e^x$)
- $\lim_{x \rightarrow \infty} \ln x = \infty$ and $\lim_{x \rightarrow 0^+} \ln x = -\infty$
($x = 0$ is a vertical asymptote of $y = \ln x$)



3.5 $y = e^x$ and $y = \ln x$ are each other's inverse function

When $a > 0$: $y = a^x \Leftrightarrow x = \log_a y$

which means that $y = \log_a x$ is the inverse function of $y = a^x$.

$$f(x) = 10^x \Rightarrow f^{-1}(x) = \log x$$

$$f(x) = e^x \Rightarrow f^{-1}(x) = \ln x$$

3.6 Exponential and logarithmic equations

3.6.1 The equation $a^x = b$

Although the solution of this equation in base a is $x = \log_a b$, usually, as long as the base is not explicitly stated, it is normal to solve these using the natural logarithm $\ln$.

Generally:

$$a^x = b \Rightarrow \ln(a^x) = \ln b \Rightarrow x \cdot \ln a = \ln b \Rightarrow x = \frac{\ln a}{\ln b}$$

Example 3.3

Solve: $3^x = 7$

Solution:

$$3^x = 7 \Rightarrow \ln(3^x) = \ln 7 \Rightarrow x \cdot \ln 3 = \ln 7 \Rightarrow x = \frac{\ln 3}{\ln 7}$$

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3.6.2 Some exponential equations

Here we will see some techniques that can be useful to solve exponential equations.

Example 3.4

Solve the equations: **a)** $3e^{2x+1} = 5$ **b)** $e^{2x} = 3e^x + 2 = 0$

Solution:

$$\text{a)} \quad 3e^{2x+1} = 5 \Rightarrow e^{2x+1} = \frac{5}{3} \Rightarrow 2x + 1 = \ln \frac{5}{3} \Rightarrow x = \frac{1}{2}(\ln \frac{5}{3} - 1)$$

b) The equation contains $e^{2x} = (e^x)^2$ and e^x : we introduce an auxiliary variable: $u = e^x$:

$$u^2 - 3u + 2 = 0 \Rightarrow u = 1 \text{ and } u = 2$$

$$\text{for } u = 1: e^x = 1 \Rightarrow x = \ln 1 = \underline{0}$$

$$\text{for } u = 2: e^x = 2 \Rightarrow x = \underline{\underline{\ln 2}}$$

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$ae^{2x} = be^x + c = 0$ can be solved by choosing $u = e^x$: $au^2 + bu + c = 0$ (quadratic equation)

3.6.3 Some logarithmic equations

Example 3.5

Solve the equations:

$$\text{a)} \quad \ln(x-1) = 2 \quad \text{b)} \quad 3 \ln x + \ln x^2 = \ln(32) \quad \text{c)} \quad \ln x + \ln(2x-1) = 0$$

Solution:

NOTE: The argument of the logarithm must be positive! Recall to check that the solution satisfies this (otherwise it must be discarded).

a) $\ln(x-1) = 2 \Rightarrow x-1 = e^2 \Rightarrow \underline{\underline{x = e^2 + 1}}$ (OK)

b) $3 \ln x + \ln x^2 = \ln 32 \Rightarrow \ln x^3 + \ln x^2 = \ln 32 \Rightarrow \ln(x^3 \cdot x^2) = \ln 32 \Rightarrow x^5 = 32 = 2^5 \Rightarrow \underline{\underline{x = 2}}$ (OK)

c) $\ln x + \ln(2x-1) = 0 \Rightarrow \ln(x \cdot (2x-1)) = 0 \Rightarrow x(2x-1) = 1 \Rightarrow 2x^2 - x - 1 = 0$ that has solution $x = -\frac{1}{2}$ and $x = 1$.

$x = -\frac{1}{2}$ makes the argument of the first logarithm negative, hence it must be discarded.

$x = 1$ is OK. The solution is therefore $\underline{\underline{x = 1}}$.

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Exercises Chapter 3

Exercise 3.1

Write the expressions as simple as possible:

a) $\frac{e^{2x} + 2e^x}{e^x}$ b) $\frac{5e^{2x} + (2e^x)^2}{(3e^x)^2}$

Exercise 3.2

The exponential function given by: $f(x) = c \cdot a^x$ can be written as $f(x) = c \cdot e^{kx}$. Determine k .

Exercise 3.3

The population number in a town on January 1, 2000 was 1500.

We assume that the town's population grows with approximately 3% each year.

- State a function that describes the population number t years after the year 2000.
- What is the population number of the town by the end of 2005 according to this model?

Exercise 3.4

t years after an organism died, the amount of radium is reduced to p % of the amount in the living organism. The half-life is approximately 1620 years.

State an exponential function that describes $p(t)$ given in %.

Exercise 3.5

Write the expressions as simple as possible without using a calculator:

- | | | |
|-------------------------------------|--------------|--|
| a) $\ln e + 2 \ln 1$ | b) $\ln e^2$ | c) $\ln a + 2 \ln \sqrt{a} - \ln a^3$ |
| d) $\ln(3a^2) + 2 \ln a - \ln(a^4)$ | e) $\ln e^x$ | f) $\ln a^3 + 3 \ln \sqrt[3]{a} - 2 \ln a^2$ |

Exercise 3.6

Write the expressions as simple as possible:

- a) $e^{3 \ln x}$ b) $e^{2 \ln \sqrt{x}}$ c) $e^{2 \ln x + \ln 5}$

Exercise 3.7

Solve the equations:

- a) $e^{2x} = 5$ b) $5e^{2x} = 11$ c) $3e^{5x} - 7 = 0$

Exercise 3.8

Solve the equations:

- a) $e^{2x} - 3e^x + 2 = 0$ b) $e^x + 2e^{-x} - 3 = 0$

Exercise 3.9

Solve the equations:

a) $2 \ln x - 1 = 0$ b) $3 \ln x + \ln x^2 = 9$
c) $2^x = 6$ d) $5 \cdot 3^x = 7$

Exercise 3.10

Show that $a^x = b \Rightarrow x = \log_a(b) = \frac{\log(b)}{\log(a)} = \frac{\ln(b)}{\ln(a)}$

Solution to the Exercises Chapter 3

3.1 a) $e^x + 2$ b) 1

3.2 We know that: $a = e^{\ln a}$. $a^x = (e^{\ln a})^x = e^{\ln a \cdot x}$ hence, $k = \ln a$.

3.3 $N(t) = 1500 \cdot (1.03)^t$ b) $N(6) = 1500 \cdot (1.03)^6 \approx 1791$

3.4 $p(t) = 100 \cdot \left(\frac{1}{2}\right)^{\frac{t}{1620}}$ or $p(t) = 100 \cdot e^{\frac{-t}{1620} \ln 2}$

3.5 a) 1 b) 2 c) 0 d) $\ln 3$ e) x f) 0

3.6 a) x^3 b) c) $5x^2$

3.7 a) $x = \frac{1}{2} \ln 5$ b) $x = \frac{1}{2} \ln \frac{11}{5}$ c) $x = \frac{1}{5} \ln \frac{7}{3}$

3.8 a) $x = 0 \vee x = \ln 2$

b) By multiplying both sides by e^x , we obtain the same equation as in a) $x = 0 \vee x = \ln 2$

3.9 a) $x = e^{\frac{1}{2}}$ b) $x = e^{\frac{9}{5}}$
 c) $x = \frac{\ln 6}{\ln 2}$ d) $x = \frac{\ln \frac{7}{5}}{\ln 3} = \frac{\ln 7 - \ln 5}{\ln 3}$

3.10 $a^x = b \Rightarrow \ln(a^x) = \ln(b) \Rightarrow x \cdot \ln a = \ln b \Rightarrow x = \frac{\ln b}{\ln a}$ Def. $a^x = b \Rightarrow x = \log_a(b)$
 $a^x = b \Rightarrow \log(a^x) = \log(b) \Rightarrow x \cdot \log a = \log b \Rightarrow x = \frac{\log b}{\log a}$

Chapter 4

Trigonometry in degrees and radians

Trigonometry (from the greek *trigonon* = triangle and *metro* = measure) is a branch of mathematics that studies relationships between side lengths and angles of triangles. The first applications of trigonometry in Ancient Greece were to geometry and astronomical studies. Throughout the years, it has been applied to astronomy, navigation, surveying, periodic functions, optics and acoustics, and many more fields.

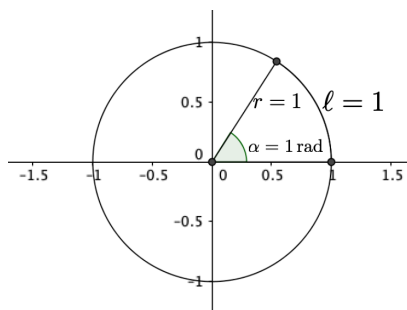
4.1 Angle measurement: degrees and radians

The angle measurement *radian* is a SI unit defined as arclength of the subtended arc on a circle divided by its radius. Thus, for a circle of radius one, 1 radian is the angle to a arc of length 1.

Radians are also referred to as absolute angle measure. Other angle measures are *degrees*, that is perhaps more known. 360° corresponds to 2π radians. The conversion formula is:

$$\frac{\text{rad}}{\pi} = \frac{\text{deg}}{180} \Rightarrow \text{rad} = \frac{\pi}{180} \text{deg}$$

where “rad” is the angle measured in radians and “deg” is the angle in degrees.



Example 4.1

Transform 30° , 45° , 60° , 90° and 360° to radians.

Solution:

Use the conversion formula:

$$30^\circ = \frac{\pi^{\text{rad}}}{6}, 45^\circ = \frac{\pi^{\text{rad}}}{4}, 60^\circ = \frac{\pi^{\text{rad}}}{3}, 90^\circ = \frac{\pi^{\text{rad}}}{2}, 360^\circ = 2\pi^{\text{rad}}$$

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REMARK: In mathematics, it is normal to use radians rather than degrees. Because of this, the “rad” is usually omitted. Therefore, $\frac{\pi}{6}$ means $\frac{\pi^{\text{rad}}}{6}$, when referred to the measure of an angle.

4.2 Right-angle triangles

A right-angle triangle is a triangle where one of the angles is 90° (or $\frac{\pi}{2}$).

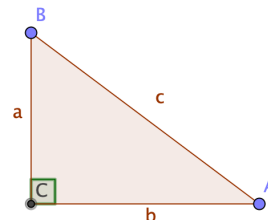
For these triangles, Pythagora's theorem holds:

$$a^2 + b^2 = c^2$$

The sides adjacent to the right angle a, b are also called legs or *catheti*, while c is the *hypotenuse*.

The relation between the catheti and the hypotenuse can be defined with help of sine and cosine:

$$\sin A = \frac{a}{c} \quad \cos A = \frac{b}{c}, \quad \tan A = \frac{a}{b} = \frac{\sin A}{\cos A}$$

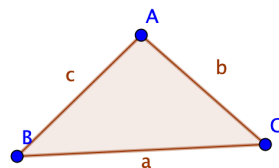


4.3 Computations with triangles

For a generic triangle, Pythagora's theorem does not hold, but can be generalized using the cosine law:

$$\text{Cosine law: } a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Sine law: } \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

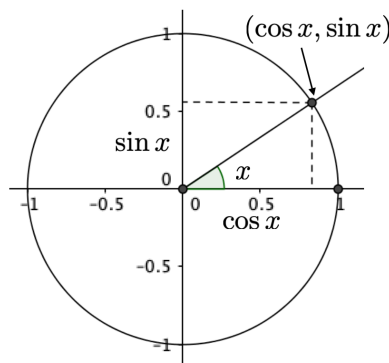


The sine law can be used to compute the remaining sides of a triangle when two angles and a side are known.

4.4 Trigonometry (in radians)

The picture shows the fundamental definitions of the trigonometric functions.

- The *trigonometric circle*: a circle with radius 1 and center at the origin
- The cosine axis: horizontal axis
- The sine axis: vertical axis
- The intersection between the line and the unit circle has coordinates $(\cos x, \sin x)$
- $\tan x = \frac{\sin x}{\cos x}$



By these definitions, and Pythagora's theorem, we have:

$$\sin^2 x + \cos^2 x = 1$$

Also with the help of the unit circle we have:

$$\begin{aligned}\sin(\pi - x) &= \sin x & \cos(\pi - x) &= -\cos x & \tan(\pi - x) &= -\tan x \\ \sin(\pi + x) &= -\sin x & \cos(\pi + x) &= -\cos x & \tan(\pi + x) &= \tan x\end{aligned}$$

One has also that:

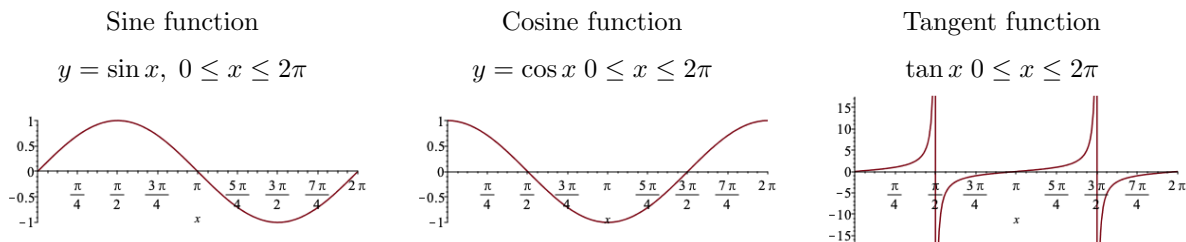
$$\cos x = \sin\left(\frac{\pi}{2} + x\right), \quad \sin x = \cos\left(\frac{\pi}{2} - x\right)$$

which means that the graph of a sine function and a cosine function are the same up to a phase shift of $\frac{\pi}{2}$.

4.5 Some familiar angles

Degree	Radian	$\sin x$	$\cos x$	$\tan x$
0	0	0	1	0
30°	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
45°	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90°	$\frac{\pi}{2}$	1	0	∞
180°	π	0	-1	0
270°	$\frac{3}{2}\pi$	-1	0	$-\infty$
360°	2π	0	1	1

4.6 The graphs of sin, cos and tan

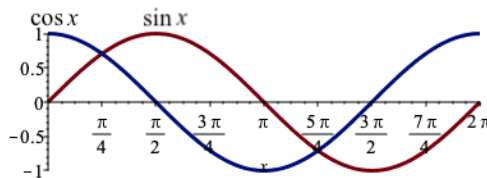


The functions $\sin x$, $\cos x$ and $\tan x$ are periodic. This means that after an interval (period) they repeat themselves. Note that the period of $\sin x$ and $\cos x$ is 2π while the period of $\tan x$ is π :

$$\sin(x + 2\pi) = \sin x, \quad \cos(x + 2\pi) = \cos x, \quad \tan(x + \pi) = \tan x$$

Recall: the graph of sine and cosine have a phase shift of $\frac{\pi}{2}$:

$$\cos x = \sin\left(\frac{\pi}{2} + x\right) \quad \sin x = \cos\left(\frac{\pi}{2} - x\right)$$



4.7 Computations with triangles (in degrees)

Is exactly the same as in Section 4.3. Recall, if you are using a calculator, to choose the degree mode.

4.8 Trigonometric formulae

The basics:

$$\sin^2 x + \cos^2 x = 1 \quad \tan x = \frac{\sin x}{\cos x} \quad \cot x = \frac{1}{\tan x}$$

Sum and difference of angles:

- $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$
- $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$
- $\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$

Double-angle formulae

- $\sin 2x = 2 \sin x \cos x$
- $\cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$

Other useful formulae:

- $\cos u \cos v = \frac{1}{2}(\cos(u + v) + \cos(u - v))$
- $\sin u \sin v = \frac{1}{2}(\cos(u - v) - \cos(u + v))$
- $\sin u \cos v = \frac{1}{2}(\sin(u + v) + \sin(u - v))$

Example 4.2

Compute: a) $\sin \frac{5\pi}{6}$ b) $\tan \frac{5\pi}{5}$ c) $\cos \frac{5\pi}{3}$

Solution:

Hint: Use the table in Section 4.5

- a) $\frac{1}{2}$ b) 1 c) $\frac{1}{2}$

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4.9 Modeling of periodic behaviour with sine and cosine functions

$$y = C_0 + C_1 \cos \frac{2\pi}{T}(t - t_0) \quad \text{and} \quad y = C_0 + C_1 \sin \frac{2\pi}{T}(t - t_0)$$

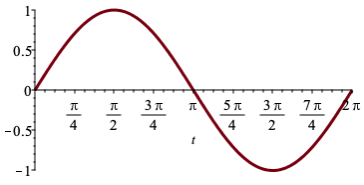
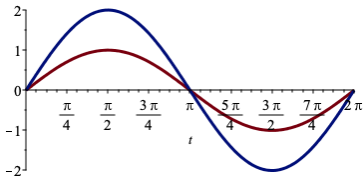
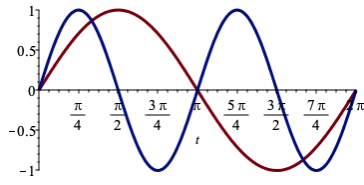
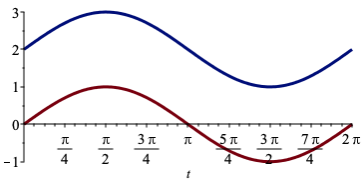
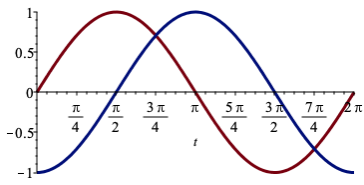
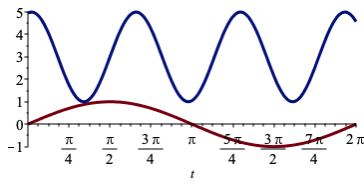
C_0 is the *mean value*, C_1 is the *amplitude*

T is the *period* and $\omega = \frac{2\pi}{T}$ is the *frequency*.

t_0 is the *phase shift*.

NOTE: Since these functions are usually functions of time, therefore we use the variable t instead of x .

Some examples to study these parameters: $y = C_0 + C_1 \sin \omega(t - t_0)$

$C_0 = 0, C_1 = 1, T = 2\pi, t_0 = 0$ $y = \sin t$ (red) 	Amplitude C_1 is doubled $y = 2 \sin t$ (blue) 	Frequency ω is doubled $y = \sin 2t$ (blue) 
Mean value $C_0 = 2$ $y = 2 + \sin t$ (blue) 	Phase shift $t_0 = \pi/2$ $y = \sin(t - \frac{\pi}{2})$ (blue) 	$C_0 = 3, C_1 = 2, \omega = \pi, t_0 = \pi/2$ $y = 3 + 2 \sin \pi(t - \frac{\pi}{2})$ (blue) 

Example 4.3

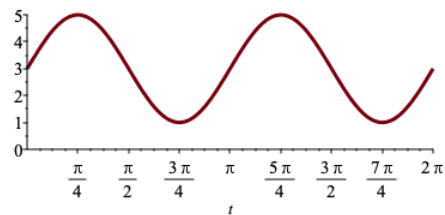
Here you see the graph of a periodic function that can be written in the form $y = C_0 + C_1 \sin \omega(t - t_0)$.

Find C_0, C_1, T, ω and t_0 .

Solution:

It is a sinus function, and we see that the mean value is $C_0 = 3$. The amplitude is $C_1 = 2$ and there is no phase shift, hence $t_0 = 0$. In the interval $[0, 2\pi]$ the graph repeats twice, hence the frequency is $\omega = 2$, and the period is $T = \pi$.

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4.10 The periodic function $f(t) = a \cos \omega t + b \sin \omega t$

This function can be written as a cosine function:

$$\begin{array}{l} a \cos \omega t + b \sin \omega t = C \cos \omega(t - t_0) \\ \text{where } C = \sqrt{a^2 + b^2} \text{ and } \tan \omega t_0 = \frac{b}{a} \end{array}$$

Note that $(C, \omega t_0)$ are the polar coordinates of the point (a, b) , therefore $(C, \omega t_0)$ belongs to the same quadrant of (a, b) . This is useful when solving $\tan \omega t_0 = \frac{b}{a}$ (finding the correct angle).

Example 4.4

Given the functions: **a)** $f(t) = \sqrt{3} \cos 2t + \sin 2t$ **b)** $f(t) = \cos \pi t + \sqrt{3} \sin \pi t$

Write them in the form $C \cos \omega(t - t_0)$ and draw their graph.

Solution:

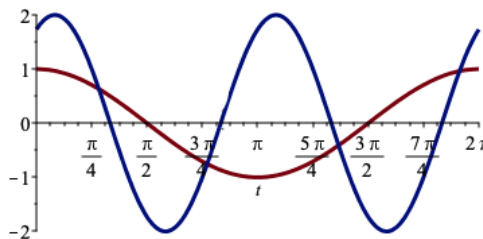
a) $C = \sqrt{a^2 + b^2} = \sqrt{3 + 1} = 2$ (amplitude)

$\tan(2t_0) = \frac{b}{a} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3} \Rightarrow 2t_0 = \frac{\pi}{6} \Rightarrow t_0 = \frac{\pi}{12}$
(phase shift, (a, b) in first quadrant).

$$\underline{\underline{f(t) = 2 \cos 2(t - \frac{\pi}{12})}}$$

So we need to draw a cosine function, with amplitude $C = 2$, frequency $\omega = 2$ and phase shift

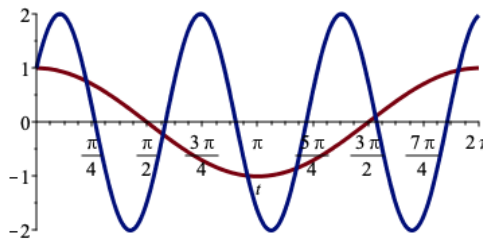
$t_0 = \frac{\pi}{12}$. (See previous section, red: $\cos t$, blue: $f(t)$)



b) $C = \sqrt{a^2 + b^2} = \sqrt{1 + 3} = 2$ (amplitude)

$\tan(\pi t_0) = \frac{b}{a} = \sqrt{3} \Rightarrow \pi t_0 = \frac{\pi}{3} \Rightarrow t_0 = \frac{1}{3}$ (phase shift, (a, b) in first quadrant).

$$\underline{\underline{f(t) = 2 \cos \pi(t - \frac{1}{3})}} \text{ (blue function)}$$



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4.11 Equations of type $a \sin x = b$, $0 \leq x \leq 2\pi$

The equation: $a \sin x = b$, or equivalently $\sin x = \frac{b}{a}$ has solution $x = \sin^{-1}(\frac{b}{a})$.

(On your calculator, just press “SHIFT $\sin(\frac{b}{a})$ ” or “INV $\sin(\frac{b}{a})$ ”. Check that your angles are in “radians” and not degrees!)

Recall that:

- $-1 \leq \sin x \leq 1$ always!

- $\sin x = \frac{b}{a}$ is satisfied by two angles: x_1 (the solution from your calculator) and $\pi - x_1$ for $0 \leq x \leq 2\pi$

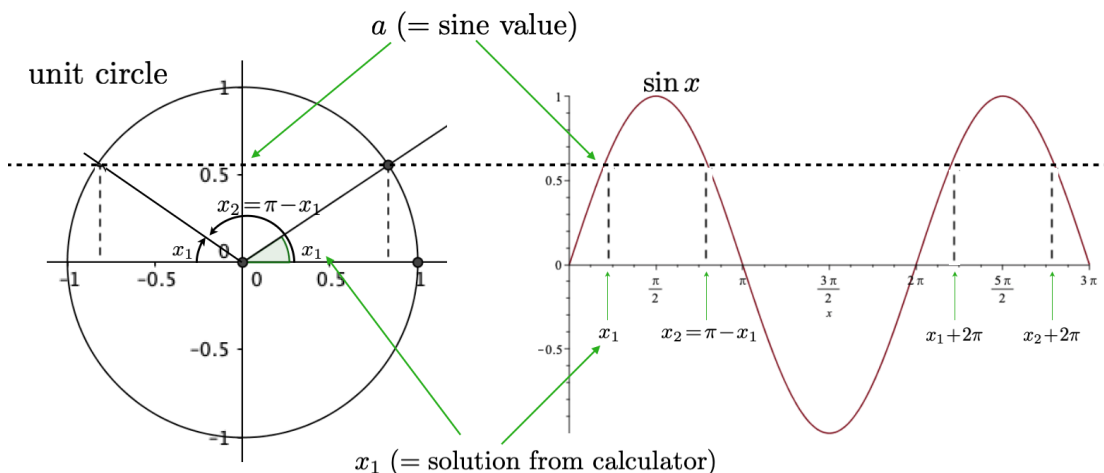
Therefore:

$$x_1 = \sin^{-1}\left(\frac{b}{a}\right) \quad \text{and} \quad x_2 = \pi - \sin^{-1}\left(\frac{b}{a}\right)$$

When we want the *general solution* on the whole real line, we obtain infinitely many angles that repeat every 2π :

$$x_1 = \sin^{-1}\left(\frac{b}{a}\right) + 2n\pi \quad \text{and} \quad x_2 = \pi - \sin^{-1}\left(\frac{b}{a}\right) + 2n\pi, \quad n = 0, \pm 1, \pm 2, \dots$$

See the illustration below using the unit circle and the sine function for an explanation for the solution of $\sin x = a$:



The equation $\sin x = a$ in degrees $0 \leq x \leq 360$ solves exactly in the same manner. Recall to put the calculator in mode “degree”.

Example 4.5

Solve the equations: a) $6 \sin x = 3$, b) $7 \sin x = 2$ b) $7 \sin x = -2$

Solution:

$$\text{a) } 6 \sin x = 3 \Rightarrow \sin x = \frac{3}{6} = \frac{1}{2}$$

$$x_1 = \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6} \quad (\sin^{-1}\left(\frac{1}{2}\right) \text{ is a known angle, see § 4.5})$$

$$x_2 = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

The general solution is: $x_1 = \frac{\pi}{6} + 2n\pi$ and $x_2 = \frac{5\pi}{6} + 2n\pi$, $n = 0, \pm 1, \pm 2, \dots$

$$\text{b) } 7 \sin x = 2 \Rightarrow \sin x = \frac{2}{7}$$

$$x_1 = \sin^{-1}\left(\frac{2}{7}\right) \approx 0.2897 \approx \underline{0.29}$$

$$x_2 = 2\pi - x_1 \approx \pi - 0.29 = \underline{2.85}$$

The general solution is: $x_1 \approx \underline{0.29 + 2n\pi}$ and $x_2 \approx \underline{2.85 + 2n\pi}$, for $n = 0, \pm 1, \pm 2, \dots$

$$\text{c) } 7 \sin x = -2$$

$x_1 = \sin^{-1}\left(-\frac{2}{7}\right) \approx -0.2897 \approx -0.29$ If the answer is required to be in $0 \leq x \leq 2\pi$, we need to add 2π to this value:

$$x_1 \approx -0.29 + 2\pi = \underline{5.99}$$

$$x_2 \approx \pi - (-0.29) = \underline{3.43}$$

The general solution is: $x_1 \approx \underline{5.99 + 2n\pi}$ and $x_2 \approx \underline{3.43 + 2n\pi}$, for $n = 0, \pm 1, \pm 2, \dots$

◇

4.12 Equation of the form: $a \cos x = b$ in $0 \leq x \leq 2\pi$

The equation: $a \cos x = b$, or equivalently $\cos x = \frac{b}{a}$ has solution $x = \cos^{-1}\left(\frac{b}{a}\right)$.

(On your calculator, just press “SHIFT COS($\frac{b}{a}$)” or “INV COS($\frac{b}{a}$)”. Check that your angles are in “radians” and not degrees!) Recall that:

- $-1 \leq \cos x \leq 1$ always!
- $\cos x = \frac{b}{a}$ is satisfied by two angles: x_1 (the solution from your calculator) and $x_2 = 2\pi - x_1$ for $0 \leq x \leq 2\pi$

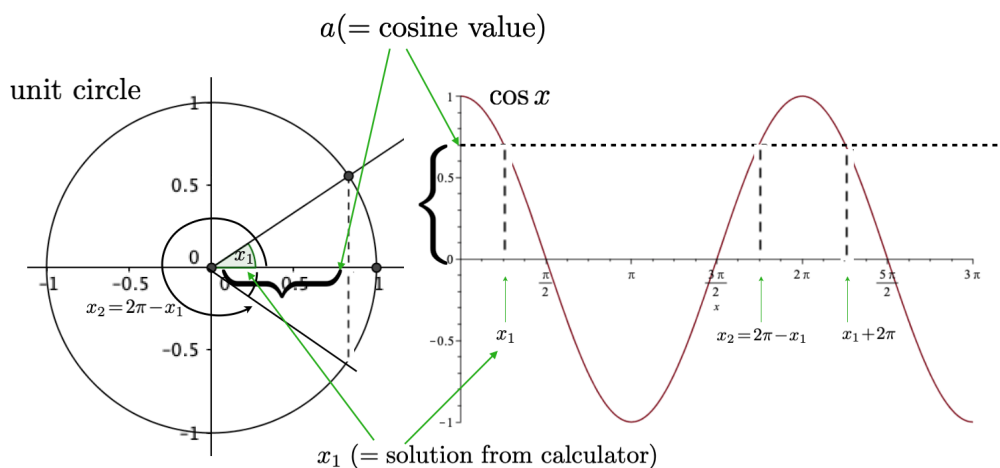
The *general solution* on the whole real line, we obtain infinitely many angles that repeat every 2π :

$$x_1 = \cos^{-1}\left(\frac{b}{a}\right) + 2n\pi \quad \text{and} \quad x_2 = 2n\pi - \cos^{-1}\left(\frac{b}{a}\right), \quad n = 0, \pm 1, \pm 2, \dots$$

The two expressions above differ only for one sign. Both solutions can thus be written in the compact form

$$x_{1,2} = 2n\pi \pm \cos^{-1}\left(\frac{b}{a}\right), \quad n = 0, \pm 1, \pm 2, \dots$$

See the illustration below using the unit circle and the sine function for an explanation for the solution of $\cos x = a$:

**Example 4.6**

Solve the equations: a) $6 \cos x = 3$, b) $7 \cos x = 3$ b) $7 \cos x = -3$

Solution:

$$\text{a) } 6 \cos x = 3 \Rightarrow \cos x = \frac{3}{6} = \frac{1}{2}$$

$$x_1 = \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3} \quad (\cos^{-1}\left(\frac{1}{2}\right) \text{ is a known angle, see § 4.5})$$

$$x_2 = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$$

The general solution is: $\underline{\underline{x_{1,2} = 2n\pi \pm \frac{\pi}{3}}}$, $n = 0, \pm 1, \pm 2, \dots$

$$\text{b) } 7 \cos x = 3 \Rightarrow \cos x = \frac{3}{7}$$

$$x_1 = \cos^{-1}\left(\frac{3}{7}\right) \approx 1.12788 \approx \underline{\underline{1.13}}, \quad x_2 = 2\pi - x_1 \approx 5.16$$

The general solution is: $\underline{\underline{x_1 \approx 2n\pi \pm 1.13}}$ for $n = 0, \pm 1, \pm 2, \dots$

$$\text{c) } 7 \cos x = -3$$

$$x_1 = \cos^{-1}\left(-\frac{3}{7}\right) \approx 2.0134 \approx 2.01 \text{ and } x_2 \approx 4.27.$$

The general solution is: $\underline{\underline{x_{1,2} \approx 2n\pi \pm 2.01}}$ for $n = 0, \pm 1, \pm 2, \dots$

◇

Exercises Chapter 4

Exercise 4.1

Let x be an angle in the first quadrant. Find (without using a calculator) $\sin x$ and $\cos x$ when:

a) $\tan x = \sqrt{3}$ b) $\tan x = 1$ c) $\tan x = \frac{5}{12}$

Exercise 4.2

Solve the equations:

a) $5 \cos x - 2 = 0$, $x \in [0, 2\pi)$
 b) $2 \sin x + \sqrt{3} = 0$, $x \in [0, 2\pi)$
 c) $4 \tan x + 11 = 2$, $x \in [0, 2\pi)$

Exercise 4.3

- a) Given $\cos v = \frac{7}{25}$ and $v \in [270^\circ, 360^\circ]$, determine $\sin v$ and $\tan v$.
 b) Show that $2 \cos(x + \frac{2\pi}{3}) - 4 \sin(x - \frac{\pi}{6}) = \cos x - 3\sqrt{3} \sin x$.
 c) Given $\cos v = -\frac{8}{17}$ and $v \in [\frac{\pi}{2}, \pi]$, determine $\sin v$ and $\sin 2v$, $\cos 2v$ and $\tan 2v$.

Exercise 4.4

In a triangle ABC , the angle $B = 82.5^\circ$, $a = 74.0$ and $b = 92.5$.

- a) Find the remaining side lengths and angles of the triangle.
 b) Find the length of the angle bisector of angle C .

Exercise 4.5

We know the following information about triangle ABC . In each exercise, find all the side lengths and angles of the triangle.

- a) $a = 4.7$ cm, $c = 6.9$ cm and $\angle C = 56^\circ$.
 b) $c = 7.2$ cm, $\angle A = 51^\circ$ and $\angle C = 72^\circ$
 c) $\angle B = 48^\circ$, $a = 8.0$ cm and $c = 6.3$ cm

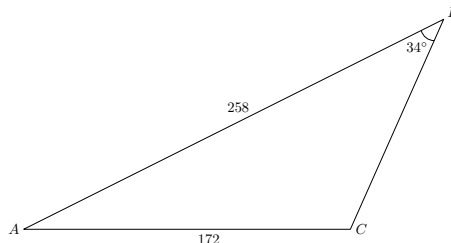
Exercise 4.6

In triangle ABC , we have $|AB| = 11.9$, $|BC| = 26.4$, the angle $C = 23.5^\circ$ and the angle A is obtuse. Determine angle A .

Exercise 4.7

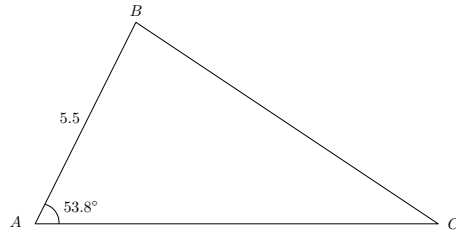
The figure shows the triangle ABC where the angle C is obtuse. The measures are shown in the figure.

- a) Find angle C .
 b) Determine the area of the triangle ABC .



Exercise 4.8

The figure shows a triangle ABC where the angle $A = 53.8^\circ$ and $|AB| = 5.5$. The area of the triangle is 24.4. Determine $|AC|$ and $|BC|$.

**Exercise 4.9**

A building site has the shape of a quadrilateral $ABCD$ where the angle $A = 80^\circ$, the angle $B = 60^\circ$, the angle $C = 105^\circ$, $|AD| = 21$ m and $|AB| = 50$ m.

- Draw a sketch of the building site and determine the length of the diagonal BD .
- Determine the area of the building site.

Exercise 4.10

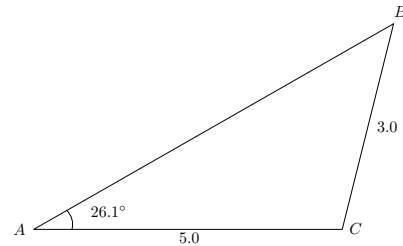
A triangle ABC is determined by $a = 9$, $b = 12$ and $c = 10$.

- Determine the angle A .
- Determine the area of the triangle ABC .

Exercise 4.11

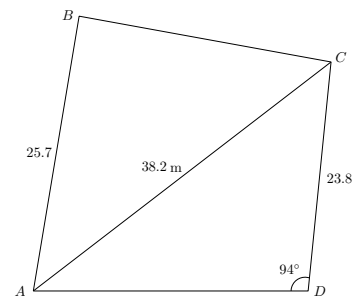
In a triangle ABC , the angle $A = 26.1^\circ$, $|AC| = 5.0$ and $|BC| = 3.0$. The angle C is obtuse.

- Determine the angles B and C .
- Determine $|AB|$.
- Determine the length of the median m_c .

**Exercise 4.12**

A building site has the shape of a quadrilateral $ABCD$. Three of the sides have the following lengths: $|AB| = 25.7$ m, $|BC| = 25.1$ m and $|CD| = 23.8$ m. The length of the diagonal AC is 38.2 m. The angle $D = 94^\circ$.

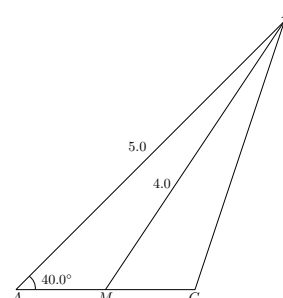
- Determine the angle B .
- Determine the area of the building site.



Exercise 4.13

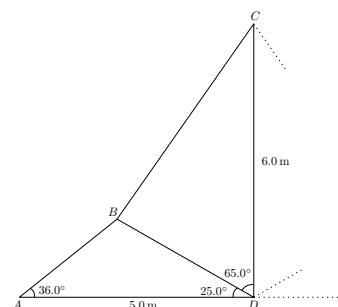
The figure shows a triangle ABC where M is the mid point of the side AC . The known measures are shown in the figure, and we know that the angle ABM is obtuse.

- Determine the angle ABM and $|AC|$.
- Determine $|BC|$ and the angle B in the triangle ABC .
- Determine the area of the triangle ABC .

**Exercise 4.14**

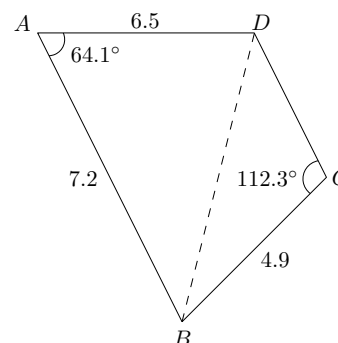
The figures shows a sketch of a gable construction for a summer house. The known measures are shown in the figure.

- Determine the length of the beams AB and BD .
- Determine the length of the beam BC and the angle BCD .

**Exercise 4.15**

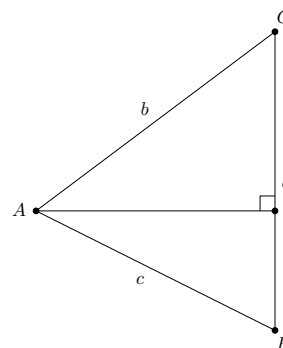
The figure shows a quadrilateral $ABCD$ where the diagonal BD is drawn. The known measures of the quadrilateral are shown in the figure.

- Determine $|BD|$.
- Determine the angle D in the triangle BDC .
- Determine $|CD|$.
- Determine $|AC|$.

**Exercise 4.16**

Consider the triangle to the right.

- Show that $a = b \cdot \cos C + c \cdot \cos B$.
- Use the sine law to show that $\sin A = \sin B \cdot \cos C + \sin C \cdot \cos B$.



Exercise 4.17

In the triangle ABC , we have $AB = 7$ cm, $BC = 4$ cm and $AC = 6$ cm. The angle bisector of angle C divides AB into the two parts, x and y . Determine the lengths of these two parts. Do the same calculations but now with $AB = c$, $BC = a$ and $AC = b$.

Exercise 4.18

In the triangle ABC we have $\angle C = 90^\circ$, $\angle A = 30^\circ$ and $AB = s$. The angle bisector for $\angle C$ divides AB into two parts. Determine the lengths of these two parts without using approximate values.

Exercise 4.19

Given a triangle $\triangle ABC$ with side lengths 4, 5.5 and 8 units, find the angles of the triangle.

Exercise 4.20

a) Use the graphs of the functions $\sin x$, $\cos x$ and $\tan x$ to show that

$$\begin{aligned}\sin(\pi - x) &= \sin x & \cos(\pi - x) &= -\cos x \\ \sin(\pi + x) &= -\sin x & \cos(\pi + x) &= -\cos x\end{aligned}$$

b) Show the same relations by using the unit circle and the definitions of $\sin x$ and $\cos x$

Exercise 4.21

- a) Use that $45^\circ + 30^\circ = 75^\circ$ to find exact values for $\sin 75^\circ$ and $\cos 75^\circ$.
b) Use that $45^\circ - 30^\circ = 15^\circ$ to find exact values for $\sin 15^\circ$ and $\cos 15^\circ$.

Exercise 4.22

a) Draw the graph of the function $f(x) = 24 - 22 \cos(\frac{\pi}{12}x)$, $x \in [0, 24]$.

The function describes how high in the sky the sun is during a summer day somewhere north of the polar circle. We call this height over the horizon for the sun height and measure it in degrees.

- b) Determine the sun height at the times 04.00 and 16.00.
c) When was the sun height 2° ?
d) When was the sun height the largest? And what was the sun height at this time?

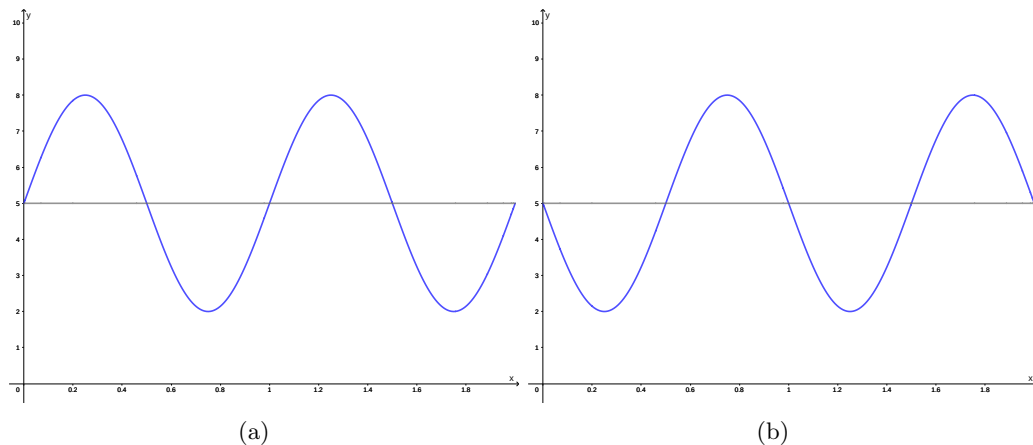
Exercise 4.23

Write the following functions on the form $f(x) = C \cos(x - v)$:

- a) $f(x) = 3 \sin x - 4 \cos x$ b) $f(x) = \cos x - \sqrt{3} \sin x$

Exercise 4.24

The graph of a function on the form $y = c + a \sin(\omega(x - x_0))$ is given below:
Determine c , a , ω and x_0 .



Solution to the Exercises Chapter 4

4.1 a) Imagine a right angled triangle where the length of the opposite side of angle x is $\sqrt{3}$ and the length of the adjacent side is 1, i.e., $\tan x = \sqrt{3}$. The length of the hypotenuse is then $\sqrt{1 + (\sqrt{3})^2} = 2$. This further gives $\sin x = \frac{\sqrt{3}}{2}$ and $\cos x = \frac{1}{2}$. We can use the same approach for b) and c).

b) $\sin x = \frac{\sqrt{2}}{2}$ and $\cos x = \frac{\sqrt{2}}{2}$.

c) $\sin x = \frac{5}{13}$ and $\cos x = \frac{12}{13}$.

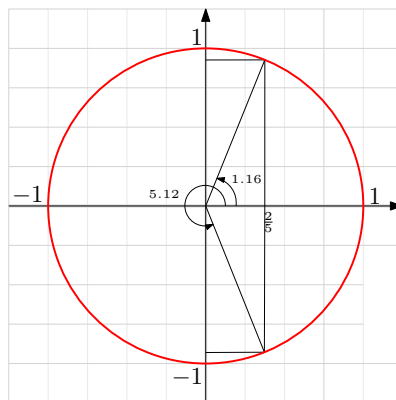
4.2 a)

$$5 \cos x - 2 = 0$$

$$5 \cos x = 2$$

$$\cos x = \frac{2}{5}$$

We use the calculator to find one of the solutions: $x = 1.16$. The other solution can be found by drawing the unit circle, $x = 2\pi - 1.16 = 5.12$. Hence the solutions to the equation are $x = 1.16$ and $x = 5.12$.

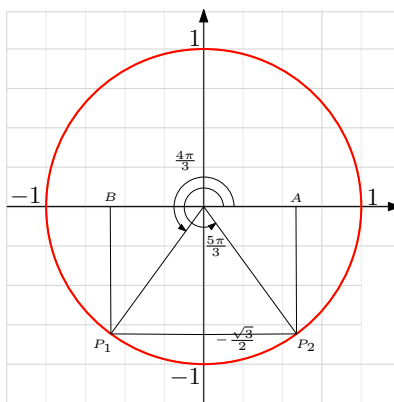


b)

$$\begin{aligned}
 2 \sin x + \sqrt{3} &= 0, & x &\in [0, 2\pi) \\
 2 \sin x &= -\sqrt{3} \\
 \sin x &= -\frac{\sqrt{3}}{2}
 \end{aligned}$$

We draw the unit circle and find two solutions.

We obtain two $30-60-90$ -triangles where $\angle AOP_2 = \frac{\pi}{3}$ and $\angle BOP_1 = \frac{\pi}{3}$. By symmetry, we obtain the solutions $x = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$ and $x = \pi + \frac{\pi}{3} = \frac{4\pi}{3}$. Hence, the two solutions to the equation are: $x = \frac{4\pi}{3}$ and $x = \frac{5\pi}{3}$.



c)

$$\begin{aligned}
 4 \tan x + 11 &= 2, & x &\in [0, 2\pi) \\
 4 \tan x &= 2 - 11 \\
 \tan x &= -\frac{9}{4}
 \end{aligned}$$

The calculator gives the solution $x = -1.15$. The general solution is thus $x = -1.15 + n \cdot \pi$. Since the solution is supposed to lie in the first revolution, we require $x \in [0, 2\pi)$.

$$n = 1 \Rightarrow x = -1.15 + 1 \cdot 3.14 = 1.99$$

$$n = 2 \Rightarrow x = -1.15 + 2 \cdot 3.14 = 5.13$$

Thus, the solutions are $x = 1.99$ and $x = 5.13$.

4.3 a) $\cos v = \frac{7}{25}$ and $v \in [270^\circ, 360^\circ]$. The Pythagorean identity gives $\cos^2 v + \sin^2 v = 1$ or $\sin^2 v = 1 - \cos^2 v$.

$$\sin^2 v = 1 - \left(\frac{7}{25}\right)^2 = 1 - \frac{49}{625} = \frac{576}{625} \quad \sin v = \pm \frac{24}{25}$$

Since $v \in [270^\circ, 360^\circ]$, $\sin v$ is negative. Thus $\sin v = -\frac{24}{25}$.

The definition of $\tan v$ gives

$$\tan v = \frac{\sin v}{\cos v} = \frac{-\frac{24}{25}}{\frac{7}{25}} = \frac{-\frac{24}{25} \cdot 25}{\frac{7}{25} \cdot 25} = -\frac{24}{7}.$$

b)

$$\begin{aligned} & 2 \cos\left(x + \frac{2\pi}{3}\right) - 4 \sin\left(x - \frac{\pi}{6}\right) \\ &= 2 \left(\cos x \cdot \cos \frac{2\pi}{3} - \sin x \cdot \sin \frac{2\pi}{3} \right) - 4 \left(\sin x \cdot \cos \frac{\pi}{6} - \cos x \cdot \sin \frac{\pi}{6} \right) \\ &= 2 \left(\cos x \cdot \left(-\frac{1}{2}\right) - \sin x \cdot \frac{1}{2}\sqrt{3} \right) - 4 \left(\sin x \cdot \frac{1}{2}\sqrt{3} - \cos x \cdot \frac{1}{2} \right) \\ &= -\cos x - \sqrt{3} \sin x - 2\sqrt{3} \sin x + 2 \cos x \\ &= \cos x - 3\sqrt{3} \sin x \end{aligned}$$

c) $\cos v = -\frac{8}{17}$ and $v \in [\frac{\pi}{2}, \pi]$

$$\cos^2 v + \sin^2 v = 1 \text{ or } \sin^2 v = 1 - \cos^2 v$$

$$\sin^2 v = 1 - \left(-\frac{8}{17}\right)^2 = 1 - \frac{64}{289} = \frac{225}{289}$$

$\sin v = \pm \frac{15}{17}$. Since $v \in [\frac{\pi}{2}, \pi]$, $\sin v$ is positive. This gives $\sin v = \frac{15}{17}$.

$$\sin 2v = 2 \sin v \cdot \cos v = 2 \frac{15}{17} \cdot \left(-\frac{8}{17}\right) = -\frac{240}{289}.$$

$$\cos 2v = 1 - 2 \sin^2 v = 1 - 2 \left(\frac{15}{17}\right)^2 = 1 - \frac{2 \cdot 225}{289} = -\frac{161}{289}$$

$$\tan 2v = \frac{\sin 2v}{\cos 2v} = \frac{-\frac{240}{289}}{-\frac{161}{289}} = \frac{240}{161}.$$

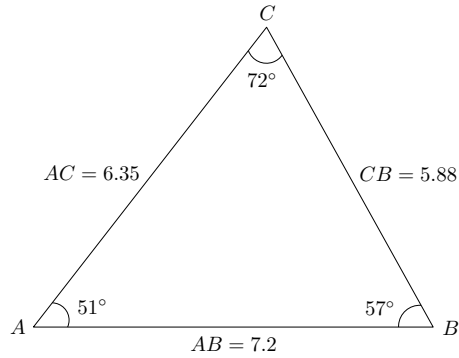
4.4 a) $A = 52.5^\circ$, $C = 45.1^\circ$, $c = 66.0$ b) $v_c = 76.0$

4.5 a) $\frac{c}{\sin C} = \frac{a}{\sin A} \Rightarrow \sin A = \frac{a \cdot \sin C}{c} = \frac{4.7}{6.9} \cdot \sin 56^\circ = 0.5647$, $A = 34.4^\circ$.

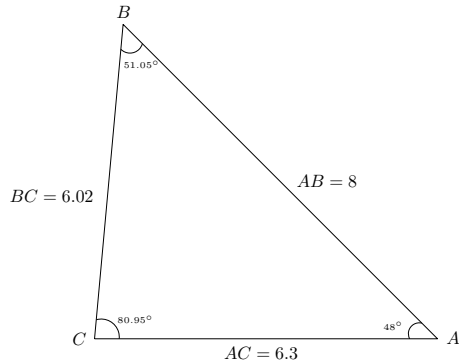
$$\angle B = 180^\circ - 56^\circ - 34.4^\circ = 89.6^\circ$$

$$\frac{b}{\sin B} = \frac{c}{\sin C} \Rightarrow b = \frac{c}{\sin C} \cdot \sin B = \frac{6.9 \text{ cm}}{\sin 56^\circ} \cdot \sin 89.6^\circ = 8.3 \text{ cm}.$$

$$\begin{aligned} \text{b) } \frac{a}{\sin A} &= \frac{c}{\sin C} \Rightarrow a = \frac{c}{\sin C} \cdot \sin A = \frac{7.2 \text{ cm}}{\sin 72^\circ} \cdot \sin 51^\circ \\ B &= 180^\circ - 51^\circ - 72^\circ = 57^\circ \\ b &= \frac{c \cdot \sin B}{\sin C} = \frac{7.2 \text{ cm} \cdot \sin 57^\circ}{\sin 72^\circ} = 6.3 \text{ cm.} \end{aligned}$$



$$\begin{aligned} \text{c) } b &= \sqrt{a^2 + c^2 - 2ac \cos B} = \sqrt{8^2 + 6.3^2 - 2 \cdot 8 \cdot 6.3 \cdot \cos 48^\circ} = 6.0 \text{ cm} \\ \frac{a}{\sin A} &= \frac{b}{\sin B} \Rightarrow \sin A = \frac{a \cdot \sin B}{b} = \frac{8.0 \cdot \sin 48^\circ}{6.0} = 0.99, A = 82^\circ \\ C &= 180^\circ - 48^\circ - 82^\circ = 50^\circ. \end{aligned}$$



4.6 The sine law: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ gives $b = \frac{a \cdot \sin B}{\sin A}$, $c = \frac{a \cdot \sin C}{\sin A}$. Thus, we have
 $a = c \cdot \cos B + b \cdot \cos C = \frac{a \cdot \sin B}{\sin A} \cdot \cos C + \frac{a \cdot \sin C}{\sin A} \cdot \cos B$. We divide this equality by a and multiply by $\sin A$: $\sin A = \sin B \cdot \cos C + \sin C \cdot \cos B$.

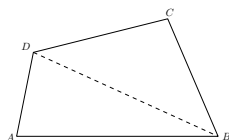
Since $\sin(B + C) = \sin(180^\circ - B - C) = \sin A$, it follows that $\sin(B + C) = \sin B \cos C + \sin C \cos B$

4.6 a) $A = 117.8^\circ$

4.7 a) $C = 123^\circ$ **b)** area 8674

4.8 a) $|AC| = 11.0$, $|BC| = 8.9$

4.9 a) $|BD| = 50.8$ m b) 1010 m².



4.10 a) $A = 47.2^\circ$ b) area 44.0

4.11 a) $B = 47.2^\circ$, $C = 106.7^\circ$ b) $|AB| = 6.5$ c) $m_c = 2.5$

4.12 a) $B = 97.5^\circ$ b) 718.3

4.13 a) angle $AMB = 126.5^\circ$, $|AC| = 2.9$

b) $|BC| = 5.1$, $B = 33.4^\circ$

c) 4.66

4.14 a) $|AB| = 2.4$ m, $|BD| = 3.4$ m

b) $|BC| = 5.5$ m, angle $BCD = 33.62^\circ$

4.15 a) $|BD| = 7.3$ b) angle $BDC = 38.4^\circ$

c) $|CD| = 3.9$ d) $|AC| = 8.2$

4.16 a) It is easy to show using the figure $a = b \cdot \cos C + c \cdot \cos B$

b) We can find this by dividing both sides by b : $\frac{a}{b} = \cos C + \frac{c}{b} \cdot \cos B$.

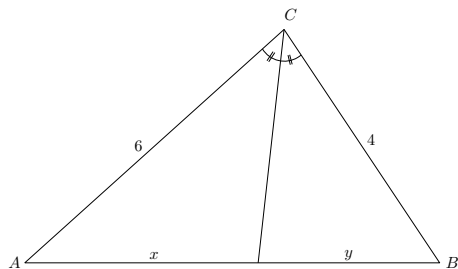
We know that $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$, and thus we have $\frac{a}{b} = \frac{\sin A}{\sin B}$ and $\frac{c}{b} = \frac{\sin C}{\sin B}$.

$\frac{\sin A}{\sin B} = \cos C + \frac{\sin C}{\sin B} \cdot \cos B$. By multiplying both sides by $\sin B$, we obtain the answer:
 $\sin A = \sin B \cdot \cos C + \sin C \cdot \cos B$.

4.17 See the figure below.

According to the angle bisector theorem, we have $\frac{x}{y} = \frac{6}{4}$. We thus obtain $x = \frac{3}{2}y$. We also have that $x + y = 7$. We therefore have $\frac{3}{2}y + y = 7$ and we obtain $\frac{5}{2}y = 7$ and thus $y = \frac{14}{5}$, $x = \frac{3}{2} \cdot \frac{14}{5} = \frac{21}{5}$.

When $AB = c$, $BC = a$ and $AC = b$, we have $x + y = c$ and $\frac{x}{y} = \frac{AC}{BC} = \frac{b}{a}$ and $x = \frac{b}{a}y$. Thus $(1 + \frac{b}{a})y = c$ and $y = \frac{c}{1 + \frac{b}{a}} = \frac{a \cdot c}{a + b}$, $x = \frac{b \cdot c}{a + b}$.



4.18 In this triangle, $AB = s$, $BC = \frac{1}{2}s$, $AC = \frac{1}{2}s\sqrt{3}$. The angle bisector theorem gives

$$\frac{\sqrt{3}/2}{1/2} = \sqrt{3} = \frac{AC}{BC} = \frac{AD}{BD} = \frac{AD}{s-AD} \text{ and thus } AD = \sqrt{3} \cdot (s-AD), AD \cdot (1+\sqrt{3}) = \sqrt{3}s$$

$$AD = \frac{\sqrt{3} \cdot (\sqrt{3}-1) \cdot s}{(\sqrt{3}+1)(\sqrt{3}-1)} = \frac{1}{2}(3-\sqrt{3}) \cdot s, \quad BD = AB-AD = s \cdot (1-\frac{1}{2} \cdot 3 + \frac{\sqrt{3}}{2}) = \frac{s}{2}(\sqrt{3}-1).$$

4.19 Let $a = 5.5$, $b = 8$ and $c = 4$. We determine $\angle \alpha$ using the cosine law:

Here, it is advantageous to rewrite:

$$a^2 = b^2 + c^2 - 2bc \cos \alpha \text{ such that } \cos \alpha \text{ is alone on one side:}$$

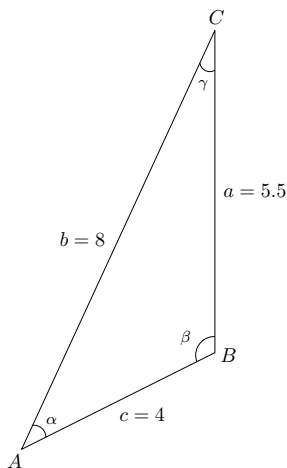
$$\cos \alpha = \frac{b^2 + c^2 - a^2}{2bc} = \frac{8^2 + 4^2 - 5.5^2}{2 \cdot 8 \cdot 4} = \frac{49.75}{64}$$

$$\alpha = \cos^{-1}(0.78) = 38.74^\circ.$$

We use the sine law to find $\angle \gamma$

$$\frac{\sin \alpha}{a} = \frac{\sin \gamma}{c} \Rightarrow \sin \gamma = \frac{c \cdot \sin \alpha}{a} = \frac{4 \sin(38.74)}{5.5} \approx 0.46, \sin \gamma \approx 0.46.$$

$$\gamma = \sin^{-1} 0.46 \approx 27.4^\circ, \text{ and thus: } \beta = 180^\circ - (38.74^\circ + 27.40^\circ) = 113.86^\circ.$$



4.20 a) This can be seen by drawing the graphs.

b) This can be seen by drawing the unit circle and with the angles and see that their sine and cosine values are equal.

$$\mathbf{4.21 \ a)} \quad \sin 75^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}, \cos 75^\circ = \frac{\sqrt{6} - \sqrt{2}}{4} \quad \mathbf{b)} \quad \sin 15^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}, \cos 15^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}$$

4.22 b) At 00.00 **c)** At 00.00, 06.00 and 18.00. **d)** Highest at 12.00.

$$\mathbf{4.23 \ a)} \quad C = 5, v = \tan^{-1} \frac{3}{-4} \approx \pi - 0.6435 \approx 2.49 \quad \mathbf{b)} \quad C = 2, v = \tan^{-1} \frac{-\sqrt{3}}{1} \approx 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}.$$

$$\mathbf{4.24 \ a)} \quad y = 5 + 3 \sin 2\pi x \quad \mathbf{b)} \quad y = 5 + 3 \sin 2\pi(x - 0.5) \text{ or } y = 5 - 3 \sin 2\pi x$$

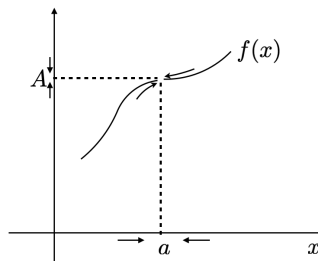
Chapter 5

Limits and continuity

5.1 Limits

Let $y = f(x)$ be a function, defined in a neighborhood of a point a but not necessarily defined for $x = a$. If, when the variable x goes to a , the function value $f(x)$ approaches a value A , then we write:

$$\lim_{x \rightarrow a} f(x) = A$$



Some rules for limits:

Assume $\lim_{x \rightarrow a} f(x) = A$ and $\lim_{x \rightarrow a} g(x) = B$. Then the following holds:

- $\lim_{x \rightarrow a} [f(x) \pm g(x)] = A \pm B$
- $\lim_{x \rightarrow a} [f(x) \cdot g(x)] = A \cdot B$
- $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{A}{B} \quad (B \neq 0)$

5.2 The limit $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0}$

In this section, we will look at the case when $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0}$. This should be indented as

$\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$ as the fraction $\frac{0}{0}$ itself is not defined. Later we will look at

cases $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \frac{\infty}{\infty}$ (same consideration about the meaning of the fraction).

The basic technique is to attempt to factor out $(x - a)$ in both numerator and denominator. As long as $x \neq a$, these simplify out and we try to compute the limit.

Another technique is to use *L'Hôpital's rule*: $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0} \Rightarrow \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$

This rule might need be applied several times.

Example 5.1

Find the following limits:

a) $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$

b) $\lim_{x \rightarrow 1} \frac{x^2 + 2x - 3}{x^2 - 1}$

c) $\lim_{x \rightarrow 1} \frac{x - 1}{\sqrt{x} - 1}$

d) $\lim_{x \rightarrow 0} \frac{\sin 5x}{x}$

e) $\lim_{x \rightarrow 9} \frac{x - 9}{\sqrt{x} - 3}$

f) $\lim_{x \rightarrow 1} \frac{x - 1}{x - \sqrt{x}}$

Solution:

a) $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} \frac{(x + 2)(x - 2)}{x - 2} = \lim_{x \rightarrow 2} (x + 2) = \underline{4}$

b) $x^2 + 2x - 3 = (x + 3)(x - 1)$, $x^2 - 1 = (x + 1)(x - 1)$. Hence $\lim_{x \rightarrow 1} \frac{x^2 + 2x - 3}{x^2 - 1} = \lim_{x \rightarrow 1} \frac{x + 3}{x + 1} = \underline{2}$

c) $\lim_{x \rightarrow 1} \frac{x - 1}{\sqrt{x} - 1} = \underline{2}$ (*Hint*: factor out $(\sqrt{x} - 1)$)

d) $\lim_{x \rightarrow 0} \frac{\sin 5x}{x} = \lim_{x \rightarrow 0} \frac{5 \cdot \sin 5x}{5x} = \underline{5}$ (*Hint*: use L'Hôpital's rule on $\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\cos x}{1} = 1$)

e) $\lim_{x \rightarrow 9} \frac{x - 9}{\sqrt{x} - 3} = \underline{6}$ (*Hint*: factor out $(\sqrt{x} - 3)$)

f) $\lim_{x \rightarrow 1} \frac{x - 1}{x - \sqrt{x}} = \underline{2}$ (*Hint*: $x - \sqrt{x} = \sqrt{x}(\sqrt{x} - 1)$, $x - 1 = \dots$)

◇

Some useful limits

$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$	$\lim_{x \rightarrow 0} \frac{\sin ax}{bx} = \frac{a}{b}$	$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$	$\lim_{x \rightarrow 0} \frac{\tan ax}{bx} = \frac{a}{b}$
---	---	---	---

Note that $\lim_{x \rightarrow 0} \frac{\sin ax}{bx} = \lim_{x \rightarrow 0} \frac{a}{b} \frac{\sin ax}{ax} = \frac{a}{b}$

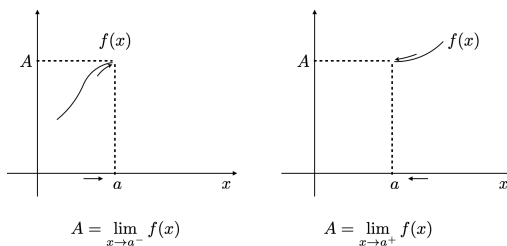
5.3 One-sided limits $\lim_{x \rightarrow a^+}$ and $\lim_{x \rightarrow a^-}$

For a given function $y = f(x)$, it is not always possible approach $x = a$ both from the left and the right side, as, for instance the function might not be defined on the other side.

If a function is defined in a neighborhood of a at least on one of the sides, then we have a one-sided limit:

$\lim_{x \rightarrow a^+} f(x)$, if we can approach $x = a$ from the right

$\lim_{x \rightarrow a^-} f(x)$ if we can approach $x = a$ from the left.



One-sided limits can be computed also if the function is defined both on the left and the right of $x = a$.

Moreover:

$\lim_{x \rightarrow a} f(x) = A$ exists if and only if $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = A$

One-sided limits are relevant to study the continuity of some functions, like for instance

$$f(x) = \frac{|x-2|}{x-2} \text{ or } f(x) = \begin{cases} \sqrt{x-2} & x \geq 2 \\ x^2 - 4 & x < 2 \end{cases}$$

Example 5.2

Compute the following limits: **a)** $\lim_{x \rightarrow 3+} \frac{|x-3|}{x-3}$ **b)** $\lim_{x \rightarrow 2} \frac{|x-2|}{x-2}$ **c)** $\lim_{x \rightarrow 0} (x^2 + \frac{x}{|x|})$

Solution:

a) For $x > 3$ is $|x-3| = x-3$. Hence $\lim_{x \rightarrow 3+} \frac{|x-3|}{x-3} = \underline{1}$

b) For $x > 2$ is $|x-2| = x-2$, for $x < 2$: $|x-2| = -(x-2)$, hence

$\lim_{x \rightarrow 2+} \frac{|x-2|}{x-2} = 1 \neq -1 = \lim_{x \rightarrow 2-} \frac{|x-2|}{x-2}$. The limit does not exist.

c) The limit does not exist (left and right limits are different)

◇

5.4 Continuity

A function $y = f(x)$ is *continuous* at $x = a$ if and only if the function is defined in a and the limit in this point exists and is equal to the function value:

$$f(x) \text{ continuous at } x = a \Leftrightarrow \lim_{x \rightarrow a} f(x) = f(a)$$

Example 5.3

Study the continuity of the following functions:

a) At $x = 2$ for: $f(x) = \begin{cases} \sqrt{x-2} & x \geq 2 \\ x^2 - 4 & x < 2 \end{cases}$ **b)** At $x = 5$ for: $f(x) = \begin{cases} \frac{|x-5|}{x-5} & x \neq 5 \\ 1 & x = 5 \end{cases}$

Solution:

a) $\lim_{x \rightarrow 2+} f(x) = \lim_{x \rightarrow 2+} \sqrt{x-2} = 0 = \lim_{x \rightarrow 2-} f(x) = \lim_{x \rightarrow 2-} (x^2 - 4)$. Thus $\lim_{x \rightarrow 2} f(x) = 0$ exists and equals the function value: $f(2) = 0$. $f(x)$ is continuous at $x = 2$.

b) $\lim_{x \rightarrow 5+} f(x) = 1$, $\lim_{x \rightarrow 5-} f(x) = -1$. The limit at $x = 5$ does not exist. The function is not continuous at $x = 5$.

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5.5 About the limit $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \frac{\infty}{\infty}$

$$\lim_{x \rightarrow \infty} \frac{a_n x^n + a_{n-1} x^{n-1} + \dots + a_0}{b_m x^m + b_{m-1} x^{m-1} + \dots + b_0} = \frac{\infty}{\infty} = \lim_{x \rightarrow \infty} \frac{x^n (a_n + \frac{a_{n-1}}{x} + \dots + \frac{a_0}{x^n})}{x^m (b_m + \frac{b_{m-1}}{x} + \dots + \frac{b_0}{x^m})}$$

$$= \lim_{x \rightarrow \infty} \frac{a_n}{b_m} x^{n-m} = \begin{cases} 0 & n < m \\ \frac{a_n}{b_m} & n = m \\ \infty & n > m \end{cases}$$

When the limit goes towards ∞ then we say that the limit does not exist.

Example 5.4

Compute the following limits:

a) $\lim_{x \rightarrow \infty} \frac{x^2 - 3x - 1}{x^2 + 2}$ b) $\lim_{x \rightarrow \infty} \frac{x^3 + 7x^2 - 5}{x^4 + 2}$ c) $\lim_{x \rightarrow \infty} \frac{x^5 - x - 1}{x^4 + 8x}$ d) $\lim_{x \rightarrow \infty} \frac{\sqrt{9x+1}}{\sqrt{x+1}}$

Solution:

a) 1 b) 0 c) ∞ (limit does not exist) d) 3

◇

5.6 Asymptotes

When a function $y = f(x)$ approaches closer and closer a straight line but never quite reaches it, this line is called an *asymptote*.

We will look further into three types of asymptotes: horizontal, vertical and oblique.

$$f(x) = \frac{x+2}{x-2}$$

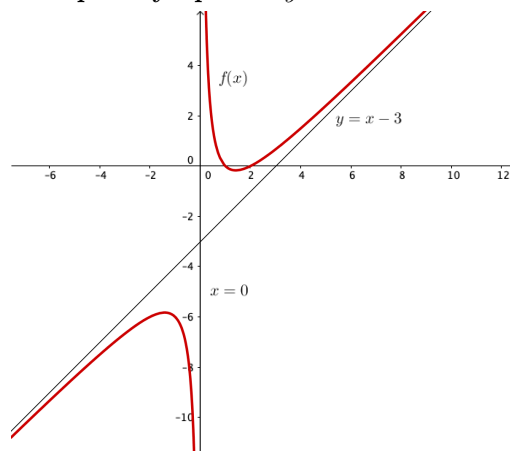
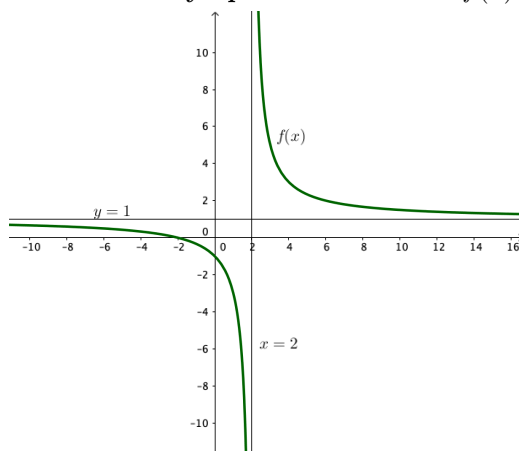
$$f(x) = \frac{x^2 - 3x + 2}{x} = x - 3 + \frac{2}{x}$$

Vertical asymptote: $x = 2 \Rightarrow f(x) \rightarrow \pm\infty$

Vertical asymptote: $x = 0 \Rightarrow f(x) \rightarrow \pm\infty$

Horizontal asymptote: $x \rightarrow \pm\infty \Rightarrow f(x) \rightarrow 1$

Oblique asymptote: $y = x - 3$



Example 5.5

Find the asymptotes of the following functions:

$$\text{a) } y = \frac{x^2 - 5x + 1}{x^2 + 1} \quad \text{b) } y = \frac{x - 1}{|x| + 3} \quad \text{c) } y = \frac{x}{x^2 - 4} \quad \text{d) } y = \frac{x^2 + |x| + 1}{x}$$

Solution:

$$\text{a) } x = \pm 1, y = 1 \quad \text{b) } y = \pm 1 \quad \text{c) } x = \pm 2, y = 0 \quad \text{d) } x = 0, y = x \pm 1$$

◇

Example 5.6

Find the following limits:

$$\text{a) } \lim_{x \rightarrow \infty} \left(\frac{2}{3}\right)^x + \left(\frac{3}{2}\right)^x \quad \text{b) } \lim_{x \rightarrow \infty} \frac{2^x - 3}{3^x + 5} \quad \text{c) } \lim_{x \rightarrow \infty} \frac{3 \cdot s^x + 9}{2^x - 3}$$

Solution:

$$\text{a) } \infty \text{ (does not exist)} \quad \text{b) } 0 \quad \text{c) } 3$$

◇

5.7 The number e

The number e is called *Euler's constant* after the swiss mathematician Leonhard Euler and *Napier's constant* after the scottish mathematician John Napier. Napier is the inventor of logarithms. The basic idea of what logarithms were to achieve is straightforward: to replace the wearisome task of multiplying two numbers by the simpler task of adding together two other numbers. Napier's original work on logarithms (published in 1614) involved, though, base $1/e$ and not e itself. The first known use of the constant, then represented by a b , was in a letter exchange between Goffried Leibniz and Christiaan Huygens in 1690 and 1691.

Some mean that the name choice “ e ” comes from “exponential” as e appears naturally as choice of base for the exponential function. Leonhard Euler started using the letter e for the constant in 1727. It's not known whether he called it after himself or not – in any case, it was first used as e , with approximate value

$$e \approx 2.718281 \dots$$

in Euler's *Mechanica*, published in 1736.

The actual value of the constant e itself is credited to Jacob Bernoulli in 1683, who attempted to find the value of the following expression

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n \quad (= e)$$

while studying compound interest.

The constant e can also be calculated as the sum of the infinite series:

$$e = \sum_{n=0}^{\infty} \frac{1}{n!} = 1 + \frac{1}{1} + \frac{1}{1 \cdot 2} + \frac{1}{1 \cdot 2 \cdot 3} + \dots$$

Exercises Chapter 5

Exercise 5.1

- I) Determine the limits: a) $\lim_{x \rightarrow 1} f(x) = \frac{x^2 + 1}{x}$ b) $\lim_{x \rightarrow 0} \frac{\cos x}{x + 1}$
- II) For the functions f and g , we know that $\lim_{x \rightarrow 1} f(x) = 3$ and $\lim_{x \rightarrow 1} g(x) = 5$. Determine the limits:
- a) $\lim_{x \rightarrow 1} [f(x) + g(x)]$ b) $\lim_{x \rightarrow 1} [f(x) \cdot g(x)]$ c) $\lim_{x \rightarrow 1} [f(x)]^2$

Exercise 5.2

Determine the limits:

- a) $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$ b) $\lim_{x \rightarrow \sqrt{3}} \frac{x^2 - 3}{x - \sqrt{3}}$ c) $\lim_{x \rightarrow 9} \frac{x - 9}{\sqrt{x} - 3}$
- d) $\lim_{x \rightarrow 1} \frac{x^2 - x}{x^2 - 3x + 2}$ e) $\lim_{x \rightarrow 1} \frac{x^2 - 2x + 1}{x^2 - 1}$ f) $\lim_{x \rightarrow \infty} \frac{x^2 - 2x + 1}{x^2 - 1}$
- g) $\lim_{x \rightarrow \infty} \frac{x^2 - 3x + 1}{x^2 + 1}$ h) $\lim_{x \rightarrow \infty} \frac{7x + 1}{x^2 + 1}$ i) $\lim_{x \rightarrow \infty} \frac{x^3 + 5x + 1}{x^2 + 1}$
- j) $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$ k) $\lim_{x \rightarrow 0} \frac{\tan 3x}{\sin 2x}$ l) $\lim_{x \rightarrow 1} \frac{x - \sqrt{x}}{\sqrt{x} - 1}$

Exercise 5.3

The functions f , g and h are given by:

$$f(x) = 5x + 7 \qquad g(x) = 2x^3 + x \qquad h(x) = \frac{2x}{x - 2}$$

- a) Show that the functions f , g and h are continuous at $x = 3$.
- b) Show that the functions f and g are continuous at all points $x = a$.
- c) Which conditions must h satisfy to be continuous at $x = a$?

Exercise 5.4

A function is defined by

$$f(x) = \begin{cases} \frac{\sin 2x}{x} & \text{when } x \neq 0, \\ 2 & \text{when } x = 0 \end{cases}$$

Determine $\lim_{x \rightarrow 0} f(x)$. Is f continuous at $x = 0$?

Exercise 5.5

Find the vertical asymptotes of the following functions.

- a) $f(x) = \frac{x + 1}{x^2 - 25}$ b) $f(x) = \frac{2x + 3}{x^2 - 5x + 4}$

Exercise 5.6

Find the horizontal asymptotes of f , when:

a) $f(x) = \frac{3x-10}{x}$ b) $f(t) = \frac{2t+9}{t^2-7}$

Exercise 5.7

For each of the following functions, find:

- 1) The roots
- 2) The vertical asymptotes
- 3) The horizontal asymptotes

a) $f(x) = \frac{3x-5}{5x-4}$ b) $f(x) = \frac{x^2-7x+6}{2(x^2-5x+6)}$

Exercise 5.8

Find the oblique asymptote of the functions:

a) $y = 2x - 1 - \frac{3}{x-8}$ b) $y = 3x - 2 + \frac{7}{8x+3}$

Exercise 5.9

Determine the smallest term in the following expression as $n \rightarrow \infty$: $f(n) = 3^n + (0.99)^n + (1.05)^{-n}$

Exercise 5.10

Determine the limits:

a) $\lim_{x \rightarrow \infty} \left(\frac{3}{4}\right)^x + \left(\frac{4}{3}\right)^x$ b) $\lim_{x \rightarrow \infty} \frac{3^x-5}{4^x+7}$ c) $\lim_{x \rightarrow \infty} \frac{8 \cdot 3^x + 9}{5 \cdot 3^x - 7}$

d) $\lim_{x \rightarrow \infty} \frac{5^x + 3^x}{7^x + 2^x}$ e) $\lim_{x \rightarrow \infty} \frac{2 \cdot 5^x + 3^x}{7^x + 2^x}$ f) $\lim_{n \rightarrow \infty} \left(\frac{1}{2}\right)^n + \left(\frac{2}{3}\right)^{2n} + 5^{-n}$

Exercise 5.11

A function is defined by:

$$f(x) = \begin{cases} \frac{x^2-9}{x-3} & \text{when } x \neq 3 \\ a & \text{when } x = 3 \end{cases}$$

Determine $\lim_{x \rightarrow 3} f(x)$. Determine a such that f is continuous at $x = 3$

Exercise 5.12 ▲

A function is defined by: $f(x) = \begin{cases} \frac{x^2-3}{|x-\sqrt{3}|} & \text{when } x \neq \sqrt{3} \\ 2\sqrt{3} & \text{when } x = \sqrt{3} \end{cases}$

Determine $\lim_{x \rightarrow \sqrt{3}} f(x)$. Is f continuous at $x = \sqrt{3}$?

Solution to the Exercises Chapter 5

5.1 I) a) 2 b) 1 II) a) 8 b) 15 c) 9

5.2 a) 6 b) $2\sqrt{3}$ c) 6
 d) -1 e) 0 f) 1
 g) 1 h) 0 i) Does not exist
 j) 3 k) $3/2$ l) 1

5.3 c) $a \neq 2$

5.4 Yes, 2.

5.5 a) $x = 5, x = -5$ b) $x = 1, x = 4$

5.6 a) $y = 3$ b) $y = 0$

5.7 a) 1. $x = \frac{5}{3}$ 2. $x = \frac{4}{5}$ 3. $y = \frac{3}{5}$
 b) 1. $x = 1, x = 6$ 2. $x = 2, x = 3$ 3. $y = \frac{1}{2}$

5.8 a) $y = 2x - 1$ b) $y = 3x - 2$

5.9 $(1.05)^{-n}$ is the smallest term. The limit does not exist.

5.10 a) Does not exist b) 0 c) $\frac{8}{5}$
 d) 0 e) 0 f) 0

5.11 6, $a = 6$

5.12 The limit does not exist ($\pm 2\sqrt{3}$):

$$\begin{aligned} \lim_{x \rightarrow \sqrt{3}^+} \frac{x^2 - 3}{(x - \sqrt{3})} &= \frac{0}{0} = \lim_{x \rightarrow \sqrt{3}} \frac{(x - \sqrt{3})(x + \sqrt{3})}{(x - \sqrt{3})} = \lim_{x \rightarrow \sqrt{3}^+} (x + \sqrt{3}) = 2\sqrt{3} \\ \lim_{x \rightarrow \sqrt{3}^-} \frac{x^2 - 3}{-(x - \sqrt{3})} &= \frac{0}{0} = \lim_{x \rightarrow \sqrt{3}} \frac{(x - \sqrt{3})(x + \sqrt{3})}{-(x - \sqrt{3})} = \lim_{x \rightarrow \sqrt{3}^-} -(x + \sqrt{3}) = -2\sqrt{3} \end{aligned}$$

Chapter 6

Derivation and differentiation

Derivatives and differentiation are useful tools to study how a function changes. To study change, we need to study *differences* (hence the name differentiation) of function values with respect to changes to the variable. The term *derivative* was most likely introduced by Lagrange, and related to the fact that we construct a new function $f'(x)$ that is implied (“derive=obtain”) from $f(x)$.

6.1 Incremental rate of a function

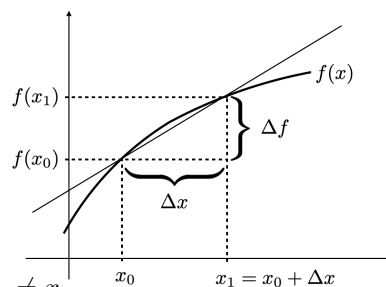
Given a function $y = f(x)$:

- *function increment*: is the change in f . It is denoted with Δy or Δf (Δ = greek D, initial for “difference”)

$$\Delta y = \Delta f = f(x_1) - f(x_0), \quad x_1 \neq x_0$$

- *incremental rate*: is the ratio of the change in f (function increment) with respect to the change in the independent variable x (independent variable increment):

$$\frac{\Delta y}{\Delta x} = \frac{\Delta f}{\Delta x} = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}, \quad \Delta x = x_1 - x_0, \quad x_1 \neq x_0.$$



The incremental rate is the slope of the straight line passing through $(x_0, f(x_0))$ and $(x_1, f(x_1))$.

6.2 Derivative of a function

The *derivative* of a function f at x is the instantaneous (=limit for $\Delta x \rightarrow 0$) incremental rate at a generic point x :

$$y' = f'(x) = \frac{dy}{dx} = \frac{df}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Example 6.1

Find the derivative of $y = x^2$ using the definition of derivative.

Solution:

$$\begin{aligned}
 y' &= \lim_{\Delta x \rightarrow 0} \frac{(x + \Delta x)^2 - x^2}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{x^2 + 2x\Delta x + (\Delta x)^2 - x^2}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\Delta x(2x + \Delta x)}{\Delta x} \\
 &= \lim_{\Delta x \rightarrow 0} (2x + \Delta x) = \underline{\underline{2x}}
 \end{aligned}$$

◇

6.3 Interpretation of the derivative

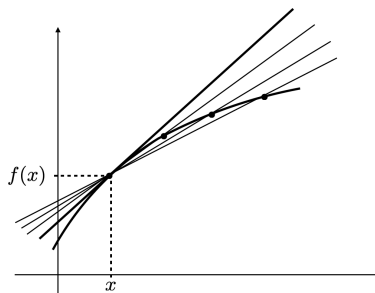
Derivative vs. incremental rate:

- The incremental rate

$$\frac{\Delta y}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

is the slope of the line passing through the points $(x, f(x))$ and $(x + \Delta x, f(x + \Delta x))$, also called *secant*, for a finite Δx .

If $f(x)$ is the curve of position of an object, then $\frac{\Delta f}{\Delta x}$ can be interpreted as *averaged velocity* of the object.



- The derivative

$$y' = f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

is the slope of the *tangent* to f at the point $(x, f(x))$.

If $f(x)$ is the curve of position of an object, then $f'(x)$ can be interpreted as the *instantaneous velocity* of the object.

6.4 Derivation of fundamental functions

	$y = f(x)$	$y' = f'(x)$	Some special cases
Constant	k	0	
Power function $y = x^r$	x^r	rx^{r-1}	$r = -1 \Rightarrow \left(\frac{1}{x}\right)' = -\frac{1}{x^2}$ $r = \frac{1}{2} \Rightarrow (\sqrt{x})' = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$
Exponential function in base e	e^x	e^x	$(e^{kx})' = ke^{kx}$
Exponential function in base a	a^x	$a^x \ln a$	
Logarithm	$\ln x$	$\frac{1}{x}$	$(\log x)' = \frac{1}{x \cdot \ln 10}$
Absolute value, $x \neq 0$	$ x $	$\frac{ x }{x}$	
Sine	$\sin x$	$\cos x$	
Cosine	$\cos x$	$-\sin x$	
Tangent	$\tan x$	$\frac{1}{\cos^2 x}$	

6.5 Important derivation rules

$$(ay)' = ay'$$

Basic rules (a, b , constants, y, u, v functions): $(u \pm v)' = u' \pm v'$

$$(au + bv)' = au' + bv'$$

Important rules:

Product rule	$(u \cdot v)' = u' \cdot v + u \cdot v'$
Quotient rule	$\left(\frac{u}{v}\right)' = \frac{u' \cdot v - u \cdot v'}{v^2}$
Chain rule	$y = y(u(x)) \Rightarrow \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

Example 6.2

Product rule: $(u \cdot v)' = u' \cdot v + u \cdot v'$

$$y = x^3 \cos x \Rightarrow y' = 3x^2 \cos x + x^3(-\sin x) = \underline{\underline{x^2(3 \cos x - x \sin x)}} \quad \diamond$$

Example 6.3

Quotient rule: $\left(\frac{u}{v}\right)' = \frac{u' \cdot v - u \cdot v'}{v^2}$

$$y = \frac{\cos x}{x} \Rightarrow y' = \frac{(-\sin x)x - 1(\cos x)}{x^2} = \underline{\underline{-\frac{x \sin x + \cos x}{x^2}}} \quad \diamond$$

Example 6.4

Chain rule: $y = y(u(x)) \Rightarrow \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

$$y = \ln(\cos x) \Rightarrow y = \ln u, u = \cos x \Rightarrow y' = \frac{1}{u} \cdot u' = \frac{1}{\cos x}(-\sin x) = \underline{\underline{-\tan x}} \quad \diamond$$

Example 6.5

Compute the derivatives $\frac{d}{dx}(\sin 2x)$ and $\frac{d}{dt}(k \cos \omega t)$

Solution:

$$(\sin 2x)' = \underline{\underline{2 \cos 2x}} \quad (k \cos \omega t)' = \underline{\underline{-k\omega \sin \omega t}}$$

$\diamond$

6.6 The derivative of composite functions of e^x and x^r

Plain case	Composite case (with chain rule) and auxiliary function $u = u(x)$	Example
$(e^x)' = e^x$	$(e^u)' = e^u \cdot u'$	$(e^{3x})' = 3e^{3x}$ (with $u = 3x$)
$(x^r)' = r \cdot x^{r-1}$	$(u^r)' = r \cdot u^{r-1} \cdot u'$	$((x^2+1)^3)' = 3 \cdot (x^2+1)^2 \cdot 2x = 6x(x^2+1)^2$ (with $u = x^2+1$)

The derivative of a^x : $(a^x)' = \ln a \cdot a^x$

We first rewrite the expression in base e : $a^x = e^{\ln a \cdot x}$

Use the rule for the composite exponential (see above): $(a^x)' = (e^{\ln a \cdot x})' = \ln a \cdot e^{\ln a \cdot x} = \ln a \cdot a^x$.

Example 6.6

Differentiate (= calculate the derivative) the following functions:

a) $y = (\cos x)^7$ **b)** $y = (2 + x^5)^9$ **c)** $y = 3x^{7x}$ **d)** $y = \pi^x$ **e)** $y = k \cdot 5^x$

Solution:

a) $y' = -7 \cos^6 x \sin x$ **b)** $y' = 45x^4(2 + x^5)^8$ **c)** $y' = 21e^{7x}$ **d)** $y' = \pi^x \ln \pi$

e) $y' = k \cdot 5^x \ln 5$

◇

6.7 The Leibniz notation $\frac{d}{dx}$

The notation $\frac{df}{dx}$ (also called *Leibniz's notation*) should be intended as the the differentiation of f with respect to x :

$$y' = f'(x) = \frac{df}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta f}{\Delta x}$$

The chain rule in Leibniz's notation can be formulated as follows:

$$y = y(u(x)) \Rightarrow \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

While Leibniz's notation is less comfortable than f' or y' (also called *Lagrange notation*) for derivatives, it becomes very useful when dealing with integral calculus, multivariate calculus as well as differential equations.

Example 6.7

$$\frac{d}{dr} \left(\frac{4}{3} \pi r^3 \right) = 4\pi r^2$$

◇

Example 6.8

Compute $\frac{d}{dt}(\frac{4}{3}\pi r^3)$ when the radius r changes with time.

Solution:

We have a composite function. Applying the chain rule: $\frac{d}{dt}(\frac{4}{3}\pi r^3) = 4\pi r^2 \frac{dr}{dt}$

◇

The example above can also be formulated in the following way:

Find the rate of change of the volume $V = \frac{4}{3}\pi r^3$ of a sphere, when the radius r changes with time.

The volume V is a composite function: $V = V(r(t))$ and $\frac{dV}{dt} = \frac{dV}{dr} \frac{dr}{dt} = 4\pi r^2 \frac{dr}{dt}$ means that the volume change is proportional to the surface $S = 4\pi r^2$ of the sphere.

6.8 Overview of the main derivation formulae and rules

$y = f(x)$	$y' = f'(x) = \frac{dy}{dx}$	Chain rule (for composite functions)	Example
k	0		
x^n	nx^{n-1}	$(u^n)' = n \cdot u^{n-1} \cdot u'$	$(\sin^5 x)' = 5 \sin^4 x \cos x$
$ x $	$\frac{ x }{x}, x \neq 0$		
e^x	e^x	$(e^u)' = e^u \cdot u'$	$(e^{kx})' = ke^{kx}$
a^x	$a^x \ln a$	$(a^u)' = a^u \cdot \ln a \cdot u'$	
$\ln x$	$\frac{1}{x}$	$(\ln u)' = \frac{1}{u} \cdot u'$	$(\ln(ax + b))' = \frac{a}{ax + b}$
$\sin x$	$\cos x$	$(\sin u)' = \cos u \cdot u'$	$(\sin(ax + b))' = a \cos(ax + b)$
$\cos x$	$-\sin x$	$(\cos u)' = -\sin u \cdot u'$	$(\cos(ax + b))' = -a \sin(ax + b)$
$\tan x$	$\frac{1}{\cos^2 x} = 1 + \tan^2 x$	$(\tan u)' = \frac{1}{\cos^2 u} \cdot u' = (1 + \tan^2 u) \cdot u'$	$(\tan(ax + b))' = \frac{a}{\cos^2(ax + b)}$

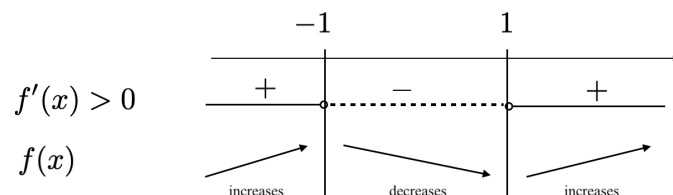
Example 6.9

Given the function $f(x) = x^3 - 3x$:

- Compute $f'(x)$ and find where the function increases and decreases
- Compute $f''(x)$ and find where the function is concave and convex
- Draw the graph of the function

Solution:

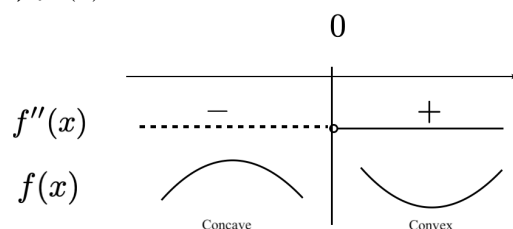
a) $f'(x) = 3x^2 - 3 = 3(x^2 - 1)$



$x = -1$ is a maximum, with value $f(-1) = 2$

$x = 1$ is a minimum, with value $f(1) = -2$

b) $f''(x) = 6x$



$x = 0$ is a flex (or inflection point)

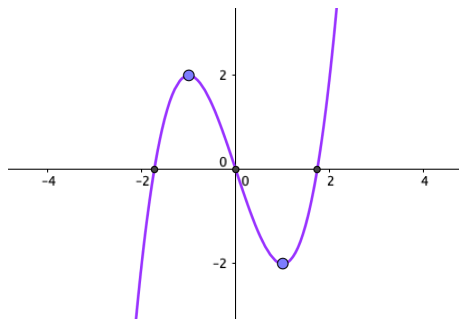
c) To draw the graph we look at the intersection with the axes:

x -axis: $f(x) = 0 \Leftrightarrow x(x^2 - 3) = 0 \Rightarrow x = 0, x = \pm\sqrt{3}$

y -axis: $x = 0 \Rightarrow f(x) = 0$ the function intersects the y axis at the origin.

The function does not have asymptotes

graph:



◇

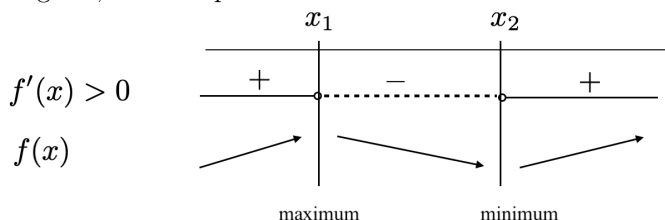
6.9 Derivatives and monotonicity properties

Let $y = f(x)$ defined in an interval I .

- $f'(x) \geq 0$ in the interval $I \Rightarrow f(x)$ is *increasing* in I
- $f'(x) > 0$ in the interval $I \Rightarrow f(x)$ is *strictly increasing* in I
- $f'(x) \leq 0$ in the interval $I \Rightarrow f(x)$ is *decreasing* in I

- $f'(x) < 0$ in the interval $I \Rightarrow f(x)$ is *strictly decreasing* in I

To find the intervals in which a function is increasing or decreasing, it is useful to use a sign diagram, for example:



The points where $f'(x) = 0$ in the *interior* of the interval I are local extremal points (local max/local min). If the interval is *closed*, a global max or global min can occur at the endpoints of the interval.

Max/min can also occur in points of the domain where the derivative does not exist: In other words, all the critical points of f must be checked for the occurrence of max/min.

$$\text{critical points of } f : \{x \in I \mid f'(x) = 0 \text{ or } f'(x) \text{ does not exist}\}$$

6.10 Max and min of a function, extremal points, studying a function

The collection of local minima and local maxima of f are the *local extremal points*.

In addition to local extremal points, there might be global extremal points, like global maxima: $f(x_0) \geq f(x)$ for all $x \in D_f$, or global minima, that is $f(x_0) \leq f(x)$ for all $x \in D_f$. The collection of local and global extremal points is simply called the *extremal points* of f . A point in this set is an *extremum*.

The points in which $f'(x) = 0$ are called *stationary points*. The collection of points in which $f'(x) = 0$ or $f'(x)$ does not exist, constitutes the set of *critical points* of f . Consequently, stationary points are a subset of critical point.

Necessary condition for an extremum

x_0 is extremum of $f(x) \Rightarrow f'(x_0) = 0$ or $f'(x_0)$ does not exist.

In other words, the extrema of a function are contained among its critical points.

Existence of maximum and minimum

A continuous function f defined in a closed interval $I = [a, b]$, has both maximum and minimum in I .

Characterization of extremal points

Assume f is a continuous function defined in a closed interval $I = [a, b]$.

1. Compute the stationary points of f ($f'(x) = 0$) and their function values
2. Compute $f(a)$ and $f(b)$
3. Find the points in which f' does not exist. (For instance $f(x) = |x|$ has no derivative for $x = 0$, thus $x = 0$ is a critical point)
Compute the function values in these points.

4. Compare these values to find eventual local/global maximum and minimum points.

Example f defined in $I = [a, b]$

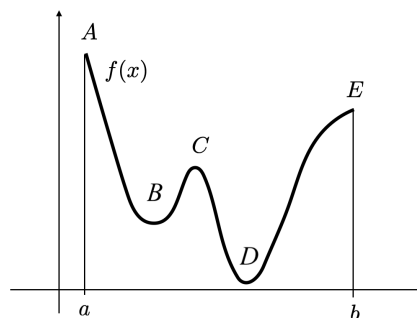
A global max $(f' \text{ does not exist})$

B local min $(f' = 0)$

C local max $(f' = 0)$

D local min $(f' = 0)$

E neither max nor min $(f' \text{ does not exist})$



Note: If $I = (a, b)$, then f is not defined in a or b . The point A is then an *supremum* (sup), but not a maximum. (A, E are not in the range of f).

By a similar token, a straight line with slope 1 in (a, b) doesn't have either maximum or minimum, but has a sup and inf (infimum).

Example 6.10

Consider the function $f(x) = x^3 - 3x$, for $-2.5 \leq x \leq 3$. Find eventual extremal points.

Solution:

We have $I = [-2.5, 3]$ is a closed interval: f has max and min in I . In example 6.9 we found that:

The stationary points of f ($f' = 0$) are $x = \pm 1 \Rightarrow x = \pm 1$ local min or local max

The other critical points are $a = -2.5$ and $b = 3$ (f has no derivative at these points)

Function values:

$$f(a) = -8.125 \quad f(-1) = 2 \quad f(1) = -2 \quad f(b) = 18$$

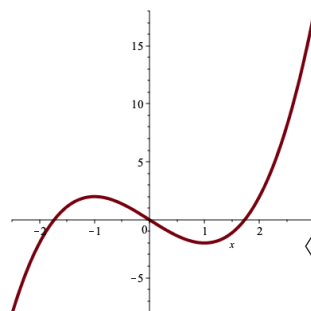
Extrema:

$x = a = -2.5$ global minimum

$x = -1$ local maximum

$x = 1$ local minimum

$x = b = 3$ global maximum



Methods for studying and drawing a function

1. Find eventual intersections with x - and y -axis
2. Find eventual asymptotes
3. Compute $f'(x)$. Find critical points and stationary points ($f' = 0$). Set up a sign diagram to study monotonicity intervals and extremal points.
4. Compute $f''(x)$ and find $f'' = 0$. Set up a sign diagram to study convexity, concavity and eventual inflection points.

The second derivative of f : concavity and convexity

If $f'(x)$ itself has a derivative, this is called *second derivative* of f and is denoted as $f''(x)$. In Leibnitz's notation:

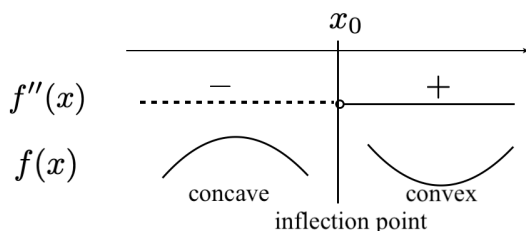
$$f''(x) = \frac{d^2 f}{dx^2} = \frac{d}{dx} \left(\frac{df}{dx} \right)$$

Assume $f''(x)$ exists in an interval I .

- $f''(x) \geq 0$ in the interval $I \Rightarrow f(x)$ is *convex* in I
- $f''(x) \leq 0$ in the interval $I \Rightarrow f(x)$ is *concave* in I
- Points in which $f''(x) = 0$ and f'' changes of sign are called *inflections points*.

Inflection points are points in which f goes from being concave to being convex or viceversa.

To find the intervals in which a function is concave or convex, it is useful to use a sign diagram, for example:



Example 6.11

Show that $f(x) = x^3$ has an inflection point.

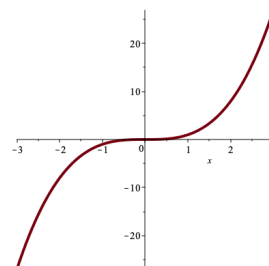
Solution:

$f'(x) = 3x^2 \geq 0$, hence f is increasing.

$x = 0$ is a stationary point

$f''(x) = 6x$: $6x > 0$ for $x > 0$ (function is convex) and $6x < 0$ for $x < 0$ (function is concave). At $x = 0$ $f''(x)$ changes sign, therefore $x = 0$ is an inflection point.

◇



6.11 The equation of the tangent and linearization

The equation of the *tangent* to a differentiable function $y = f(x)$ at a point a is

$$y = f(a) + f'(a)(x - a)$$

To *linearize* a differentiable function f at a point $x = a$ is equivalent to approximate $f(x)$ with the tangent at this point:

$$f(x) \approx f(a) + f'(a)(x - a)$$

In other words, in a small neighborhood of the point $(a, f(a))$, the function f looks almost like a straight line (the tangent).

Example 6.12

Find the equation of the tangent line of f at the given points and linearize.

a) $f(x) = \sin x, \quad x = 0$ **b)** $f(x) = e^x, \quad x = 0$

Solution:

a) $f'(x) = \cos x \Rightarrow f'(0) = 1 \Rightarrow y = x$

a) $f'(x) = e^x \Rightarrow f'(0) = 1 \Rightarrow y = 1 + x = x + 1$

◇

Increment of a function

Consider the linearized form of $y = f(x)$. In a nearby point of x , $x + \Delta x$, the linearized function gives $f(x + \Delta x) \approx f(x) + f'(x)(x + \Delta x - x)$, which is equivalent to:

$$\Delta f = f(x + \Delta x) - f(x) \approx f'(x) \cdot \Delta x$$

Namely, in a neighborhood of x , the increment Δf of f is proportional to the increment Δx of x with factor $f'(x)$.

Example 6.13

Estimate how much the volume of a sphere grows if the radius is increased from 10 to 10.1 cm and from 10 to 10.01 cm. The volume of a sphere of radius r is given as $V = \frac{4}{3}\pi r^3$

Solution:

We apply the formula for the increment: $\Delta V = V' \cdot \Delta r$.

We have already computed $V' = 4\pi r^2$ in a previous exercise

$\Delta r = 10.1 - 10 = 0.1$: $\Delta V = 4\pi r^2 \cdot \Delta r = 4\pi \cdot 10^2 \cdot 0.1 = 4\pi \cdot 10 \approx \underline{\underline{125.66\text{cm}^3}}$

$\Delta r = 10.01 - 10 = 0.01$: $\Delta V = 4\pi \cdot 10^2 \cdot 0.01 = 4\pi \approx \underline{\underline{12.57\text{cm}^3}}$

◇

In the previous example, we could have computed the *exact* value of the increments:

$$\Delta V = V(10 + 0.1) - V(10) = \frac{4}{3}\pi(10.1^3 - 10^3) = 126.92$$

$$\Delta V = V(10 + 0.01) - V(10) = \frac{4}{3}\pi(10.01^3 - 10^3) = 12.58$$

The example shows that the smaller Δr , the more precise is the approximation.

Exercises Chapter 6

Exercise 6.1

Differentiate the functions.

a) $f(x) = x^2 - 2x + 5$ b) $g(x) = 3x^2 + 5x - 2$

Exercise 6.2

Differentiate the functions.

a) $f(x) = x^{1.2}$ b) $f(x) = 2x + 3x^{2.5}$

Exercise 6.3

Differentiate the expressions with respect to x .

a) $x + \frac{1}{x}$ b) $x^2 + \frac{1}{x^2}$ c) $x + \sqrt{x}$

Exercise 6.4

Differentiate the functions.

a) $f(x) = (2x - 3)^3$ b) $f(x) = (x^2 + 1)^2$ c) $f(x) = (x^3 + 5x)^2$

Exercise 6.5

Differentiate the functions using the product rule.

a) $f(x) = x^2(x^4 + x)$ b) $f(x) = (x^2 + 1)\cos x$

Exercise 6.6

Determine $y' = \frac{dy}{dx}$ when

a) $y = x^5 + 2x + 7$	b) $y = 5x + 2\cos x + 1$	c) $y = \frac{\sin x}{x}$
d) $y = (x^3 + 1)\sin x$	e) $y = \frac{\cos x}{x^3}$	f) $y = \frac{x+1}{3x+1}$
g) $y = \sqrt{x} + \frac{1}{\sqrt[3]{x}} + \frac{1}{x}$	h) $y = \frac{1}{x^2+1}$	i) $y = (2x + 5)^{100}$
j) $y = \sqrt{x^3 - 2x}$	k) $y = \frac{x^2-3}{x^2+1}$	l) $y = (x^3 + 5x)^{\frac{1}{4}}$
m) $y = \sqrt[4]{x^3 + 5x}$	n) $y = x^2 \ln x$	o) $y = \frac{\cos 2x}{x}$

Exercise 6.7

Determine $y' = \frac{dy}{dx}$ when **a)** $y = \frac{\sin x}{\sqrt{x}}$ **b)** $y = \sqrt{x} \cos x$

Exercise 6.8

Differentiate the functions:

a) $f(x) = 3x^5 - 4x + 5$ **b)** $f(x) = \sqrt{x} + \frac{1}{x} + \frac{1}{\sqrt{x}}$ **c)** $y = (x^2 - x + 1)^{15}$

Exercise 6.9

- a)** Determine the instantaneous incremental rate for the function: $f(x) = x^5 - 4x^2 + 3$ at a point with $x = -1$.
b) Determine the tangent line for the function: $g(x) = x^3 - 2x + 3$ at a point with $x = -1$, which lies on the curve of the function.

Exercise 6.10

- a)** Determine the extremal points of the function $y = x^3 - 12x + 12$. Determine the inflection point.
b) Determine the extremal points of the function $y = x^3 - 6x^2 + 4$. Determine the inflection point.

Exercise 6.11 ▲

Determine y' when:

a) $y = \frac{x^2 + 3x - 2}{\sqrt[3]{x}}$ **b)** $y = \frac{1}{\sqrt[3]{x}} + \frac{1}{\sqrt[3]{x}} + x^3$ **c)** $y = \sqrt[3]{x^3 - 5x - 3}$
d) Use the formula $\lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$ to determine the derivative of $f(x) = \frac{1}{x+2}$

Exercise 6.12

Determine the limits:

a) $\lim_{x \rightarrow 4} \frac{x^3 - 64}{x - 4}$ **b)** $\lim_{x \rightarrow -3} \frac{x^3 + 2x^2 - 3x}{x + 3}$ **c)** $\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9}$ **d)** $\lim_{x \rightarrow 9} \frac{x - 3}{|x - 3|}$

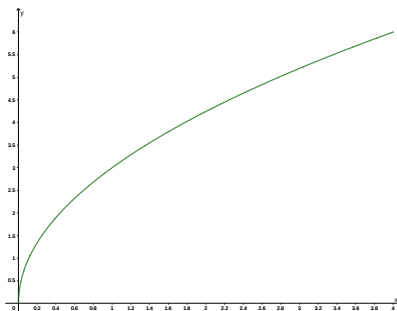
Exercise 6.13

- a)** Given the function $f(x) = 2xe^{-\frac{x^2}{2}}$. Determine $\lim_{x \rightarrow +\infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$
b) Show that $f'(x) = 2(1 - x^2)e^{-\frac{x^2}{2}}$ and $f''(x) = 2x(x^2 - 3)e^{-\frac{x^2}{2}}$
c) Determine the roots, extremal points and inflection points.

Exercise 6.14

We are given a curve $f(x) = 3\sqrt{x}$, $0 \leq x \leq 4$. The point $P_1(a, f(a))$ lies on the curve.

- 1) Find the equation for the tangent line at the point P_1 .



- 2) This tangent intersects the x -axis at the point P_2 . Determine the coordinates of P_2 .

Exercise 6.15

A function $y = \frac{x^2 - x + 1}{x}$, $x \in (0, 10]$ is given.

- Determine the extremal points of the function.
- Determine the point on the curve where the distance between the curve and the point $(0, -1)$ is the smallest.

Exercise 6.16

A snow ball with the shape of a sphere is melting with a constant speed $3 \frac{\text{mm}^3}{\text{min}}$, and the spherical form is kept throughout the melting. How fast is the radius of the snow ball decreasing at the moment when the radius is 6 mm?

How fast is the surface area of the snow ball decreasing at the same moment?

Exercise 6.17

The equation of a function is given by: $f(x) = x^4 - 4x^3 + 10$; $x \in [-5, 5]$

Determine:

- Where the function is increasing and decreasing
- Extremal values
- Inflection points and the concavity of the function
- Sketch the curve.

Exercise 6.18

The equation of a function is given by: $f(x) = \frac{x^2 + 1}{x^2 - 4}$

Determine:

- Extremal values
- Where the function is increasing and decreasing
- Inflection points and the concavity of the function
- Sketch the graph.

Solution to the Exercises Chapter 6

6.1 a) $2x - 2$ b) $6x + 5$

6.2 a) $1.2x^{0.2}$ b) $2 + 7.5x^{1.5}$

6.3 a) $1 - \frac{1}{x^2}$ b) $2x - \frac{2}{x^3}$ c) $1 + \frac{1}{2\sqrt{x}}$

6.4 a) $6(2x - 3)^2$ b) $4x(x^2 + 1)$ c) $2(x^3 + 5x)(3x^2 + 5)$

6.5 a) $6x^5 + 3x^2$ b) $f(x) = 2x \cos x - (x^2 + 1) \sin x$

6.6 a) $5x^4 + 2$ b) $5 - 2 \sin x$ c) $\frac{x \cos x - \sin x}{x^2}$

d) $3x^2 \sin x + (x^3 + 1) \cos x$ e) $-\frac{x \sin x + 3 \cos x}{x^4}$ f) $-\frac{2}{(3x + 1)^2}$

g) $\frac{1}{2\sqrt{x}} - \frac{1}{3x\sqrt[3]{x}} - \frac{1}{x^2}$ h) $-\frac{2x}{(x^2 + 1)^2}$ i) $200(2x + 5)^{99}$

j) $\frac{3x^2 - 2}{2\sqrt{x^3 - 2x}}$ k) $\frac{8x}{(x^2 + 1)^2}$ l) $\frac{3x^2 + 5}{4\sqrt[4]{(x^3 + 5x)^3}}$

m) $\frac{3x^2 + 5}{4\sqrt[4]{(x^3 + 5x)^3}}$ n) $x(2 \ln x + 1)$ o) $-\frac{2x \sin 2x + \cos 2x}{x^2}$

6.7 a) $y = \frac{\cos x \cdot \sqrt{x} - \sin x \cdot \frac{1}{2\sqrt{x}}}{x} = \frac{2x \cos x - \sin x}{2x\sqrt{x}}$

b) $y = \frac{1}{2\sqrt{x}} \cos x - \sqrt{x} \sin x = \frac{\cos x - 2x \sin x}{2\sqrt{x}}$

6.8 a) $f'(x) = 15x^4 - 4$ b) $f(x) = \frac{1}{2\sqrt{x}} - \frac{1}{x^2} - \frac{1}{2x\sqrt{x}}$

c) $y = 15(x^3 - x + 1)^{14}(3x^2 - 1)$

6.9 a) $f'(x) = 5x^4 - 8x$, and therefore the instantaneous increment at $x = -1$ is: $f'(-1) = 13$.

b) The equation for the tangent in $x = a$: $f(x) - f(a) = f'(a)(x - a)$

$y - f(-1) = f'(-1)(x + 1) \Rightarrow y - 4 = x + 1 \Rightarrow y = x + 5$

$f'(x) = 3x^2 - 2$, $f'(-1) = 1$

6.10 a) Extremal points: $(-2, 28)$, $(2, -4)$ and inflection point: $(0, 12)$

b) Extremal points: $(0, 4)$, $(4, -28)$ and inflection point: $(2, -12)$

6.11 a) $y = \frac{x^2 + 3x - 2}{\sqrt[3]{x}} = x^{2-\frac{1}{3}} + 3x^{1-\frac{1}{3}} - 2x^{-\frac{1}{3}} = x^{\frac{5}{3}} + 2x^{\frac{2}{3}} - 2x^{-\frac{1}{3}}$

$y' = \frac{5}{3}x^{\frac{2}{3}} + 2x^{-\frac{1}{3}} + \frac{2}{3}x^{-\frac{4}{3}}$

b) $y = \frac{1}{\sqrt[5]{x}} + \frac{1}{\sqrt[3]{x}} + x^3 = x^{-\frac{1}{5}} + x^{-\frac{1}{3}} + x^3 \Rightarrow y' = -\frac{1}{5}x^{-\frac{6}{5}} - \frac{1}{3}x^{-\frac{4}{3}} + 3x^2$

c) $y = \sqrt[3]{x^3 - 5x - 3} = (x^3 - 5x - 3)^{\frac{1}{3}} \Rightarrow y' = \frac{1}{3}(x^3 - 5x - 3)^{-\frac{2}{3}}(3x^2 - 5) = \frac{(3x^2 - 5)}{3(x^3 - 5x - 3)^{\frac{2}{3}}}$

d) $f(x) = \frac{1}{x+1}$ $f(x + \Delta x) = \frac{1}{x + \Delta x + 2}$ $f(x + \Delta x) - f(x) = \frac{1}{x + \Delta x + 2} - \frac{1}{x + 2}$

$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{x + 2 - x - \Delta x - 2}{\Delta x(x + \Delta x + 2)(x + 2)} = \lim_{\Delta x \rightarrow 0} \frac{-1}{(x + \Delta x + 2)(x + 2)} =$

6.12 Note that you can also use L'Hospital's rule.

$$\text{a) } \lim_{x \rightarrow 4} \frac{x^3 - 64}{x - 4} = \frac{0}{0} = \lim_{x \rightarrow 4} \frac{(x - 4)(x^2 + 4x + 16)}{x - 4} = \lim_{x \rightarrow 4} (x^2 + 4x + 16) = 16 + 16 + 16 = 48$$

$$\text{b) } \lim_{x \rightarrow -3} \frac{x^3 + 2x^2 - 3x}{x + 3} = \frac{0}{0} = \lim_{x \rightarrow -3} \frac{x(x^2 + 2x - 3)}{x + 3} = \lim_{x \rightarrow -3} \frac{x(x + 3)(x - 1)}{x + 3} \\ = \lim_{x \rightarrow -3} x(x - 1) = 12$$

$$\text{c) } \lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9} = \frac{0}{0} = \lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{(\sqrt{x} - 3)(\sqrt{x} + 3)} = \lim_{x \rightarrow 9} \frac{1}{\sqrt{x} + 3} = \frac{1}{6}$$

$$\text{d) } \lim_{x \rightarrow 3} \frac{x - 3}{|x - 3|} = \begin{cases} \lim_{x \rightarrow 3^+} \frac{x - 3}{(x - 3)} = 1 \\ \lim_{x \rightarrow 3^-} \frac{x - 3}{-(x - 3)} = -1 \end{cases} \quad \text{Thus, the limit does not exist.}$$

6.13 a) 0, 0 **b)** Root: (0, 0). Extremal points: $(1, \frac{2}{\sqrt{e}})$ and $(-1, -\frac{2}{\sqrt{e}})$. Inflection points: $(0, 0)$, $(\sqrt{3}, \frac{2\sqrt{3}}{e\sqrt{e}})$ and $(-\sqrt{3}, -\frac{2\sqrt{3}}{e\sqrt{e}})$

$$\text{6.14 1) } f(x) = 3\sqrt{x} \quad f'(x) = \frac{3}{2\sqrt{x}} \\ y - 3\sqrt{a} = \frac{3}{2\sqrt{a}}(x - a) \Rightarrow y = \frac{3}{2\sqrt{a}}x + \frac{3}{2}\sqrt{a}. \\ \text{2) } y = 0 \Rightarrow \frac{3}{2\sqrt{ax}} + \frac{3}{2}\sqrt{a} = 0 \Rightarrow x = -a, (-a, 0)$$

6.15 a) $y = \frac{x^2 - x + 1}{x}$; $x \in \langle 0, 10 \rangle$; $y = x - 1 + \frac{1}{x} \Rightarrow \frac{dy}{dx} = 1 - 0 - \frac{1}{x^2} = 0$; this gives us $x = 1$. y has absolute minimum at $x = 1$. Control via sign chart.

$$\text{b) Distance } S = \sqrt{(a - 0)^2 + \left(\frac{a^2 - a + 1}{a} + 1\right)^2} = \sqrt{(a)^2 + \left(a - 1 + \frac{1}{a} + 1\right)^2} = \sqrt{(a)^2 + \left(a + \frac{1}{a}\right)^2} \\ \frac{dS}{da} = \frac{1}{2} \left(a^2 + \left(a + \frac{1}{a} \right)^2 \right)^{-\frac{1}{2}} \left(2a + 2 \left(a + \frac{1}{a} \right) \left(1 - \frac{1}{a^2} \right) \right) = 0 \\ \Rightarrow \frac{(2a + 2(a + \frac{1}{a})(1 - \frac{1}{a^2}))}{2 \cdot (a^2 + (a + \frac{1}{a})^2)^{\frac{1}{2}}} = 0$$

$$a + a - \frac{1}{a} + \frac{1}{a} - \frac{1}{a^3} = 0 \Rightarrow 2a - \frac{1}{a^3} = 0 \Rightarrow 2a^4 - 1 = 0 \Rightarrow a^4 = \frac{1}{2} \Rightarrow a = \left(\frac{1}{2}\right)^{\frac{1}{4}} = \sqrt[4]{\frac{1}{2}} = \frac{1}{2}\sqrt[4]{8} \\ \text{The distance is smallest at } a = \frac{1}{2}\sqrt[4]{8}$$

$$\text{6.16 } dr/dt = r'(t) = -1/48\pi \\ O(t) = 4\pi \cdot r^2 \quad \frac{dO}{dt} = 8\pi \cdot r \cdot \frac{dr}{dt} \\ \left[\frac{dO}{dt} \right]_{r=6\text{mm}} = -8\pi \cdot 6\text{mm} \cdot \frac{1}{4\pi} \cdot \frac{3}{36} \frac{\text{mm}}{\text{min}} = 1 \frac{\text{mm}^2}{\text{min}}$$

6.17 a) $f(x) = x^4 - 4x^3 + 10$; $x \in [-5, 5]$, $f'(x) = 4x^3 - 12x^2 + 0 = 0$, $4x^2(x - 3) = 0$
 $x = 0$, $x = 3$, by drawing a sign chart, we find that f increases for $x \in \langle 3, 5 \rangle$ and decreases for $x \in \langle -5, 3 \rangle$ and has a local minimum at $x = 3$.
 $f''(x) = 12x^2 - 24x = 0 \Rightarrow 12x(x - 2) = 0 \Rightarrow x = 0$, $x = 2$. By drawing a sign chart, we find that f is convex for $x \in \langle -5, 0 \rangle$ and concave for $x \in \langle 0, 2 \rangle$ and has inflection points in $(0, 10)$ and $(2, -6)$

$$\text{6.18 } f(x) = \frac{x^2 + 1}{x^2 - 4} \quad f'(x) = \frac{(x^2 - 4)2x - (x^2 + 1)2x}{(x^2 - 4)^2} = \frac{-10x}{(x^2 - 4)^2} = 0, -10x = 0, x = 0 \\ f \text{ increases when } x \in \langle -\infty, -2 \rangle \text{ and when } x \in \langle -2, 0 \rangle, \text{ decreases when } x \in \langle 0, 2 \rangle \text{ and when } x \in \langle 2, \infty \rangle, \text{ has a local maximum at } x = 0. \\ f''(x) = \frac{(x^2 - 4)^2(-10) + 10x(2(x^2 - 4)(2x))}{(x^2 - 4)^4} = \frac{10x^2 + 4}{(x^2 - 4)^3} = 0, 10x^2 + 4 = 0. \text{ We obtain}$$

no real values, and hence there are no inflection points. The function is convex when $x \in \langle -\infty, -2 \rangle$ and when $x \in \langle 2, \infty \rangle$ and concave when $x \in \langle -2, 2 \rangle$.

Chapter 7

Integration

Integration, along with differentiation, is a fundamental tool in calculus, for instance to solve problems in mathematics and physics like finding the area of an arbitrary shape, the length of a curve, the volume of a solid, the average of a function, just to mention a few examples.

7.1 The definite integral as area under a function

The definite integral (also called *Riemann integral*) is the most common type of integration used by physicists and engineers.

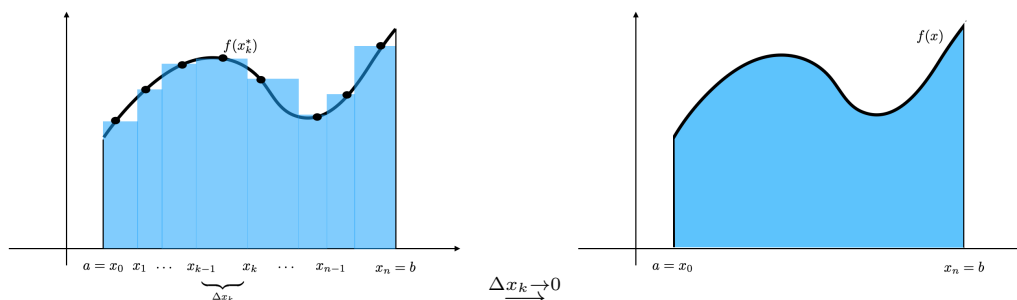
The Riemann integral is defined as taking a limit of a Riemann sum:
Consider a mesh on the interval $[a, b]$, $a = x_0 < x_1 < x_2 < \dots < x_n = b$, partitioning $[a, b]$ in subintervals $[x_{k-1}, x_k]$ of size $\Delta x_k = x_k - x_{k-1}$, $k = 1, \dots, n$ (not necessarily equal). Then,

$$S = \int_a^b f(x)dx = \lim_{\max \Delta x_k \rightarrow 0} \sum_{k=1}^n f(x_k^*) \Delta x_k$$

where $a \leq x \leq b$, x_k^* are arbitrary points in the interval $[x_{k-1}, x_k]$ and $\max \Delta x_k$ (*mesh size*) is the maximum size of all the subintervals.

Note that the integral symbol $\int$ is a “S” for “sum” (of infinitesimal terms) and that dx is a kind of infinitesimal subinterval width (infinitesimal Δx)

As the Riemann integral is defined as the limit of the mesh size going to zero, the Riemann integral S can be interpreted as the *area* under the curve $f(x)$ in the interval $[a, b]$:



Fundamental theorem of calculus

Assume f to be a *continuous* function on an open interval I and $a \in I$. If the function F is defined as:

$$F(x) = \int_a^x f(t)dt$$

then $F'(x) = f(x)$ for all $x \in I$, where $F'(x)$ is the derivative of $F(x)$. The function $F(x)$ is also called an *antiderivative* of f .

Since the derivative of a constant is zero, if $F'(x) = f(x)$ then also $(F(x) + C)' = f(x)$. In other words, antiderivatives, if they exist, are defined up to a constant. Another way of stating this is that the antiderivative is a nonunique inverse of the derivative.

7.2 The definite integral

As seen in the previous section, the area under the function $y = f(x)$ for $a \leq x \leq b$ is then computed with help of the definite integral

$$A = \int_a^b f(x)dx = F(x)\Big|_a^b = F(b) - F(a)$$

where F is any antiderivative of f .

Example 7.1

Find the area under the curve $y = x^2$ in the interval $[0, 2]$

Solution:

$$F(x) = \frac{1}{3}x^3 \text{ is an antiderivative of } x^2, \text{ as } F'(x) = x^2 = f(x). \text{ Hence:}$$

$$\int_0^2 x^2 dx = \frac{1}{3}x^3 \Big|_0^2 = \frac{1}{3}2^3 - \frac{1}{3}0^3 = \underline{\underline{\frac{8}{3}}}$$

◇

7.3 The indefinite integral

Recall: the antiderivative is defined up to a constant. For this reason, an integral without specified extremes is called *indefinite integral* and is defined as:

$$\begin{array}{ccc} & \text{antiderivative} & \\ & \downarrow & \\ \int f(x)dx & = & F(x) + C \\ \uparrow & & \uparrow \\ \text{integrand} & & \text{constant} \end{array} \qquad F'(x) = f(x)$$

7.4 Integration formulae

$\int x^n dx = \frac{1}{n+1}x^{n+1} + C, \quad n \neq -1$	$\int \cos x dx = \sin x + C$
$\int \frac{1}{x} dx = \ln x + C$	$\int \sin x dx = -\cos x + C$
$\int e^x dx = e^x + C$	$\int \tan x dx = -\ln \cos x + C$
$\int a^x dx = \frac{1}{\ln a} a^x + C$	$\int \frac{1}{\cos^2 x} dx = \tan x + C$

7.5 Rules for definite and indefinite integral

Definite integral	Indefinite integral
$\int_a^b f(x) dx = F(x) \Big _a^b = F(b) - F(a)$	$\int f(x) dx = F(x) + C$
$\int_a^a f(x) dx = 0$ $\int_b^a f(x) dx = -\int_a^b f(x) dx$ $\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx$	Linearity: $\int k f(x) dx = k \int f(x) dx, \quad k \text{ constant}$ $\int (A f(x) + B g(x)) dx = A \int f(x) dx + B \int g(x) dx,$ $A, B \text{ constants}$
Improper integral: $\int_a^\infty f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx$ If the limit exists, then the improper integral converges. Otherwise, the integral is divergent.	

Example 7.2

$$\int (1 + 4x + 3\sqrt{x}) dx = x + 4 \cdot \frac{1}{2} x^2 + 3 \cdot \frac{1}{\frac{1}{2} + 1} x^{\frac{1}{2} + 1} + C = x + 2x^2 + 3 \cdot \frac{2}{3} x^{\frac{3}{2}} + C = \underline{\underline{x + 2x^2 + 2x\sqrt{x} + C}}$$

◇

Example 7.3

$$\int \frac{1}{\sqrt{x}} dx = \int x^{-\frac{1}{2}} dx = \frac{1}{-\frac{1}{2} + 1} x^{-\frac{1}{2} + 1} + C = 2x^{\frac{1}{2}} + C = \underline{\underline{2\sqrt{x} + C}} \quad \diamond$$

Example 7.4

$$\int \left(2 + \frac{1}{x} + \frac{1}{x^2}\right) dx = 2x + \ln |x| + \frac{1}{-2 + 1} x^{-2 + 1} = \underline{\underline{2x + \ln |x| - \frac{1}{x} + C}} \quad \diamond$$

7.6 Integration by substitution

If the integrand is a composite function $f(u(x))$ multiplied by the derivative $u'(x)$ of the kernel, then one can use a *change of variables* $x \rightarrow u$ to simplify the integration. To do this, du must be found as function of dx .

Example 7.5

Compute $\int x e^{x^2} dx$.

Solution:

The integrand is the product of $e^{x^2} = e^{u(x)}$ and x , which is proportional to $u'(x)$.

Apply the change of variables: $u = x^2$, $du = 2x dx \Rightarrow dx = \frac{1}{2x} du$:

$$\int x e^{x^2} dx = \int x e^u \frac{du}{2x} = \frac{1}{2} \int e^u du = \frac{1}{2} e^u + C = \underline{\underline{\frac{1}{2} e^{x^2} + C}}$$

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Example 7.6

Compute $\int \cos(3x) dx$.

Solution:

Use $u = 3x$, $du = 3 dx \Rightarrow dx = \frac{1}{3} du$:

$$\int \cos(3x) dx = \int \cos u \frac{1}{3} du = \frac{1}{3} \int \cos u du = \frac{1}{3} \sin u + C = \underline{\underline{\frac{1}{3} \sin(3x) + C}}$$

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Proceeding in a similar way as in the previous examples, we can therefore conclude that for a constant k , one has:

- $\int \sin(kx) dx = -\frac{1}{k} \cos(kx) + C$
- $\int \cos(kx) dx = \frac{1}{k} \sin(kx) + C$
- $\int e^{kx} dx = \frac{1}{k} e^{kx} + C$

7.7 Integration by parts

If the integrand is of the form $u \cdot v'$ and it is easier to integrate $u' \cdot v$, then it is possible to “push” the integral from $u \cdot v'$ to $u' \cdot v$ using the following formula:

$$\int u \cdot v' dx = uv - \int u' \cdot v dx$$

The integration by parts is a consequence of the derivation rule for products: $(uv)' = u'v + uv'$. Since the indefinite integral is an antiderivative, we have $uv = \int (uv)' dx = \int u'v dx + \int uv' dx \Rightarrow$

$$\int uv' = uv - \int u'v \, dx.$$

Integration by parts is useful for integrals of the form:

$$\int xe^{ax} \, dx, \int x \cos(ax) \, dx, \int x \sin(ax) \, dx, \int x \ln x \, dx, \int \ln x \, dx, \text{ and many more.}$$

Example 7.7

Compute $\int xe^{7x} \, dx$

Solution:

$$\text{Set } u = x \ (\Rightarrow u' = 1) \text{ and } v' = e^{7x} \ (\Rightarrow v = \frac{1}{7}e^{7x})$$

$$\int \overset{u}{\downarrow} \overset{v'}{\downarrow} x e^{7x} \, dx = x \cdot \frac{1}{7}e^{7x} - \int 1 \cdot \frac{1}{7}e^{7x} \, dx = \frac{1}{7}xe^{7x} - \frac{1}{7} \int e^{7x} \, dx = \frac{1}{7}xe^{7x} - \frac{1}{49}e^{7x} + C$$

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Example 7.8

Compute $\int x \cos(3x) \, dx$

Solution:

$$\text{Set } u = x \ (\Rightarrow u' = 1) \text{ and } v' = \cos(3x) \ (\Rightarrow v = \frac{1}{3} \sin(3x))$$

$$\begin{aligned} \int \overset{u}{\downarrow} \overset{v'}{\downarrow} x \cos(3x) \, dx &= x \cdot \left(\frac{1}{3} \sin(3x)\right) - \int 1 \cdot \frac{1}{3} \sin(3x) \, dx = \frac{1}{3}x \sin(3x) - \frac{1}{3} \int \sin(3x) \, dx \\ &= \underline{\underline{\frac{1}{3}x \sin(3x) + \frac{1}{9} \cos(3x) + C}} \end{aligned}$$

◇

Example 7.9

Compute $\int x \ln x \, dx$

Solution:

$$\text{Set } u = \ln x \ (\Rightarrow u' = \frac{1}{x}) \text{ and } v' = x \ (\Rightarrow v = \frac{1}{2}x^2)$$

$$\int \overset{v'}{\downarrow} \overset{u}{\downarrow} x \ln x \, dx = \frac{1}{2}x^2 \cdot \ln x - \int \frac{1}{2}x^2 \cdot \frac{1}{x} \, dx = \frac{1}{2}x^2 \ln x - \frac{1}{2} \int x \, dx = \underline{\underline{\frac{1}{2}x^2 \ln x - \frac{1}{4}x^2 + C}}$$

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7.8 Some application of the definite integral

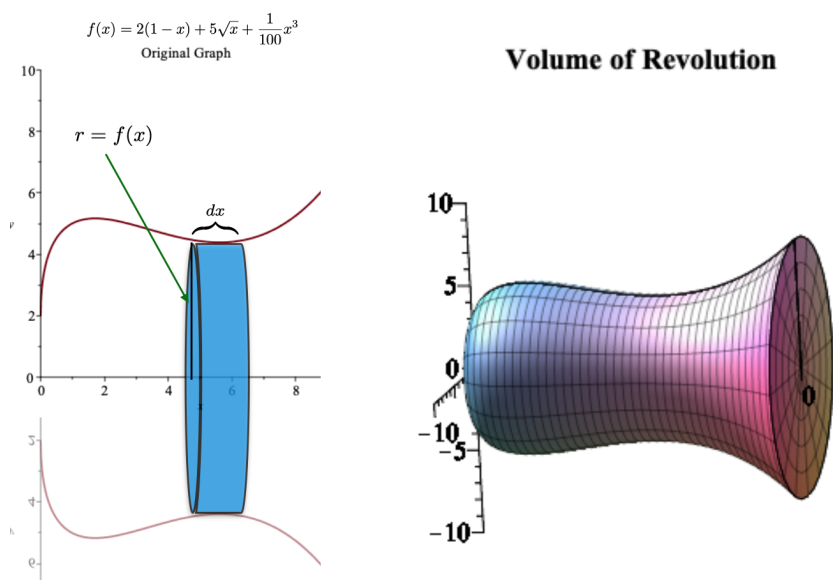
The definite integral has many applications. It can be used, among others, to compute:

- **Areas** (See example 7.1)

- **Volumes**

For instance, when the area under the curve to the function $y = f(x)$, $a \leq x \leq b$, is rotated around the x axis, we divide the object in infinitesimal cylinders of height $h = dx$ and base with radius $r = f(x)$. The area of the base is then $A = \pi r^2 = \pi f^2(x)$, and the volume of the cylinder is $dV = A \cdot h = A dx$. The total volume of the object is obtained by summing over all these infinitesimal volumes dV :

$$V = \int_a^b dV = \int_a^b A dx = \boxed{\int_a^b \pi f^2(x) dx}$$



- **The total consumption**

Assume $y = f(t)$ is the consumption of a good per time unity, as a function of time. The infinitesimal consumption, in a time dt is $f(t)dt$. The total consumption in the time interval $[a, b]$ is then

$$\int_a^b f(t) dt \quad (= \text{area under the curve})$$

- **Average value of a function**

If $y_1, y_2, \dots, y_n$ are n values, the average value of these is $\frac{1}{n} \sum_{i=1}^n y_i = \frac{1}{n} \sum_{i=1}^n y_i \cdot 1$, where 1 is the mesh size and n is the interval size.

Similarly, for a function $y = f(x)$:

$$\bar{y} = \bar{f} = \frac{1}{b-a} \int_a^b f(x) dx$$

where $b - a$ is the mesh width (length of interval) and dx is the infinitesimal mesh size.

Example 7.10

In Norheimesund, the water temperature is measured every day during the summer. The June temperature in Celsius degree, on a specific year, is given by the function

$$T(x) = 13 - 4 \sin\left(\frac{\pi}{16}(x + 2)\right), \quad x \in [0, 30]$$

Compute the average temperature of the water in June that year.

Solution:

$$\begin{aligned} \bar{T} &= \frac{1}{30 - 0} \int_0^{30} T(x) dx = \frac{1}{30} \int_0^{30} (13 - 4 \sin(\frac{\pi}{16}(x + 2))) dx \\ &= \frac{1}{30} \left(13x + 4 \frac{16}{\pi} \cos(\frac{\pi}{16}(x + 2)) \right) \Big|_0^{30} = \frac{1}{30} \left(390 + \frac{64}{\pi} (\cos(2\pi) - \cos(\pi/8)) \right) \approx 13.1^\circ\text{C} \end{aligned}$$

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Exercises Chapter 7

Exercise 7.1

Find the integrals.

- | | | |
|---|----------------------------------|--|
| a) $\int dx$ | b) $\int (x+2) dx$ | c) $\int \left(\sqrt{x} + \frac{1}{\sqrt[3]{x}} + \frac{1}{x} \right) dx$ |
| d) $\int (3 \cos x + 5e^x) dx$ | e) $\int (\cos 2x + 3e^{7x}) dx$ | f) $\int (2x-1)^{99} dx$ |
| g) $\int x^2(x^3+2)^7 dx$ | h) $\int xe^{4x} dx$ | i) $\int \cos x(2+\sin x)^8 dx$ |
| j) $\int \frac{1}{2x+3} dx$ | k) $\int \frac{1}{2x+3} dx$ | l) $\int \frac{1}{(2x+3)^2} dx$ |
| m) $\int x^5 \ln x dx$ | n) $\int \frac{(\ln x)^5}{x} dx$ | o) $\int (1+2x)^8 dx$ |
| p) $\int x \sin x dx$ | q) $\int x \sin 3x dx$ | r) $\int \frac{x^2}{\sqrt{x^3+1}} dx$ |
| s) Using differentiation, show that $\int \frac{3x^2}{x^3+1} dx = \ln x^3+1 + C$ | | |

Exercise 7.2

Find the area of the region defined by the curve of the function $f(x) = e^x + e^{-x}$ and the x -axis in the interval $[0, \ln 2]$

Exercise 7.3 ▲

a) Find the indefinite integrals:

1) $\int x \cdot \cos(x^2) dx$	2) $\int x \cdot \cos x dx$
--------------------------------	-----------------------------

b) Find the definite integrals:

1) $\int_0^{\frac{\pi}{6}} (\cos(3x) - \sin(2x)) dx$	2) $\int_1^2 \frac{x}{2x^2-1} dx$
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Solution to the Exercises Chapter 7

- 7.1** a) $x + C$ b) $\frac{1}{2}x^2 + 2x + C$ c) $\frac{2}{3}x^{\frac{3}{2}} + \frac{3}{2}x^{\frac{2}{3}} + \ln|x| + C$
 d) $3\sin x + 5e^x + C$ e) $\frac{1}{2}\sin 2x + \frac{3}{7}e^{7x} + C$ f) $\frac{1}{200}(2x-1)^{100} + C$
 g) $\frac{1}{24}(x^3+2)^8 + C$ h) $\frac{1}{4}xe^{4x} - \frac{1}{16}e^{4x} + C$ i) $\frac{1}{9}(2+\sin x)^9 + C$
 j) $\frac{1}{2}\ln|2x+3| + C$ k) $\frac{1}{2}\ln|2x+3| + C$ l) $-\frac{1}{2(2x+3)} + C$
 m) $\frac{1}{6}x^6 \ln x - \frac{1}{36}x^6 + C$ n) $\frac{1}{6}(\ln x)^6 + C$ o) $\frac{1}{18}(1+2x)^9 + C$
 p) $-x \cos x + \sin x + C$ q) $-\frac{1}{3}x \cos 3x + \frac{1}{9}\sin 3x + C$ r) $\frac{2}{3}\sqrt{x^3+1} + C$
 s) $[\ln(x^3+1)]' = \frac{3x^2}{x^3+1} \cdot [\ln u]' = \frac{u'}{u}, u = x^3+1$

$$\mathbf{7.2} \quad \int_0^{\ln 2} (e^x + e^{-x}) dx = [e^x - e^{-x}] \Big|_0^{\ln 2} = e^{\ln 2} - e^{-\ln 2} - (e^0 - e^{-0}) = 2 - \frac{1}{2} = \frac{3}{2}$$

$$\mathbf{7.3} \text{ a) } 1) \int x \cdot \cos(x^2) dx = \frac{1}{2} \sin(x^2) + C$$

$$2) \int x \cdot \cos x dx = x \sin x + \cos x + C$$

$$\begin{aligned} \mathbf{b) } 1) \int_0^{\frac{\pi}{6}} (\cos(3x) - \sin(2x)) dx &= \left[\frac{1}{3} \sin(3x) + \frac{1}{2} \cos(2x) \right]_0^{\frac{\pi}{6}} \\ &= \frac{1}{3} \sin \frac{\pi}{2} + \frac{1}{2} \cos \frac{\pi}{3} - \left(\frac{1}{3} \sin 0 + \frac{1}{2} \cos 0 \right) = \frac{1}{12} \end{aligned}$$

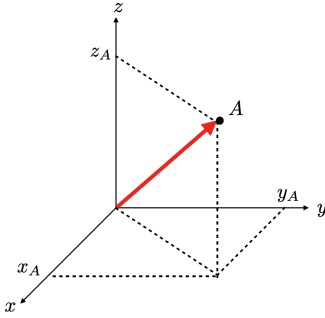
$$2) \int_1^2 \frac{x}{2x^2-1} dx = \left[\frac{1}{4} \ln|2x^2-1| \right]_1^2 = \frac{1}{4} (\ln 7 - \ln 1) = \frac{1}{4} \ln 7$$

Chapter 8

Vectors

8.1 What is a vector?

A vector is a mathematical object that has a given magnitude and a direction. In this chapter we will go over some concepts and formulae for vectors in a three-dimensional coordinate system. Beware that vectors can be higher dimensional, it is not too difficult to extend the concept to vectors with n components, though it is more difficult to visualize.



Given a point A of coordinates (x_A, y_A, z_A) , we can construct a vector from the origin O to A :

$$\vec{OA} = [x_A, y_A, z_A]$$

NOTE: Although in this compendium we write the vectors as *row vectors*, the *column vector* representation:

$$\vec{OA} = \begin{bmatrix} x_A \\ y_A \\ z_A \end{bmatrix}$$

is the most commonly used in more advanced courses.

The length of the vector $\vec{OA}$ is $|\vec{OA}| = \sqrt{x_A^2 + y_A^2 + z_A^2}$

Given two points A and B of coordinates (x_A, y_A, z_A) and (x_B, y_B, z_B) , we can construct the vector $\vec{AB}$:

$$\vec{AB} = [x_B - x_A, y_B - y_A, z_B - z_A]$$

The *length of the vector* $\vec{AB}$ is

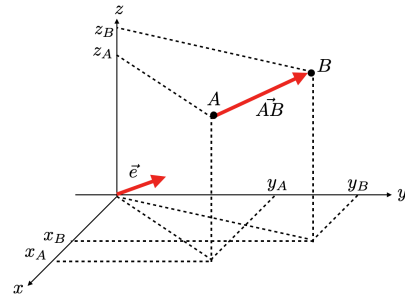
$$|\vec{AB}| = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2}$$

and is the same as the distance between the points A and B .

A vector of length 1 is called *unit vector*. The vector $\vec{e}$:

$$\vec{e} = \frac{\vec{AB}}{|\vec{AB}|}$$

is the *unit vector along* $\vec{AB}$



8.2 Vector algebra

Rather than denoting vectors as $\vec{AB}$, in mathematics it is not uncommon to denote vectors with boldface fonts, like $\mathbf{\vec{u}}$ or simply $\mathbf{u}$

Another common notation is that the component of a vector are usually denoted as u_1, u_2, u_3 rather than $x_{\mathbf{u}}, y_{\mathbf{u}}, z_{\mathbf{u}}$

Consider the vectors $\mathbf{u} = [u_1, u_2, u_3]$ and $\mathbf{v} = [v_1, v_2, v_3]$:

$$\begin{aligned}\mathbf{u} + \mathbf{v} &= [u_1, u_2, u_3] + [v_1, v_2, v_3] = [u_1 + v_1, u_2 + v_2, u_3 + v_3] \\ \mathbf{u} - \mathbf{v} &= [u_1, u_2, u_3] - [v_1, v_2, v_3] = [u_1 - v_1, u_2 - v_2, u_3 - v_3] \\ \mathbf{u} + \mathbf{0} &= \mathbf{0} + \mathbf{u} = \mathbf{u} \\ t\mathbf{u} &= t[u_1, u_2, u_3] = [tu_1, tu_2, tu_3]\end{aligned}$$

Length of a vector: $|\mathbf{u}| = \sqrt{u_1^2 + u_2^2 + u_3^2}$

8.3 Scalar product

The *scalar product* between two vectors $\mathbf{u}$ and $\mathbf{v}$ is defined as:

$$\mathbf{u} \cdot \mathbf{v} = [u_1, u_2, u_3] \cdot [v_1, v_2, v_3] = u_1v_1 + u_2v_2 + u_3v_3$$

One has:

$$\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}| |\mathbf{v}| \cos \theta$$

where θ is the angle between the two vectors, that can be computed from

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}| |\mathbf{v}|}$$

Two nonzero vectors are *orthogonal* if their scalar product is zero:

$$\mathbf{u} \perp \mathbf{v} \Leftrightarrow \mathbf{u} \cdot \mathbf{v} = 0 \Leftrightarrow u_1v_1 + u_2v_2 + u_3v_3 = 0$$

(The definition of orthogonality is a generalization of the concept of perpendicularity of f.ex. segments)

Two nonzero vectors are *parallel* if one is multiple of the other:

$$\mathbf{u} \parallel \mathbf{v} \Leftrightarrow \mathbf{v} = t\mathbf{u} \text{ for some } t \neq 0$$

Another way to check that $\mathbf{u} \parallel \mathbf{v}$ is to check whether $\frac{u_1}{v_1} = \frac{u_2}{v_2} = \frac{u_3}{v_3}$ (provided that none of these are zero).

8.4 Vector product

Denote by $\mathbf{i}, \mathbf{j}, \mathbf{k}$ the unit vectors along the x -, y -, and z -direction, respectively. Then any vector $\mathbf{a} = [a_1, a_2, a_3]$ can be written as $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$.

The *vector product* between two vectors $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$ is defined as:

$$\mathbf{a} \times \mathbf{b} = \mathbf{i}(a_2b_3 - a_3b_2) + \mathbf{j}(a_3b_1 - a_1b_3) + \mathbf{k}(a_1b_2 - a_2b_1) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

and has length $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}| |\mathbf{b}| \sin \theta$.

8.5 Equation of a straight line

i) Given a point and a direction

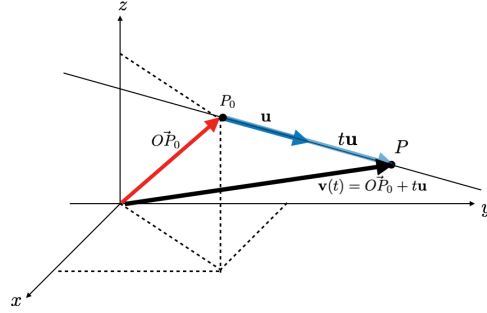
Given a point P_0 with coordinates (x_0, y_0, z_0) and a vector $\mathbf{u} = [u_1, u_2, u_3]$, the equation of the line through P_0 and parallel to $\mathbf{u}$ is $\mathbf{v}(t) = \vec{OP_0} + t\mathbf{u}$.

An arbitrary point P on such line has coordinates (x, y, z) :

$$P : \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} + t \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

or

$$P : \begin{cases} x = x_0 + tu_1 \\ y = y_0 + tu_2 \\ z = z_0 + tu_3 \end{cases}$$



ii) Given two points

Find the direction vector between the two points, and use i)

8.6 Equation of a plane

Plane through a point and normal to a vector

We shall find the equation of a *plane perpendicular* to a vector $\mathbf{n} = [a, b, c]$ (normal vector) and passing through a point $P_0(x_0, y_0, z_0)$.

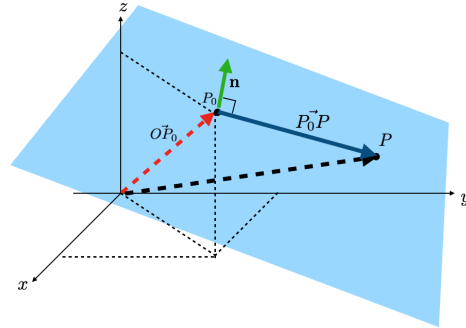
If $P(x, y, z)$ is any point on the plane, a necessary and sufficient condition is $\vec{P_0P} \perp \mathbf{n}$:

$$\vec{P_0P} \cdot \mathbf{n} = 0 \Leftrightarrow [x - x_0, y - y_0, z - z_0] \cdot [a, b, c] = 0$$

or, equivalently

$$\boxed{\vec{P_0P} \cdot \mathbf{n} = 0 \Leftrightarrow a(x - x_0) + b(y - y_0) + c(z - z_0) = 0}$$

(equation of the plane through P_0 and with normal $\mathbf{n}$)



Plane defined by two vectors and a point

Two nonzero, non-parallel vectors $\mathbf{u}$ and $\mathbf{v}$ define a plane passing through O . The points on the plane are $t\mathbf{u} + s\mathbf{v}$.

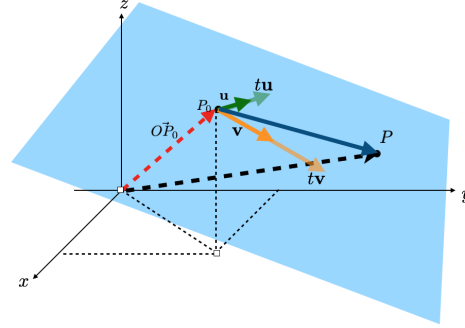
For $P(x, y, z)$ to be in the plane generated by $\mathbf{u}$ and $\mathbf{v}$ we must require $\vec{OP} = t\mathbf{u} + s\mathbf{v}$.

For the plane to pass through P_0 , we must transpose it by the vector $\vec{OP_0}$:

$$\vec{OP} = \vec{OP_0} + t\mathbf{u} + s\mathbf{v}$$

An arbitrary point P on such line has coordinates (x, y, z) :

$$P : \begin{cases} x = x_0 + tu_1 + sv_1 \\ y = y_0 + tu_2 + sv_2 \\ z = z_0 + tu_3 + sv_3 \end{cases}$$

**8.7 Distance of a point to a plane**

The distance of a point $P(x_1, y_1, z_1)$ to the plane $ax + by + cz + d = 0$ is given by:

$$l = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$$

8.8 Projection of a vector on another vector

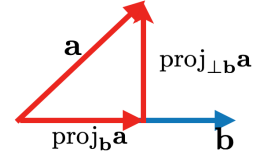
Given a vector $\mathbf{a} = [a_1, a_2, a_3]$ and a vector $\mathbf{b} = [b_1, b_2, b_3] \neq \mathbf{0}$, let $\mathbf{e}$ be the unit vector along the direction $\mathbf{b}$.

The *orthogonal projection* of $\mathbf{a}$ on $\mathbf{b}$ is the component of $\mathbf{a}$ along $\mathbf{b}$:

$$\text{proj}_{\mathbf{b}} \mathbf{a} = (|\mathbf{a}| \cos \theta) \frac{\mathbf{b}}{|\mathbf{b}|} = (|\mathbf{a}| \cos \theta) \mathbf{e} = (\mathbf{a} \cdot \mathbf{e}) \mathbf{e} \quad (\text{proj}_{\mathbf{b}} \mathbf{a} \parallel \mathbf{b})$$

As $\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|}$ one can write

$$\text{proj}_{\mathbf{b}} \mathbf{a} = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\mathbf{b} \cdot \mathbf{b}} \right) \mathbf{b}$$

**Orthogonal decomposition theorem**

Given a vector $\mathbf{b} = [b_1, b_2, b_3] \neq \mathbf{0}$, then any vector $\mathbf{a} = [a_1, a_2, a_3]$ can be decomposed in a vector $\mathbf{a}_{\parallel} \parallel \mathbf{b}$ and a vector $\mathbf{a}_{\perp} \perp \mathbf{b}$:

$$\mathbf{a} = \mathbf{a}_{\parallel} + \mathbf{a}_{\perp} = \text{proj}_{\mathbf{b}} \mathbf{a} + \text{proj}_{\perp \mathbf{b}} \mathbf{a}, \quad \text{proj}_{\mathbf{b}} \mathbf{a} = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\mathbf{b} \cdot \mathbf{b}} \right) \mathbf{b}, \quad \text{proj}_{\perp \mathbf{b}} \mathbf{a} = \mathbf{a} - \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\mathbf{b} \cdot \mathbf{b}} \right) \mathbf{b}$$

8.9 Some further useful properties in vector calculus

- The angle between two planes is the angle between the normals of the two planes.

- A quadrilateral $ABCD$ is a parallelogram if $\vec{AB} = \vec{DC}$
- Three points A, B, C are on the same line if $\vec{AB} \parallel \vec{AC}$ or $\vec{AB} \times \vec{AC} = \vec{0}$
- Four points A, B, C, D lie in the same plane if $(\vec{AB} \times \vec{AC}) \cdot \vec{AD} = 0$

Exercises Chapter 8

Exercise 8.1

Consider the vectors

$$\mathbf{u} = [a, b, c] \text{ and } \mathbf{v} = [d, e, f]$$

- a) Write a formula for the scalar product between $\mathbf{u}$ and $\mathbf{v}$ using the components.
- b) Write down the formula for the length of these vectors.
- c) How can we find the angle between these vectors?
- d) What can we say about the angle between these two vectors if the scalar product is a negative number?
- e) How can we determine if these two vectors are parallel?

Exercise 8.2

Consider the points $A(2, 0, 0)$, $B(0, 3, 0)$ and $C(0, 0, 5)$.

- a) Draw the points in a coordinate system.
- b) Show, using the sketch, that a vector from the origin to the point A is perpendicular to the unit vectors along the y -axis and the z -axis. That is: $\vec{OA} \perp \vec{j}$ and $\vec{OA} \perp \vec{k}$
- c) Find the coordinates of $\vec{AB}$, $\vec{AC}$, $\vec{CA}$, $\vec{BC}$
- d) Determine the length of $\vec{AB}$ and $\vec{AC}$. Determine the angle between the vectors $\vec{AB}$ and $\vec{AC}$.
- e) Determine the sum $\vec{AB} + 2\vec{AC} + \vec{BC}$.
- f) Determine the vector $\vec{x}$ such that $\vec{AB} + \vec{BC} + 2\vec{x} = 0$.

Exercise 8.3

- a) Consider the points $A(1, 3, -1)$, $B(2, 1, 2)$ and $C(4, 3, k)$.
Determine k such that $\vec{AB} \perp \vec{AC}$.
- b) Determine t such that the vectors $[t, -1, 2]$ and $[2, 2, -4]$ are parallel.

Exercise 8.4

In a parallelogram $ABCD$, AC is a diagonal. The corners have the coordinates $A(-3, 1, -1)$, $B(2, 3, 0)$ and $C(5, 1, 1)$. Determine the coordinates of D .

Solution to the Exercises Chapter 8

8.1 d) The angle between the vectors is larger than 90° .

8.2 b) $\vec{OA} \cdot \vec{j} = [2, 0, 0] \cdot [0, 1, 0] = 0$ $\vec{OA} \cdot \vec{k} = [2, 0, 0] \cdot [0, 0, 1] = 0$

c) $\vec{AB} = [-2, 3, 0]$, $\vec{AC} = [-2, 0, 5]$, $\vec{CA} = -\vec{AC} = [2, 0, -5]$ and $\vec{BC} = [0, -3, 5]$.

d) $|\vec{AB}| = \sqrt{13}$, $|\vec{AC}| = \sqrt{29}$. $\cos A = \frac{\vec{AB} \cdot \vec{AC}}{|\vec{AB}| |\vec{AC}|} = \frac{4}{\sqrt{13 \cdot 29}} \Rightarrow A \approx 78.1$ degrees.

e) $\vec{AB} + 2\vec{AC} + \vec{BC} = [-6, 0, 15]$ or

$\vec{AB} + 2\vec{AC} + \vec{BC} = \vec{AB} + \vec{BC} + 2\vec{AC} = \vec{AC} + 2\vec{AC} = 3\vec{AC} = 3[-2, 0, 5] = [-6, 0, 15]$

f) $\vec{AB} + \vec{BC} + 2\vec{x} = 0 \Rightarrow 2\vec{x} = -\vec{AC} \Rightarrow \vec{x} = -\frac{1}{2}\vec{AC} = [1, 0, -\frac{5}{2}]$

8.3 a) $\vec{AB} \cdot \vec{AC} = [1, -2, 3] \cdot [3, 0, k+1] = 0 \Rightarrow 3 + 0 + 3(k+1) = 0 \Rightarrow k = -2$

b) $t = -1$

8.4 We can use that $\vec{AB} = \vec{DC}$, where $D(x, y, z)$. From this, we obtain $D(0, -1, 0)$.